TRAINING REFERENCE · NCVT / DGT

ITI Tool Compendium

Tools, equipment and training aids used across the Craftsmen Training Scheme (CTS / ITI) and the Crafts Instructor Training Scheme (CITS / NSTI)

With identification points, operating essentials, care routines and safety precautions for every entry

Engineering & non-engineering trades · minor and major tools · hand tools, machine tools, metrology, welding, electrical, electronics, automotive, refrigeration, construction, textile, printing, food, beauty, health and instructional aids

PREPARED FOR	COVERAGE	SOURCE BASIS
ITIs, NSTIs, CITS instructor trainees and instructors under NCVT / DGT	226 numbered tool entries in 11 groups, plus framing and reference chapters	DGT / NCVT trade curricula, Annexure-I tool and equipment lists, and standard workshop safety practice

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A research compilation, not an official DGT publication. Verify specifications against the current curriculum PDF for the trade before use in an indent or an audit.

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The register in Part B is indexed by tool group. Each entry within a group carries a code in the form group-letter plus serial number.

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About This Compendium

This compendium consolidates every class of tool, instrument, machine, test equipment and training aid that appears in the Craftsmen Training Scheme (CTS / ITI) and Crafts Instructor Training Scheme (CITS / NSTI) curricula published by the Directorate General of Training (DGT) under the National Council for Vocational Training (NCVT). Tools are grouped by engineering practice rather than by trade, so a single instrument appears once with every trade that uses it. For each tool the document records what the trainee or instructor must be able to identify, the operation(s) they must be able to perform, the inspection and care routine, and the specific safety precautions and personal protective equipment required. The scope covers engineering and non-engineering trades, minor and major tools, hand tools, precision metrology, machine tools, electrical and electronic test instruments, welding and gas equipment, automotive, refrigeration, fabrication, ICT, and the Training Methodology (TM) aids used by CITS instructor trainees.

How to use it

	Part / feature	What it gives you
1	Part A	Framing chapters. Read Ch. 1–6 once to understand the CTS/CITS structure, NSQF levels, and the universal safety hierarchy that applies to every tool in the document.
2	Part B	The master tool register (Ch. 7), 216 numbered entries in eleven practice-wise groups, each with identification points, operating essentials, care, safety precautions and PPE. This is the core deliverable.
3	Part C	Instructor and reference material: the CITS Training Methodology aid list (Ch. 8), tool competence checklists (Ch. 9), a trade-to-tool index (Ch. 10), glossary and bibliography (Ch. 11) and the limitations of this compilation (Ch. 12).
4	Numbering	Every tool carries a stable code of the form group-letter + serial (e.g. M-014 is the fourteenth measuring and marking tool). Codes can be quoted in job sheets, store indents and inspection checklists.
5	Columns	Name / Identification = what to look at to recognise and verify the tool. Operate = the minimum competence expected. Safety / PPE = the non-negotiable controls. Care = what keeps the tool accurate and serviceable.

A note on conventions: quantities and specifications are quoted as they appear in the DGT source curriculum where available, and as typical workshop practice otherwise. Every parameter that a trainee might act on — speed, pressure, torque, test voltage, temperature — must be confirmed against the machine manual and the shop standard operating procedure before use.

PART A

Framework

What CTS and CITS are, how tools enter the curriculum, and the safety hierarchy that applies to every tool in the register.

- 1. Scope, sources and method
- 2. The CTS and CITS curriculum framework
- 3. Universal safety framework for workshop tools
- 5. PPE and safety signage matrix
- 6. Power tool and machine guarding rules
- 7. Calibration, care, storage and 5S

1. Scope, Sources and Method

1.1 What this document is

This is a research compilation, not an official DGT publication. It gathers into one place the tool and equipment content that is normally scattered across several hundred trade curriculum PDFs on dgt.gov.in, bharatskills.gov.in and the NSTI/ITI websites. The intent is that an instructor preparing a demonstration, a trainee preparing for the All India Trade Test (AITT), or an institute preparing a store indent can find the tool, its identification points, its operation and its hazards without opening a dozen documents.

1.2 What CTS and CITS are

The Craftsmen Training Scheme (CTS) was introduced in 1950 and is delivered through roughly 15,000 ITIs (about 2,700 government and 12,300 private) and 33 National Skill Training Institutes. Duration ranges from six months to two years, with engineering trades normally two years and non-engineering trades normally one year. Training is approximately 70 per cent practical and 30 per cent theory, examined by the All India Trade Test and certified by NCVT as the National Trade Certificate (NTC). The Crafts Instructor Training Scheme (CITS) is the one-year instructor qualification delivered at NSTIs; it combines Trade Technology, Engineering Technology / Vocational Calculation and Science, and Training Methodology, and leads to the National Craft Instructor Certificate (NCIC).

1.3 Source hierarchy used

Wherever possible, tool names follow the Annexure-I 'List of Tools and Equipment' of the current DGT competency-based curriculum for the trade, because that annexure is the document an institute is actually audited against. Primary sources consulted include the CTS 3.0 and revised CTS curricula for Fitter and Electrician, the CITS NSQF-5/6 curricula for Fitter and for Core Skills (Training Methodology, Workshop Calculation and Science, Engineering Drawing), and the trade curriculum index on bharatskills.gov.in and cstaricalcutta.gov.in. The full list appears in the Bibliography (Ch. 11).

1.4 Level of confidence

Statements about curriculum structure, subject names, NSQF levels, pass criteria and tool names are taken from published DGT material. Statements about operating technique and safety are compiled from standard Indian and international workshop practice (IS/BIS, OSHA, ISO and manufacturer guidance) and should be read as good practice, not as a substitute for the shop safety manual or a manufacturer's operating instructions. Where a parameter is safety-critical (for example gas welding pressures or electrical isolation), the local standard operating procedure prevails.

1.5 How the tool groups were derived

Trade curricula were read from the practical side: each Professional Skill learning outcome names the operations to be performed, and each operation implies a tool. Those tools were then de-duplicated across trades and grouped by the physical principle they exploit — marking, measuring, metal removal by hand, metal removal by machine, joining, forming, electrical, electronic, fluid, thermal, automotive, ICT and instructional. A tool that serves two groups is placed in the group of its primary function and cross-referenced. This grouping is why the same vernier caliper serves Fitter, Turner, Machinist, Tool and Die Maker, Motor Vehicle and Draughtsman, and appears once here rather than a dozen times.

1.6 Limits

The register is broad but not exhaustive. Very specialised equipment that appears in only one or two advanced trades (for example semiconductor fabrication trainers or marine engine test beds) is represented at class level rather than item level. Regional and State Council variants, and trades introduced after the source curricula were published, may add items not listed here.

2. The CTS and CITS Curriculum Framework

2.1 Subject structure in CTS

An engineering CTS trade carries Trade Practical (Professional Skill), Trade Theory (Professional Knowledge), Engineering Drawing, Workshop Calculation and Science, and Employability Skills. Non-engineering trades carry Trade Practical, Trade Theory and Employability Skills, with Vocational Calculation and Science where applicable. In a typical two-year engineering trade the notional split is 1,680 hours practical and 720 hours theory over 2,400 hours, i.e. 70:30. Employability Skills is a 60-hour component in each year. A representative one-year structure shows Trade Practical 840 hours, Trade Theory 360 hours and Employability Skills 60 hours over 1,200 hours.

2.2 Where tools enter the curriculum

Tools appear at three levels. First, the introductory block of every trade — the trainee is taught the trade's tool list, safety attitude, PPE, first aid, safe disposal of swarf and waste, hazard identification, safety signs, electrical accident prevention and fire extinguisher use before touching a machine. Second, in the practical task columns, where each job names the operation and therefore the tool. Third, in Annexure-I, which is the auditable list of trainee tool kits, shop tools and accessories, and machinery with specification and quantity for a batch (commonly 20 or 25 trainees, plus one for the instructor).

2.3 The CITS structure

CITS is a one-year, NSQF level 5 or 6 programme. Trade Technology comprises Professional Skill and Professional Knowledge at basic and advanced level. Training Methodology is common to all trades and is split into TM Practical and TM Theory (approximately 270 + 180 hours in the NSQF-6 Fitter document), with a mandatory 150 hours of On-the-Job Training or group project where industry attachment is not available. Engineering Technology (Workshop Calculation and Science, Engineering Drawing) or Vocational Calculation and Science and Soft Skills complete the programme. The instructor is assessed on all components — for example 40 per cent in Trade Theory, 40 per cent in Principles of Teaching, 40 per cent in Vocational Calculation and Science and 60 per cent in Trade Practical, with 80 per cent attendance — before being awarded the NCIC.

2.4 The two audiences for a tool list

A CTS trainee needs to identify a tool, select it correctly, use it safely and maintain it. A CITS instructor trainee needs all of that plus the ability to demonstrate it, to explain its construction and principle, to prepare the job sheet or operation sheet that uses it, to specify it in an indent, to inspect it, and to teach the safety precautions around it. This document therefore records identification, operation, care and safety for every entry, and adds a separate instructor-focused chapter (Ch. 8) for the Training Methodology aids.

2.5 NSQF level expectation

NSQF level determines the depth of tool competence. At levels 3 and 3.5 (COPA, DTP, Draughtsman variants, many one-year service trades) the expectation is correct routine use under supervision. At level 4 (most one-year engineering and two-year non-engineering trades) the trainee works independently to specification. At level 5 (Electrician, Fitter variants, advanced maintenance trades) the trainee diagnoses, sets up and inspects. CITS instructors are trained at level 5 or 6, which is why they must be able to teach, assess and improve the use of the same tools.

2.6 Representative CTS trades by discipline

The trade list below is a consolidated view of the trades currently offered under CTS across ITIs and NSTIs. It is grouped by discipline so that a reader can find the tool group that applies. A single trade may draw on several tool groups.

Machining & Fitting	Fitter, Turner, Machinist, Machinist (Grinder), Tool & Die Maker (Dies & Moulds), Tool & Die Maker (Press Tools, Jigs & Fixtures), Operator Advanced Machine Tools, Mechanic Machine Tools Maintenance
Fabrication & Welding	Welder (Gas & Electric), Welder (Pipe), Sheet Metal Worker, Structural Fabricator, Forger & Heat Treater, Foundryman Technician, Mechanic Mining Machinery
Electrical	Electrician, Wireman, Lift & Escalator Mechanic, Mechanic Power Electronics, Electroplater
Electronics & ICT	Electronics Mechanic, Mechanic Consumer Electronics Appliances, Mechanic Industrial Electronics, Mechanic Medical Electronics, Mechanic Computer Hardware, Mechanic Radio & TV, Mechanic Communication Equipment Maintenance, ICT System Maintenance, Mechanic-cum-Operator Electronics Communication System, Mechanic Mechatronics, Network Technician, Data Entry Operator, COPA, DTP Operator, Web Designing & Computer Graphics, Multimedia Animation & Special Effects
Automotive	Mechanic Motor Vehicle, Mechanic (Auto Electrical & Electronics), Mechanic Diesel Engine, Mechanic Repair & Maintenance of Light/Heavy Vehicles, Mechanic Repair & Maintenance of Two Wheelers, Mechanic Tractor, Driver-cum-Mechanic, Mechanic Agricultural Machinery, Marine Engine Fitter, Vessel Navigator
Refrigeration, HVAC & Fluid	Mechanic (Refrigeration & Air-Conditioner), Plumber, Pump Operator-cum-Mechanic, Sanitary Hardware Fitter, Attendant Operator (Chemical Plant), Instrument Mechanic (Chemical Plant), Maintenance Mechanic (Chemical Plant), Laboratory Assistant (Chemical Plant)
Building & Construction	Mason (Building Constructor), Carpenter, Painter General, Domestic Painter, Industrial Painter, Draughtsman (Civil), Architectural Draughtsman / Architectural Assistant, Interior Design & Decoration, Surveyor, Building Maintenance, Stone Mining/Processing Machine Operator, Excavator Operator (Mining)
Textile, Apparel & Leather	Dress Making, Cutting & Sewing, Sewing Technology, Fashion Design & Technology, Surface Ornamentation Techniques, Computer Aided Embroidery & Designing, Textile Mechatronics, Weaving Technician, Spinning Technician, Textile Wet Processing Technician, Leather Goods Maker, Footwear Maker
Printing & Media	Litho Offset Machine Minder, Plate Maker-cum-Impositor, Process Cameraman, Digital Photographer, Photographer, Desktop Publishing Operator
Food, Agriculture & Allied	Craftsman Food Production (General/Vegetarian), Baker & Confectioner, Catering & Hospitality Assistant, Food & Vegetable Processing, Dairying, Horticulture, Floriculture & Landscaping, Agro Processing, Cane & Bamboo Worker
Service, Health & Beauty	Basic Cosmetology, Spa Therapy, Physiotherapy Technician, Health Sanitary Inspector, Health Safety & Environment, Hospital House Keeping, Radiology Technician, Old Age Care Assistant, Creche Management, Firemen, Fire Technology, Counselling Skills
Business, Office & Emerging	Secretarial Practice (English), Stenographer & Secretarial Assistant (Hindi/English), Office Assistant-cum-Computer Operator, Finance Executive, Marketing Executive, Human Resource Executive, Front Office Assistant, Library & Information Science, Travel & Tourism, Tourist Guide, Insurance Agent, Artificial Intelligence Programming Assistant, Additive Manufacturing, Drone Technician, Semiconductor Technician, Information Technology

Table 2.1 — CTS trades grouped by discipline. Source: DGT and NSTI trade listings.

2.7 Subjects and their tool implications

Subject	Common abbreviation	What it covers	Tool implication
Trade Practical	TP / PS	Workshop jobs; the professional skill component, about 70 per cent of hours	All physical tools, machines and consumables
Trade Theory	TT / PK	Principles, construction, materials, calculations of the trade	Tool construction, selection and specification knowledge
Engineering Drawing	ED	Reading and making drawings, projection, sections, tolerances	Drawing board, instruments, CAD workstation, plotter
Workshop Calculation & Science	WCS	Applied arithmetic, mensuration, physics, materials	Measuring instruments, calculators, lab apparatus
Employability Skills	ES	Communication, IT literacy, entrepreneurship, safety, quality	Computer, projector, PPE for ES demonstrations
Training Methodology	TM (CITS only)	Educational psychology, methods, instructional technology, assessment	Training aids: boards, OHP, LCD, camera, smart board

Subject	Common abbreviation	What it covers	Tool implication
Vocational Calculation & Science	VCS (CITS)	Trade-relevant mathematics and science for instructors	Measuring instruments, demonstration apparatus
On-the-Job Training / Group Project	OJT (CITS)	150 hours in industry or a group project	Industry-standard tools and equipment, project tooling

Table 2.2 — Curriculum subjects and the tools each implies.

2.8 Assessment thresholds

Programme	Theory requirement	Practical requirement	Other
CTS (typical ITI)	33% Trade Theory; 33% Employability Skills	60% Trade Practical	Minimum 80% attendance to sit AITT
CITS (NSTI, Fitter NSQF-6)	40% Trade Theory; 40% Principles of Teaching; 40% VCS	60% Trade Practical	80% attendance

Table 2.3 — Typical pass criteria. Requirements vary by NSTI and by year; confirm against the current scheme document.

3. Universal Safety Framework for Workshop Tools

3.1 Why safety is taught before the tool

Every CTS curriculum opens with the same block before the first job: importance and list of trade tools, safety attitude development and PPE, first aid, safe disposal of waste such as cotton waste and metal chips, hazard identification and avoidance, safety signs for danger, warning, caution and personal safety, preventive measures for electrical accidents, and use of fire extinguishers. This ordering is deliberate. A trainee who cannot recognise a hazard cannot be trusted with a machine, and an instructor who cannot teach the hazard cannot demonstrate the tool. The paragraphs below summarise the common framework; each tool entry in the tool entries in Part B then add the specific controls.

3.2 Hierarchy of control applied to workshop tools

Controls are applied in order of effectiveness and only then by PPE. Elimination: do not perform the operation by hand if a guarded machine or a jig can do it. Substitution: use a file instead of a dry grinder where the finish allows, use a low-voltage tool in wet areas. Engineering control: fixed guards, interlocked chuck guards, machine vice instead of hand holding, local exhaust ventilation for fumes and dust, residual current devices, machine anchoring. Administrative control: standard operating procedure, permit to work, tool-box talk, job rotation, housekeeping and 5S, restricted access to welding and gas areas, inspection and calibration schedules. PPE: the last line, never the only line.

3.3 Hazard classes encountered with tools

Mechanical — cutting edges, entrapment in rotating parts, ejection of work or tool bits, flying swarf, crushing, burns from hot work, stored-energy release from springs and hydraulics. Electrical — shock, arc flash, fire, and the special case of stored charge in capacitors. Chemical — acids, alkalis, solvents, fluxes, electrolytes, refrigerant, paint and adhesive vapours. Physical — noise above 85 dB(A), hand-arm vibration, heat, cold, radiation (ultraviolet from arc, laser in level and alignment tools, X-ray in non-destructive testing). Ergonomic — manual handling, repetitive filing, poor posture at a bench or under a vehicle. Fire and explosion — fuel gas, solvent vapour, dust. Biological and general — confined spaces, poor ventilation, slips on oil and swarf.

3.4 The ten workshop rules that apply to every tool

One: receive instruction and demonstration before first use of any tool. Two: wear the PPE specified for that tool — no loose clothing, no loose sleeves, no rings, watches or chains near rotating equipment, hair tied back. Three: never remove or defeat a guard or an interlock. Four: hold work in a vice, clamp or fixture, never in the hand, when drilling, punching or chiselling. Five: keep hands clear of the cutting line and never leave a key, chuck key or spanner on a machine. Six: isolate, lock off and prove dead before adjusting or maintaining any electrical or powered equipment. Seven: use the right tool for the material and the size; do not extend a spanner with a pipe or hammer a wrench. Eight: keep the work area clear — swarf brushed, not blown; oily rags in a metal bin; walkways and fire points unobstructed. Nine: report every defect, crack, frayed cable, leaking regulator or missing guard and withdraw the tool from service. Ten: know the location of the first aid box, eye wash, fire extinguisher, main switch and emergency exit, and know how to raise the alarm.

3.5 Power tool safety rules

Check the tool and cable before use and use an RCD/ELCB-protected supply. Never carry a tool by its cable or pull the plug out by the cable. Use double-insulated (Class II) or low-voltage tools where possible. Match the wheel, blade or bit to the tool's rated speed and material. Abrasive wheels must be ring-tested, correctly mounted with blotting paper washers, guarded and never run above the marked maximum speed; the tool rest clearance on a bench grinder is kept within about 2 mm and the tongue guard within about 3 mm. Hold a grinder so that sparks fly away from the body and never towards another person or a gas cylinder. Do not use a cut-off wheel for side grinding. Let a moving part stop completely before setting the tool down. In confined or wet locations use a 110 V site transformer or battery tools.

3.6 Machine tool rules

Only a trained and authorised person operates a machine. Confirm the guard is in place and functional, the emergency stop works, and the workholding is tight. Remove the chuck key before starting a lathe or drill — treat it as a reflex. Set

the correct spindle speed and feed for the material; never change speed or gearing while the spindle is moving. Do not reach over or lean on a rotating workpiece, and never measure, adjust, clean or clear chips while it turns. Stop the machine before changing a tool or a workpiece. Use a brush or hook, not fingers or compressed air, to clear chips. Keep the floor around the machine dry and free of swarf and coolant. On CNC equipment, verify the programme, the tool offsets and the simulated path in single block and with the rapid override reduced before the first cut, and know where the feed hold and emergency stop are.

3.7 Electrical safety essentials

Treat every conductor as live until proved dead with a tester of the correct voltage range. Isolate at the main switch, lock and tag, then verify at the point of work. Use insulated tools with the correct voltage rating; inspect insulation for cuts before each use. Earth every metal enclosure, use three-pin plugs, and never bridge a fuse or a protective device. Work with one hand where practicable and stand on a dry insulating mat. Do not work alone on live equipment; a second person must be present and able to isolate the supply. Capacitors retain charge — discharge through a resistor and short them. Do not use water on an electrical fire; use CO2 or dry powder. Any shock victim needs medical attention even if they appear unharmed, because cardiac arrhythmia may follow.

3.8 Compressed gas cylinder rules

Store cylinders upright, secured by chain or strap, with valve caps fitted, away from heat, ignition and sunlight, and with oxygen and fuel gas separated by distance or a fire-resistant barrier. Never use oil or grease on an oxygen fitting — oxygen plus oil is an explosive combination. Before connecting, crack the cylinder valve briefly to blow out dust, standing to the side of the outlet. Use only the regulator specified for that gas, and fit a flashback arrester at both the torch and the regulator end. Open the cylinder valve slowly and fully for oxygen, about one turn for fuel gas unless the manufacturer specifies otherwise. Test every joint for leaks with soap solution, never with a flame. Keep cylinders at least about 3 m from the flame, or shield them. Close the cylinder valve and drain the hoses after work, then back off the regulator screws.

3.9 Fire, first aid and emergency response

Fires are classified A (solid), B (flammable liquid), C (flammable gas), D (metal) and F (cooking oil), with electrical fires treated as a separate risk. Water suits class A only and is never used on live electricity, liquids or metal fires. Foam suits A and B. CO2 suits electrical and liquid fires and is clean but has poor cooling and can displace oxygen in a small space. Dry powder is the general-purpose choice. Sand and dry powder are used for metal fires. In first aid, the priorities are to make the scene safe, then airway, breathing and circulation. For burns, cool with clean running water for at least ten to twenty minutes and do not apply ointment, ice or cotton. For eye injury or foreign body, wash from the inner to the outer corner or use an eye wash station and never rub. For electric shock, isolate before touching the casualty. For a deep cut, apply direct pressure with a clean dressing and elevate. For fainting, lay the person flat and raise the legs. All incidents, including near misses, are recorded and reviewed.

3.10 Housekeeping and 5S

5S — Sort, Set in order, Shine, Standardise, Sustain — is explicitly part of the CTS and CITS syllabus. Applied to tools it means: no tool without a place and no place without a tool; shadow boards and tool kits checked at issue and return; surfaces and machines cleaned after each job; measuring instruments wiped, oiled and cased; calibrated instruments segregated with status labels; damaged tools tagged and withdrawn; and a weekly audit of the shop floor. Good housekeeping is also the cheapest accident prevention available, since most slips, trips, fires and lost tools trace back to it.

4. Personal Protective Equipment and Safety Signage Matrix

4.1 Selection logic

PPE is selected from the hazard, not from habit. The matrix below maps the common ITI operations to the minimum PPE that should be issued. Where an operation appears in more than one row, the more protective requirement applies. Every item must be to a recognised standard, fit the wearer, and be inspected before use.

4.2 What each item protects against

Safety glasses protect against flying chips and dust from the front and the side, so side-shielded types are preferred. A face shield adds protection against larger fragments, hot slag and chemical splash but is not a substitute for glasses. A welding helmet combines a filter against ultraviolet and infrared with impact protection. Gloves must be matched to the hazard: leather for abrasion and heat, cut-resistant knit for sheet and glass, nitrile or butyl for chemicals, insulating rubber rated to the voltage for electrical work, and cotton or none near rotating machinery because a glove can be caught. Safety footwear protects the toes from impact and the sole from penetration, and anti-static or insulating types are specified for particular areas. Hearing protection is required from about 85 dB(A), with ear muffs preferred for intermittent high noise and plugs for long exposure. Respiratory protection ranges from a simple dust mask for nuisance dust through a P2/P3 particulate filter for silica and metal fume to a cartridge respirator for solvents and a supplied-air system for isocyanates and sandblasting.

4.3 Limits of PPE

PPE fails silently: a scratched lens, a cut glove, a filter past its date, a helmet with a cracked shell. It also creates a false sense of safety and can introduce new hazards — gloves caught in a drill, a mask that fogs goggles so the wearer lifts them. This is why Chapter 3 puts engineering and administrative controls ahead of PPE, and why the instructor must check PPE at the start of every practical session, not once a term.

4.4 Signage

Safety signs follow a five-colour convention that every trainee must be able to read at a glance. The table below is the minimum that a shop floor should display, and it should be taught with the tools rather than in isolation, because the sign is the last reminder before the hazard.

4.5 PPE matrix by operation

Operation / hazard	Eyes and face	Hands and arms	Body and feet	Head, hearing and respiratory	Other controls
Bench fitting, filing, chipping, hacksawing	Safety glasses	Leather or cut-resistant gloves for handling sections only, never near rotating parts	Apron or coverall; steel-toe shoes; dry floor	Cap or hair net; dust mask if dusty; ear plugs if noisy	Work clamped in a vice; chipping screen or guard
Drilling, turning, milling, grinding	Safety glasses; face shield for grinding	No gloves near rotating parts	Close-fitting coverall, no loose sleeves; safety shoes	Cap or hair net; ear protection above 85 dB(A); mask or LEV for coolant mist and dust	Guards and interlocks; chuck key on a chain; isolation for setting
Arc welding	Welding helmet or shield, shade 9–13 by current	Welding gauntlets	Leather apron or flame-retardant jacket; spats; steel-toe boots	Helmet plus cap; fume extraction or respirator for coated and stainless steel; ear protection when chipping slag	Screens around the bay; earth return close to the joint
Gas welding, brazing and cutting	Goggles shade 4–5	Leather gauntlets	Leather apron; steel-toe boots	Cap; ventilation or respirator for flux fume	Flashback arresters; extinguisher and fire blanket within reach

Operation / hazard	Eyes and face	Hands and arms	Body and feet	Head, hearing and respiratory	Other controls
Electrical work	Safety glasses for drilling and cutting; arc-rated face shield where arc flash is possible	Insulating gloves rated to the voltage	Arc-rated clothing for live or high-energy work; insulating shoes or mat	Insulated helmet; no respiratory need	Isolation, lockout tagout, voltage detector, second person present
Chemical handling, plating and battery work	Chemical goggles plus face shield	Chemical-resistant gloves	Chemical apron; rubber boots	Face shield; respirator for the specific vapour	Eye wash and safety shower; SDS available; bunded storage
Refrigeration and AC service	Safety glasses or goggles	Insulating and cut-resistant gloves	Coverall; steel-toe shoes	Cap; respirator when brazing; refrigerant monitor	Ventilation in plant rooms; no brazing near refrigerant
Painting, solvent and adhesive work	Goggles	Solvent-resistant gloves	Coverall; safety shoes	Cap; organic vapour respirator	Ventilation; no ignition source; fireproof cabinet for solvents
Spray painting	Goggles or full face shield	Solvent-resistant gloves	Coverall with hood; safety shoes	Hood; supplied-air or cartridge respirator	Spray booth with extraction and filters
Woodworking machines	Safety glasses or face shield	No gloves on rotating cutters	Close-fitting clothing; steel-toe shoes	Cap or hair net; dust mask or LEV; hearing protection	Guards, push sticks, feather boards, riving knife
ICT hardware and electronics assembly	Safety glasses when soldering or cutting	ESD-safe gloves or none; anti-static wrist strap mandatory	Anti-static coat; closed shoes	Not normally required; fume extraction for soldering	ESD mat and grounded bench; no food or drink at the bench
Lifting and material handling	As per the task	Grip gloves	High-visibility vest; steel-toe boots	Helmet in overhead work areas; dust mask if dusty	Use equipment not the back; keep the load close; ask for help

Table 4.1 — Minimum PPE by operation. Where rows overlap, the more protective requirement applies.

4.6 Safety signage

Red — prohibition	Circular red border with diagonal bar	No smoking, no entry, do not operate, no naked flame
Yellow — warning	Triangular yellow with black symbol	Electrical hazard, hot surface, slippery floor, rotating parts, laser, UV arc
Blue — mandatory	Solid blue circle with white symbol	Wear eye protection, wear helmet, use guard, earth the machine, wear gloves
Green — safe condition	Rectangular green with white symbol	First aid, eye wash, emergency exit, assembly point, safety shower
Red — fire equipment	Rectangular red with white symbol	Fire extinguisher, hose reel, fire blanket, alarm call point

Table 4.2 — The five sign categories and representative messages.

4.7 Fire classes and correct media

Class	Fuel	Correct medium	Never use
A	Wood, paper, cloth, rubber	Water, foam, dry powder	Water on live electrical parts
B	Petrol, oil, paint, solvent, LPG liquid	Foam, dry powder, CO2	Water jet
C	LPG, acetylene, hydrogen, natural gas	Isolate the gas, then dry powder or CO2	Extinguishing without isolating the leak
D	Magnesium, aluminium powder, sodium	Dry sand, dry powder for metal fires	Water, foam, CO2
F	Cooking oil and fat	Wet chemical, fire blanket	Water

Class	Fuel	Correct medium	Never use
Electrical	Live electrical equipment	CO2, dry powder (isolate supply first)	Water, foam

Table 4.3 — Fire classes, the correct extinguishing medium and the dangerous mistakes.

5. Power Tool and Machine Guarding Rules

5.1 Why a separate chapter

Power tools and machine tools are where the hand tool's local injury becomes a life-changing injury. The rules below are general across all the powered items in Groups 3, 4, 10 and 11, and they should be posted at each machine and covered in the first practical session of every trade that uses one.

5.2 Guarding principles

A guard is a physical barrier that prevents access to a danger point and, where possible, contains ejected fragments. Fixed guards are preferred; interlocked guards are used where access is needed for operation; adjustable guards follow the work; and self-adjusting guards move with the material. A guard must not be removable without a tool, must not create a new trap point, and must not obstruct the view of the work more than necessary. Guards that have been removed and not refitted are the most common finding in an ITI safety audit.

5.3 The pre-start check

Before starting any powered machine: guard in place and secure; emergency stop tested; chuck key, spanner and tools removed; workholding tight; workpiece clear of the cutter; speed and feed correct for the material; coolant and extraction working; and nobody else in the zone. Say the check out loud when learning — it becomes a habit faster than it becomes a chore.

5.4 Electrical protection for power tools

Use a residual current device of 30 mA sensitivity on all portable tools, and prefer Class II double-insulated or 110 V site tools in wet, outdoor or confined locations. Inspect the cable, plug and strain relief at the start of each shift. Never repair a cable with insulation tape, never use a tool with a cracked casing, and disconnect at the plug — not by pulling the cable — before changing an accessory.

5.5 Working alone and emergency response

No trainee works alone on a machine, and no instructor leaves a practical class unsupervised while machines are running. Every shop needs a clearly marked main isolator, a first aid box, an eye wash, and a fire extinguisher appropriate to the materials present. The emergency sequence is: stop the machine, isolate the supply, raise the alarm, give first aid within competence, and record the incident.

5.6 Fifteen rules for powered tools

- **1.** Ring-test every bonded abrasive wheel before mounting and check the maximum speed marked on it against the machine speed.
- **2.** Fit the guard and the tool rest: bench grinder rest clearance about 2 mm, tongue guard about 3 mm.
- **3.** Never mount a wheel on a machine whose spindle speed exceeds the wheel rating, and never exceed the marked speed by fitting a larger pulley.
- **4.** Use the correct flange and blotting paper washers; tighten evenly and do not over-tighten.
- **5.** Stand to the side of a grinder when starting it, and let it run for a minute before use.
- **6.** Do not grind on the side of a straight wheel, and do not force work into a wheel.
- **7.** For portable grinders, check the guard is present and unmoved, and use the side handle.
- **8.** Match the blade to the material for circular saws and cut-off machines; never remove the riving knife or the lower blade guard.
- **9.** Use a push stick and feather boards on a circular saw or a table saw for narrow work.
- **10.** Never use a drill bit larger than the chuck or the drill's rated capacity.
- **11.** Use a machine vice or clamp; never hold a workpiece in the hand under a drill press.

- **12.** Use low-voltage or battery tools in wet or confined areas, and protect all supplies with an RCD.
- **13.** Inspect the cable, plug and strain relief before each use; do not use a taped or taped-over cable.
- **14.** Disconnect from the supply before changing a blade, bit, wheel or battery.
- **15.** Keep the work area clear of flammables when grinding, cutting or welding — sparks travel far.

6. Calibration, Care, Storage and 5S

6.1 The three categories of tool

Tools divide into consumables (blades, drills, taps, electrodes, abrasives), durable tools (hammers, spanners, files, hand tools) and instruments (calipers, micrometers, gauges, test equipment). Each has a different care regime: consumables are replaced on condition, durable tools on defect, and instruments on a calibration schedule. Confusing the three is why instruments get used as hammers and why blunt drills get forced.

6.2 Calibration

A calibration is a comparison against a traceable standard, recorded with the date and the result. In an ITI the minimum set is: vernier calipers and micrometers (annually, plus a zero check at every use), dial indicators (annually), gauge blocks and slip gauges (on a longer cycle, with careful handling), torque wrenches (annually), electrical meters (annually, with a functional check before each use), meggers (annually), pressure and temperature instruments (annually or per the standard), and weighing equipment in food trades (annually). Instruments that fail are withdrawn, tagged and either adjusted by a competent person or scrapped. A calibration sticker on the instrument, with the next due date, is the simplest auditable system.

6.3 Storage

Storage protects accuracy and prevents injury. Rules that matter: measuring instruments in cases, never loose in a drawer; files and reamers individually racked so that teeth do not rub; drills in an index; taps and dies in their box; hacksaw blades with tension released; chisels and punches with edges protected; gas cylinders upright and chained with caps on; flammable liquids in a fireproof cabinet; and chemicals in labelled, bunded containers with the safety data sheet available. Tool kits are issued against a list and checked on return, which also teaches accountability.

6.4 5S applied to the tool store

Sort: remove broken, obsolete and unidentified items. Set in order: a place for every tool and a label for every place, with frequently used items at hand height. Shine: clean the store and the tools, and inspect while cleaning. Standardise: a tool list per trade, a calibration register, an issue log, a colour code for ownership. Sustain: a weekly audit, a monthly review of the register, and a place on the notice board for the results. This is the same 5S that the CITS Training Methodology syllabus requires the instructor to teach.

6.5 Records that an inspection will ask for

Tool and equipment register, purchase and specification records, calibration certificates and schedule, maintenance log for machines, statutory inspections (hoists, pressure vessels, lifting gear, scaffolds), PPE issue and inspection records, accident and near-miss register, fire equipment service records, and training records showing that each trainee was demonstrated and assessed on each tool.

PART B

The Master Tool Register

216 numbered entries in eleven practice groups, covering every class of tool, instrument, machine and test equipment in the CTS and CITS curricula — minor and major.

- Group 1 Hand tools and bench work
- Group 2 Measuring, marking-out and inspection instruments
- Group 3 Cutting, drilling, sawing, shearing and grinding
- Group 4 Machine tools and machine-shop accessories
- Group 5 Joining, welding, thermal cutting and fastening
- Group 6 Electrical tools and test instruments
- Group 7 Electronics, computer hardware and ICT
- Group 8 Automotive and vehicle service
- Group 9 Refrigeration, AC, plumbing and fluid services
- Group 10 Building, construction, woodworking and painting
- Group 11 Textile, printing, media, food, beauty and health

7. The Master Tool Register

This is the core of the document. Tools are grouped by engineering practice and numbered continuously within each group. Each entry carries a stable code, the tool name and its aliases, what to look at in order to identify it, the minimum operating competence, the specific safety precautions, the care routine, the trades that use it and the PPE.

The groups are: **H** hand tools and bench work; **M** measuring, marking-out and inspection; **C** cutting, drilling, sawing, shearing and grinding; **T** machine tools; **W** joining, welding and thermal cutting; **E** electrical tools and test instruments; **X** electronics and ICT; **A** automotive; **R** refrigeration, air-conditioning and fluid services; **B** building, construction, woodworking and painting; **S** textile, printing, media, food, beauty and health service tools.

Group	Code	Title	Entries
H	H-001 to H-033	Hand Tools and Bench Work	33
M	M-001 to M-034	Measuring, Marking-out and Inspection Instruments	34
C	C-001 to C-025	Cutting, Drilling, Sawing, Shearing and Grinding	25
T	T-001 to T-015	Machine Tools and Machine-shop Accessories	15
W	W-001 to W-014	Joining, Welding, Thermal Cutting and Fastening	14
E	E-001 to E-020	Electrical Tools, Wiring Accessories and Test Instruments	20
X	X-001 to X-012	Electronics, Computer Hardware and ICT Tools	12
A	A-001 to A-014	Automotive and Vehicle Service Tools	14
R	R-001 to R-014	Refrigeration, Air-Conditioning, Plumbing and Fluid Services	14
B	B-001 to B-016	Building, Construction, Woodworking and Painting	16
S	S-001 to S-019	Textile, Printing, Media, Food, Beauty and Health Service Tools	19
		Total entries in the register	216

Table B.1 — The register at a glance.

Anatomy of an entry

Each panel carries the same eight fields, so that an entry can be read in a fixed order and quoted in a job sheet. **Identify** tells the reader what to look at — shape, material, graduation, marking — to recognise the tool and to notice that it is worn or damaged. **Operate** states the minimum competence: the set-up, the technique and the checkpoint that separates a correct result from a plausible one. **Safety / precautions** is highlighted, because it is the field that is skipped when a lesson runs late. **Care / calibration** covers what keeps the tool accurate and serviceable. **Trades** lists where the tool appears in the CTS and CITS curricula, and **PPE** gives the minimum issue for that tool. The *Also called* line carries the alternative names a trainee will meet in a store indent or a manufacturer's catalogue.

Choosing the instrument for the tolerance

Measurement is not a single skill; it is a decision about accuracy. The table below maps a measurement need to the instrument that the curriculum expects, using the tolerances that appear in the CTS and CITS Fitter outcomes (± 0.25 mm for basic fitting, ± 0.04 mm and 30 minutes of arc for fits, ± 0.02 mm for precision fitting). Using a more precise instrument than the job needs is slow; using a less precise one is wrong.

Measurement need	Expected tolerance	Instrument	Least count / note
Rough length, marking out	± 0.5 mm	Steel rule	0.5 mm; sight vertically to avoid parallax
General external and internal size	± 0.1 mm	Vernier caliper	0.02 mm; check zero at every use
Precision diameter, journal, thickness	± 0.01 mm	Outside micrometer	0.01 mm; use the ratchet, never force

Measurement need	Expected tolerance	Instrument	Least count / note
Bores, cylinder wear	± 0.01 mm	Dial bore gauge set with a micrometer	0.01 mm; measure at three levels, two directions
Depth of a blind hole or recess	± 0.02 mm	Depth micrometer or depth gauge	0.01 mm; three readings around the feature
Angle	5 minutes	Vernier bevel protractor	5 min; or a sine bar with slip gauges for high accuracy
Runout, squareness, alignment	0.01 mm	Dial test indicator with magnetic base	0.01 mm; pre-load the stylus about half a turn
Thread pitch diameter	Class of fit	Thread micrometer or three-wire method	Best practice for precision threads
Surface finish	Ra value	Roughness comparator plates, or a profilometer	N1–N12 visual and tactile comparison; instrument for a number
Flatness and straightness	0.01 mm	Surface plate with blueing, straight edge and feeler	Marking and scraping control
Complex geometry, position, form	Micron level	Coordinate measuring machine	Temperature-controlled room; probe qualified on a reference sphere
Torque of a fastener	± 3 to 5 per cent	Torque wrench	Return to the lowest setting after use; calibrate annually
Voltage, current, resistance, continuity	Meter accuracy class	Digital multimeter	Match the CAT rating to the installation
Insulation resistance	Megohm	Insulation tester (megger)	500 V or 1000 V test voltage; discharge after the test
Earth resistance	Ohm	Earth tester	Fall-of-potential method with spikes
Rotational speed	rpm	Tachometer or stroboscope	Optical type keeps the hand away from the shaft
Temperature	± 1 degree C	Thermometer, thermocouple or infrared pyrometer	Set the emissivity for infrared types

Table B.2 — Instrument selection by accuracy requirement.

Group 1 — Hand Tools and Bench Work

Every engineering trade begins at the bench. These are the tools a trainee holds in the hand, and they are the tools on which the habit of safe working is formed. In the Annexure-I trainee tool kit of the Fitter CTS well under twenty items appear — try square, straight scriber, flat files, hacksaw frame, safety goggles — and everything else is issued from the shop store. The bench tools below are therefore listed as ‘trainee kit’ or ‘shop issue’ so that an institute can map them to its own indent.

33 entries in this group.

H-001 Steel rule (bench rule)

Also called: Scale, engineer's rule, 150/300/600 mm

Identify

A flat spring-steel strip with engraved graduations, usually 0.5 mm on one edge and 1 mm on the other, some with imperial divisions. Trainee kit item in Fitter, Electrician, Turner and most metal trades.

Operate

Measure to the nearest graduation by placing the end squarely against a true face, sighting directly above the graduations to avoid parallax, and reading with a light behind the edge. Used for setting dividers, checking lengths and marking parallel lines.

Safety / precautions

Do not use a rule as a lever, scraper or screwdriver; a bent rule throws a burr that cuts. Beware of burred ends on a used rule.

H-001 Steel rule (bench rule)

Care / calibration	Wipe off moisture and oil lightly; store flat in the tool roll so that it does not bend. Discard if the end is worn or the graduations are illegible.
Trades	All engineering trades
PPE	Safety glasses when working at the bench

H-002 Measuring tape (steel, retractable)

Also called: Tape measure, 3 m / 5 m / 7.5 m / 30 m

Identify	A coiled steel blade in a plastic or metal case with a lock and a hook at zero. The hook is deliberately loose by its own thickness for inside and outside measurement.
Operate	Pull the blade out, engage the lock, read, then release and allow it to retract under control. For long measurements hold the tape taut and level, and add the case length for internal measurements where the tape body is used.
Safety / precautions	Let the blade retract slowly — a free-running tape whips and can cut. Keep the blade away from rotating shafts.
Care / calibration	Wipe the blade dry before retracting to prevent internal rust; do not bend the blade at the hook.
Trades	All trades including Carpentry, Masonry, Fabrication, Electrical
PPE	None beyond normal shop PPE

H-003 Engineer's try square

Also called: Try square, 100/150/300 mm blade

Identify	A hardened steel blade riveted or set at exactly 90 degrees into a thicker stock, with the inner edge of the stock and the blade forming the reference faces. The knife-edge type has a bevelled blade used for precision work.
Operate	Hold the stock firmly against the datum face and bring the blade down to the work without pressing. Judge squareness against a light line, and check for square in both directions by turning the work. Never cut with it.
Safety / precautions	The blade is sharp at the corners; handle by the stock. Do not use a try square as a hammer or a straight edge for striking.
Care / calibration	Keep the stock and blade free of burrs; a single burr destroys the reading. Oil lightly, store in the tool roll, and have it checked against a reference square.
Trades	Fitter, Turner, Machinist, Tool & Die Maker, Sheet Metal, Welder
PPE	Safety glasses

H-004 Combination set

Also called: Combination square set, 300 mm

Identify	A hardened rule with a groove, carrying interchangeable heads — square head, protractor head and centre head. Shop issue in Fitter and Machinist trades.
Operate	Fit the required head, set it with the spirit level or the protractor scale, and use for 90 and 45 degree marking, finding the centre of a round bar with the centre head, and setting angles. Always lock the head before use.
Safety / precautions	Protractor and centre heads contain a small bubble level that breaks if dropped; handle as an instrument, not a striking tool.

H-004 Combination set

Care / calibration	Clean the groove before sliding a head; oil the rule; store in the case with the heads loose.
Trades	Fitter, Machinist, Tool & Die Maker, Welder, Sheet Metal
PPE	Safety glasses

H-005 Scriber (straight and bent)

Also called: Scribing needle, 150 mm straight and bent

Identify	A slender hardened steel point set in a knurled body, straight or bent at one end so that the point can reach a surface plate. Trainee kit item, 15 cm.
Operate	Draw fine lines on a chalked or blued surface, holding the scriber inclined at about 60 degrees to the surface and 30 degrees to the direction of movement, with the point dragging rather than cutting deeply. Sharpen on an oilstone.
Safety / precautions	The point is needle-sharp; carry with a cork or in the tool roll, never in a pocket. Do not use for prying.
Care / calibration	Keep the point sharp and lightly oiled; a blunt scriber requires pressure and slips.
Trades	Fitter, Turner, Machinist, Sheet Metal, Welder/Fabricator
PPE	Safety glasses; cut-resistant handling

H-006 Dividers and wing compass

Also called: Spring dividers, 150 mm; wing compass 250/300 mm

Identify	Two hardened legs joined by a spring bow with an adjusting screw, pointed at both ends (dividers) or with one leg carrying a pencil or scriber attachment (compass).
Operate	Set the gap with the screw against a rule, hold by the knurled head and scribe circles or transfer distances by rotating the tool, keeping the legs vertical. Tap lightly to fine-set.
Safety / precautions	Points are sharp; never carry open dividers in a pocket. Do not use as a punch.
Care / calibration	Oil the screw; keep points equal in length so that the tool does not tilt.
Trades	Fitter, Sheet Metal, Welder/Fabricator, Carpentry, Masonry, Draughtsman (on paper)
PPE	Safety glasses

H-007 Spring calipers — outside, inside and hermaphrodite

Also called: Odd-leg caliper, jenny caliper, 150 mm

Identify	Bow-spring legs with an adjusting nut. Outside calipers have inward-turned legs for external diameters; inside calipers have outward-turned legs for bores; the hermaphrodite or odd-leg caliper has one straight and one bent leg for finding centres and scribing parallel to an edge.
Operate	Set approximately by hand and fine-set with the screw; then either transfer the setting to a rule or use the caliper against the work with a light feel, adjusting until it just touches. Never force a caliper over a workpiece.
Safety / precautions	Spring-loaded legs can snap shut on a finger; set with the legs pointing away from the body.

H-007 Spring calipers — outside, inside and hermaphrodite

Care / calibration	Oil the nut and spring; check that the legs are not sprung out of alignment.
Trades	Fitter, Turner, Machinist, Sheet Metal, Welder
PPE	Safety glasses

H-008 Centre punch, dot punch and prick punch

Also called: Centre punch 9 x 150 mm; dot punch; prick punch

Identify	Short hardened steel rods with a point ground to 90 degrees (centre punch), about 60 degrees (dot punch) or a fine 30–40 degree point (prick punch), often knurled at the middle. Trainee kit item for Electrician and Fitter.
Operate	Locate the point on the marked intersection, hold vertically, and strike once with a light hammer to create a witness mark; deepen the centre punch mark before drilling so that the drill does not wander. Grind the point on a bench grinder with the punch rotating.
Safety / precautions	Strike squarely and keep the fingers clear; a glancing hammer blow injures the holding hand. Wear glasses — punch heads can chip and fly.
Care / calibration	Re-grind the point to the correct angle when blunt; discard if the head mushrooms, and never re-harden a cracked head.
Trades	Fitter, Turner, Machinist, Sheet Metal, Welder, Electrician (panel work)
PPE	Safety glasses; hold with pliers if the work is small

H-009 Ball peen hammer

Also called: Engineer's hammer, 0.25 / 0.45 / 0.5 / 1 kg

Identify	A forged carbon-steel head with one flat face and one rounded ball face, on a hickory or ash handle, wedged through the eye. Ball peen 0.45 kg and 0.22 kg are standard shop issue in Fitter trades.
Operate	Hold near the end of the handle, strike with the flat face for punching and chiselling and the ball face for riveting and peening, letting the wrist and forearm supply the blow. Eye protection and a firm grip on the workpiece are essential.
Safety / precautions	Check the head is tight on the handle before every use — a loose head flies off. Never strike a hardened surface, another hammer or a chisel head with another hammer. Keep the face free of oil so the hammer does not slip. Stand so that the swing is clear of other people.
Care / calibration	Replace a split or oil-soaked handle; re-wedge if the head works loose; dress a mushroomed face on a grinder only with the head removed.
Trades	All engineering trades
PPE	Safety glasses; impact-resistant gloves for the holding hand where permitted

H-010 Cross peen and sledge hammer

Also called: Cross peen 250 g; sledge 2/4 kg

Identify	Cross peen has a wedge-shaped peen at right angles to the handle; the sledge has two flat faces and a long handle for two-handed use. Shop issue in Fitter, Electrician and Forging trades.
Operate	Cross peen is used for spreading and drawing metal and for starting a hole; the sledge is used with a striker for heavy forging and for driving stakes. Keep the strike square and let the hammer do the work.
Safety / precautions	Two-person striking requires agreed signals and eye contact. Never strike a hardened tool. Check the handle and head for cracks every time.

H-010 Cross peen and sledge hammer

Care / calibration	As for the ball peen; keep the striking face slightly crowned, not flat and polished.
Trades	Fitter, Forger & Heat Treater, Fabrication, Masonry, Carpentry
PPE	Safety glasses, face shield for heavy striking, steel-toe boots

H-011 Cold chisel (flat, cross-cut, diamond, round-nose)

Also called: Flat chisel 12/19 mm; cape chisel; diamond point; round nose

Identify	A hexagonal or octagonal carbon-steel bar with a cutting edge ground at the end and a slightly mushroomed head. Cutting angles are around 60–70 degrees for mild steel, sharper for soft metals, blunter for cast iron.
Operate	Hold the chisel at about 60 degrees to the work with the cutting edge on the marked line, and strike with a ball peen hammer, cutting away from the body and towards a supported part of the work. Re-grind on a bench grinder, cooling in water so that the temper is not lost.
Safety / precautions	Wear goggles and, for sustained chipping, a face shield; chips are sharp and hot. Never chip towards another person or towards a wall. Use a chipping guard or a screen. Remove the mushroomed head — it can split and eject fragments.
Care / calibration	Grind the edge back to shape before it becomes blunt; keep the head dressed; store with the edge protected.
Trades	Fitter, Welder/Fabricator, Sheet Metal, Masonry, Electrician (wall chasing)
PPE	Safety glasses or face shield, gloves for holding, hearing protection in enclosed areas

H-012 Files — flat, hand, half-round, round, square, triangular, knife-edge and needle

Also called: Second cut and smooth files, 150–300 mm; Swiss-pattern needle file set

Identify	A hardened high-carbon steel bar with teeth cut in one direction (single cut for finishing, double cut for fast removal), a tang at one end and a safe (uncut) edge on some types. Grade is judged by tooth pitch: rough, bastard, second cut, smooth, dead smooth. Trainee kit items are flat second cut and smooth, 250 mm.
Operate	Hold the handle in the right hand and the tip of the file in the left, and stroke away from the body with a steady pressure on the forward stroke and a light return, rocking the file slightly so that the teeth cut evenly. Cross-file for flatness, draw-file for finish, and clean the teeth with a file card. Select the file for the material and the shape for the surface.
Safety / precautions	Never use a file without a handle — the tang will pierce the palm. Do not hammer a file or use it as a pry bar; hardened files snap and throw fragments. Keep fingers clear of the work edge. Do not blow chips off — brush them.
Care / calibration	Never let files rub against each other or against other tools; store separately or in a rack. Keep free of oil and water, and never oil a file. Clean with a file card in the direction of the cut.
Trades	All engineering trades; needle files in Tool & Die Maker, Instrument Mechanic
PPE	Safety glasses; gloves when filing sharp-edged sheet

H-013 Hacksaw frame and blades

Also called: Fixed 150/300 mm or adjustable frame; 18, 24 and 32 TPI blades

Identify	A tubular or pressed-steel frame holding a thin high-speed-steel or bimetal blade in tension, with a wing nut for tensioning and, on the adjustable type, a rotating front end for sawing in a confined space. Trainee kit item, fixed 300 mm frame.

H-013 Hacksaw frame and blades

Operate	Select the blade pitch for the section thickness — about 18 TPI for thick sections, 24 TPI general, 32 TPI for thin tube and sheet — mount with the teeth pointing forward, tension until the blade rings, and saw with long steady strokes using most of the blade at about 40–60 strokes per minute. Start a cut with a triangular notch or at an edge.
Safety / precautions	Keep the work clamped close to the vice jaws so that it does not spring. Keep hands and fingers clear of the cut line, and guard against a sudden breakthrough. A blade that is too loose or too tight snaps and can cut the hand.
Care / calibration	Release the tension after use so that the frame does not distort; replace a blade with missing teeth; brush the frame dry.
Trades	All engineering trades
PPE	Safety glasses; gloves for handling long sections

H-014 Bench vice and leg vice

Also called: Bench vice 100/125 mm jaws; leg vice 100 mm

Identify	A cast-iron or steel body with hardened, serrated or smooth jaws, a movable jaw on a square-thread screw, and a swivel base on some types. The leg vice has one long leg that reaches the floor to take hammer blows.
Operate	Mount at elbow height with the fixed jaw over the bench leg. Grip the work as low as possible in the jaws, use soft jaws or a clamp for finished surfaces, and tighten only enough to hold. For heavy chipping use the leg vice so the shock goes to the floor.
Safety / precautions	Never hammer on the vice body or use it as an anvil except the leg vice. Do not leave the handle protruding into a walkway. Watch the pinch point between the jaws, and do not grip a part that can spin or spring.
Care / calibration	Oil the screw and the slide; keep the jaws and the anvil area clean; replace worn jaw plates; avoid overtightening which bends the screw.
Trades	All engineering and fabrication trades
PPE	Safety glasses; gloves when handling rough sections

H-015 C-clamp, toolmaker's clamp and hand vice

Also called: G-clamp 50/100/150/200 mm; toolmaker's clamp; pin vice

Identify	A C-shaped frame with a screw and swivel pad; the toolmaker's clamp is a small parallel-jaw clamp tightened by a screw through the middle; the hand vice is held in the hand to grip small work for filing.
Operate	Place the clamp so that the pressure is square to the joint, use a packing piece to protect finished faces, and tighten firmly but not enough to distort the work. For welding, use the correct clamp material and keep the earth return close.
Safety / precautions	A clamp can slip off a tapered or oily part and release stored force; always consider which way it will fly. Do not use a bent or cracked clamp.
Care / calibration	Oil the screw; keep the swivel pad free; store with the screw partly open.
Trades	Fitter, Welder, Sheet Metal, Carpentry, Electrician
PPE	Gloves and safety glasses

H-016 Screwdrivers — slotted, Phillips, Pozidriv, Torx and insulated

Also called: Electrician's screwdriver, thin stem 4 x 100/150/250 mm; heavy duty 5 x 200 mm

H-016 Screwdrivers — slotted, Phillips, Pozidriv, Torx and insulated

Identify	A hardened blade ground to the fastener profile, set in a handle. Insulated electrician's drivers carry a rated voltage marking and a thin stem for terminal screws. Trainee kit items in the Electrician trade: 4 x 100, 4 x 150, 4 x 250 and 5 x 200 mm insulated.
Operate	Match the blade width and thickness to the slot, keep the driver in line with the screw, and apply axial pressure before turning. Choose the correct tip for Phillips, Pozidriv, Torx or hex and do not use a slotted driver in a cross recess.
Safety / precautions	Use only insulated drivers rated for the voltage when working near live parts, and check the insulation for cuts. Never use a screwdriver as a chisel, punch or pry bar. Do not hold the work in the hand — a slipping blade goes into the palm.
Care / calibration	Dress a burred tip on an oilstone; keep the handle free of oil; discard if the insulation is cut or the blade is twisted.
Trades	All trades; insulated type mandatory in Electrical trades
PPE	Safety glasses; insulating gloves for live work

H-017 Spanners — open end, ring, combination, box and adjustable

Also called: DE spanner set 6–32 mm; ring spanner; adjustable wrench 150/250/300 mm; box spanner set with Tommy bar

Identify	Forged steel with a jaw or ring sized to the nut across flats. Open end has two jaws at 15 degrees; ring spanner fully encircles the nut; box spanner is a tube with hex ends turned by a Tommy bar; adjustable wrench has a movable jaw and a worm screw.
Operate	Choose the size that fits the flats exactly, pull the spanner towards you rather than push, keep the jaw square to the nut, and use the ring end where access permits because it grips all six flats. Use a socket and ratchet for high torque.
Safety / precautions	Never extend a spanner with a pipe or hammer it; never use an open-end spanner as a hammer. A spanner that slips over a burred nut will skin the knuckles — pull, do not push, and keep the hand open.
Care / calibration	Keep jaws clean and unchamfered; oil the adjusting screw; discard a sprung or opened-out jaw.
Trades	Mechanic Motor Vehicle, Fitter, Refrigeration & AC, Plumber, Electrician, Millwright
PPE	Gloves; safety glasses when working overhead

H-018 Socket set, torque wrench and Allen keys

Also called: Interchangeable ratchet socket set 12 mm driver; torque wrench 5–250 Nm; hex key set 1.5–10 mm

Identify	The socket set has hardened sockets, extensions, universal joints and a ratchet handle; the torque wrench has a calibrated scale or preset click; hex keys are L-shaped or T-shaped hardened bars for socket-head screws.
Operate	Fit the socket fully over the fastener before turning; use a six-point socket on damaged nuts. Set the torque wrench to the specified figure, apply a steady pull to the click or break point, and never use it to loosen. Use the correct hex key size and keep it square in the socket.
Safety / precautions	Torque wrenches are precision instruments — do not drop them or use them as breakers. A cheap hex key can round out a socket screw and then slip. Beware of reaction torque when tightening large fasteners.
Care / calibration	Return the torque wrench to its lowest setting before storage; oil lightly; keep sockets in the tray so that they are not lost or chipped.
Trades	Mechanic Motor Vehicle, Two Wheeler, Diesel, Refrigeration & AC, Millwright, Lift & Escalator
PPE	Gloves; safety glasses

H-019 Pliers — combination, long nose, side cutting, water pump and circlip

Also called: Combination plier 200 mm insulated; nose plier 150 mm; diagonal cutter 150 mm; circlip plier internal/external

Identify	Two crossed levers on a pivot with a gripping jaw and, usually, a cutting edge near the pivot. Insulated types carry plastic-dipped handles rated for voltage. Circlip pliers have pins or jaws for internal and external retaining rings.
Operate	Grip, bend, twist and cut wire with the appropriate jaw; use long-nose pliers for reaching into confined spaces and side cutters for flush cutting. Keep the wire deep in the jaw near the pivot for maximum leverage.
Safety / precautions	Insulated pliers protect against accidental contact but are not a licence to work live — isolate first. Never cut hardened steel wire or a live conductor with insulated pliers and bare hands. Do not use pliers as a hammer or a spanner.
Care / calibration	Oil the pivot; keep the cutting edges dressed; discard if the insulation is cut or the jaws are sprung.
Trades	Electrician, Electronics Mechanic, Wireman, Fitter, Refrigeration, ICT
PPE	Safety glasses; insulating gloves for electrical work

H-020 Tin snips and aviation shears

Also called: Straight snips 250/300 mm; bend snips; aviation shears left/right/straight

Identify	Hand shears with short hardened blades and long handles for leverage. Straight snips cut along a line; bend or aviation shears have offset blades for curves and are colour-coded left, right and straight.
Operate	Open the blades wide so that the sheet is cut deep in the jaw, and cut just outside the marked line. Use the correct shear for the direction of curve. For internal cut-outs, start from a drilled hole.
Safety / precautions	Sheet metal edges are razor sharp — wear gloves and deburr every cut edge. Cut away from the body and never place a finger between the blades. Do not cut hardened or thick steel beyond the shear's capacity.
Care / calibration	Oil the pivot; do not cut wire or hardened material; store with the blades closed and the point protected.
Trades	Sheet Metal Worker, Welder/Fabricator, Refrigeration & AC, Electrician
PPE	Cut-resistant gloves, safety glasses

H-021 Hand reamer, tap wrench and die stock

Also called: Adjustable hand reamer 9/12/18 mm; taper reamer set; tap wrench; die stock

Identify	The reamer is a fluted cylindrical cutter with a square shank driven by a tap wrench; taps are threaded, fluted cutters in taper, second and bottom leads; the die stock holds a split die for external threads.
Operate	Ream with a slow, steady clockwise rotation and a cutting fluid, never reversing, to size a drilled hole. Tap with the tap held square, backed off a quarter turn every half turn to break the chip, using the correct tapping drill size and lubricant. Use taper, second then bottom tap in sequence for a through hole.
Safety / precautions	A tap is brittle and snaps easily — a broken tap in a hole is hard to remove and may need a spark eroder or tap extractor. Keep hands clear when the tap breaks free. Never turn a reamer backwards, which dulls the flutes.
Care / calibration	Clean and oil after use; store in the box so that the flutes are not chipped; replace a chipped reamer.
Trades	Fitter, Turner, Machinist, Tool & Die Maker, Millwright, Plumber (pipe threads)
PPE	Safety glasses; cutting fluid can irritate skin

H-022 Scrapers and hand scrapers

Also called: Flat scraper 150 mm; three-corner; half-round; bearing scraper

Identify	A hardened steel blade ground with a fine flat edge and a slight burr, used to remove minute amounts of metal and to produce bearing contact. Types are flat, three-corner, half-round and bearing.
Operate	Apply blue marking to the mating surface, rub the work on the surface plate to show high spots, and scrape only the marked spots with a short, controlled stroke. Repeat until the contact points are evenly distributed — typically 10 to 25 spots per 25 mm square for a bearing surface.
Safety / precautions	Scrapers are razor-sharp; keep both hands behind the cutting edge and wear a cut-resistant glove on the guiding hand. Never scrape towards the body.
Care / calibration	Hone the edge frequently on an oilstone; store in a leather roll with the edge protected.
Trades	Fitter, Tool & Die Maker, Mechanic Machine Tools Maintenance, Millwright
PPE	Cut-resistant gloves, safety glasses

H-023 Hand drill and breast drill

Also called: Hand drill 0–6 mm; breast drill; push drill

Identify	A crank handle driving a pinion and chuck, with a chest plate on the breast drill for pressure. Used where there is no power or where a portable electric drill is unsafe.
Operate	Centre-punch the hole, mount the correct drill, apply pressure with the chest or the free hand while turning the crank steadily, and ease off as the drill breaks through.
Safety / precautions	Clamp the work; a hand drill that grabs in thin sheet can spin and strike the face. Do not drill into a surface that hides a cable or pipe.
Care / calibration	Oil the gear and chuck; keep the jaws clean; remove the drill bit after use.
Trades	Fitter, Sheet Metal, Electrician, Plumber
PPE	Safety glasses

H-024 File card and brush

Also called: File brush

Identify	A stiff wire-bristle brush with a card back used to clean file teeth.
Operate	Brush along the direction of the cut, from the safe edge towards the tang, to lift out chips and pinning. Do not use the hand.
Safety / precautions	Wire bristles can fly; brush away from the face and wear glasses.
Care / calibration	Replace when the bristles are flattened; keep it out of oil.
Trades	All metal-cutting trades
PPE	Safety glasses

H-025 Bench brush, hand brush and swarf rake

Also called: Chip brush, dust pan and brush

Identify	Soft or stiff bristle brushes with a wooden or plastic back; a hook-ended rake for chip removal from machine trays.

H-025 Bench brush, hand brush and swarf rake

Operate	Brush chips from the bench and machine into a bin; use a rake or hook for machine swarf; never use compressed air on a machine tool.
Safety / precautions	Metal swarf is sharp and can carry coolant; wear gloves and never sweep swarf with a bare hand. Compressed air drives chips into the eyes and under the skin.
Care / calibration	Clean and store dry; replace when splayed.
Trades	All workshop trades
PPE	Gloves, safety glasses

H-026 Oil can, oil stone and abrasive stone

Also called: Oil can 0.25 litre; oilstone 150 x 50 x 25 mm; carborundum stone

Identify	The oil can is a hand pump can with a spout for precise lubrication; oilstones are rectangular natural or synthetic abrasive blocks in coarse and fine grades.
Operate	Apply a few drops of the specified lubricant to machine oil points and to the work when tapping. Hone chisels, scrapers, scribers and punches on the oil stone with the stone flooded with oil, maintaining the correct bevel angle.
Safety / precautions	Oil on the floor is a slip hazard; wipe up spills at once. Do not use an unknown oil on a machine tool — the wrong lubricant damages bearings.
Care / calibration	Keep the stone flat by rubbing on emery cloth on a flat plate; keep it covered when not in use; do not drop it.
Trades	All workshop trades
PPE	Gloves; clean up spills

H-027 V-block, angle plate, parallel block and surface gauge

Also called: V-block pair 70/150 mm with clamps; angle plate 100 x 200 mm; universal surface gauge (scribing block) 220 mm

Identify	The V-block is a hardened, ground block with a 90 degree V slot and a clamp for holding round work; angle plates are L-shaped, ground reference surfaces; the surface gauge or scribing block has a heavy base, a spindle and an adjustable scriber for marking heights.
Operate	Seat the work in the V and clamp it, or seat a block against an angle plate, and mark from a surface plate using the surface gauge set to a height rule or a vernier height gauge.
Safety / precautions	Blocks are heavy — handle with both hands and keep feet clear. A workpiece that is not clamped can topple off the plate and crush a foot.
Care / calibration	Keep the ground faces free of burrs and oil them; store on a padded shelf; never strike the block.
Trades	Fitter, Turner, Machinist, Tool & Die Maker, Millwright
PPE	Steel-toe boots, gloves

H-028 Surface plate and marking-off table

Also called: Surface plate 450 x 450 mm cast iron or granite; marking table 910 x 910 x 1220 mm

Identify	A thick, ribbed cast-iron plate with a hand-scraped or ground top, or a granite plate, mounted on a stand; the marking-off table is a tall, rigid table with a true top for marking large work.

H-028 Surface plate and marking-off table

Operate	Clean the plate, place work only on clean dry supports, and mark using a surface gauge or height gauge. Use the plate for inspection by blueing and for checking flatness.
Safety / precautions	Never hammer on the plate; put nothing hot or heavy on it; keep it clear of files and tools that can chip the surface. A granite plate can crack and a cast plate can ring loudly — use hearing protection when working on it.
Care / calibration	Wipe with a soft cloth and light oil after use; cover when not in use; have flatness re-certified periodically; lift by the frame, never by the top.
Trades	Fitter, Tool & Die Maker, Machinist, Instrument Mechanic
PPE	Safety shoes; gloves for heavy work

H-029 Punches — letter and number punch sets

Also called: Steel stamps 3 mm; figure punch set

Identify	Hardened steel stamps with a raised character at the end, held in the hand and struck with a hammer to mark serial numbers and identification on metal.
Operate	Set out a straight line with a rule and a scriber, hold the punch vertically and upright, and strike squarely once with a heavy hammer. Use a jig or a spacing block for a row of characters.
Safety / precautions	Use a holder or pliers for short punches; striking at an angle makes the punch skip. Wear glasses — the head can chip.
Care / calibration	Dress a mushroomed head; oil lightly; keep the character face clean.
Trades	Fitter, Welder/Fabricator, Mechanic Motor Vehicle, Sheet Metal
PPE	Safety glasses, gloves for the holding hand

H-030 Screw extractor (Ezy-out) and stud remover

Also called: Stud extractor; broken tap extractor

Identify	A hardened, tapered, left-hand fluted tool that is driven into a drilled hole in a broken screw or stud so that turning anticlockwise draws it out; `the stud remover is a roller or cam device for gripping a stud.
Operate	Centre-punch and drill the broken fastener with a drill slightly smaller than the minor diameter, then insert the extractor and turn anticlockwise with a tap wrench. Apply penetrating oil and heat before extraction where possible.
Safety / precautions	Extractors are very hard and brittle and snap easily, leaving an even harder obstruction. Do not use excessive torque or a cheater bar.
Care / calibration	Keep the flutes clean; store separately; discard if the tip is rounded.
Trades	Mechanic Motor Vehicle, Fitter, Millwright, Tool & Die Maker
PPE	Safety glasses, gloves

H-031 Pipe wrench, pipe vice and chain vice

Also called: Stillson wrench 300/400 mm; pipe vice 100 mm; chain wrench

Identify	The pipe wrench has serrated, self-tightening jaws on an adjustable frame; the pipe vice has V or flat serrated jaws that grip a pipe horizontally and allow it to rotate for threading.
Operate	Set the jaws to the pipe size, apply the wrench with the jaw opening facing the direction of rotation, and pull. In the pipe vice, clamp the pipe close to the work and rotate it end-for-end for threading or cutting.

H-031 Pipe wrench, pipe vice and chain vice

Safety / precautions	Serrated jaws score and weaken pipe — use a strap wrench on finished surfaces. Oil and graphite make pipe threads slippery; a slipped wrench hits the knuckles. Support long pipe lengths to prevent them falling.
Care / calibration	Oil the adjusting screw and the jaw teeth; keep the vice screw lubricated; replace worn jaws.
Trades	Plumber, Fitter, Refrigeration & AC, Millwright
PPE	Gloves, safety glasses, steel-toe boots

H-032 Pipe cutter and pipe threading die set

Also called: Wheel pipe cutter 15–50 mm; adjustable BSP die set 15–50 mm

Identify	The cutter has a hardened wheel and two or more rollers on an adjustable body with a reamer at the end; the die set holds BSP or metric dies for cutting external pipe threads.
Operate	Seat the pipe, tighten the wheel a fraction of a turn per revolution, keep the cut square, then ream the burr. Thread with cutting oil, reversing a quarter turn to break the chip, and check the thread with a gauge or a fitting.
Safety / precautions	Cut pipe ends are sharp; ream every cut and wear gloves. Clamp the pipe securely — a pipe spinning in a cutter can fracture the wheel.
Care / calibration	Oil the cutter wheel and rollers; replace a chipped wheel; clean dies and oil them.
Trades	Plumber, Refrigeration & AC, Fitter, Instrument Mechanic
PPE	Gloves, safety glasses

H-033 Punches, chisels and hammers for masonry and carpentry

Also called: Masonry chisel 19/25 mm; carbide-tipped chisel; claw hammer; mallet; wood chisel

Identify	Masonry chisels are heavy, forged or carbide-tipped bars; the claw hammer has a steel head with a claw for nail extraction; the mallet is a wooden or rubber headed hammer; the wood chisel has a flat bevel blade and a socket or tang handle.
Operate	Chisel masonry with short blows and eye protection, working towards a free edge. Drive nails with the hammer face, not the claw. Strike a wood chisel with a mallet, never a steel hammer, and cut with the bevel against the waste.
Safety / precautions	Masonry chipping throws sharp chips — wear goggles and a face shield and erect a screen. Wood chisels cut deeply; keep both hands behind the blade. Mallets used carelessly crush fingers.
Care / calibration	Hone chisels and keep the edges protected; dress mushroomed heads; replace split handles.
Trades	Mason (Building Constructor), Carpenter, Building Maintenance, Painter
PPE	Goggles, face shield for chasing, gloves, hearing protection

Group 2 — Measuring, Marking-out and Inspection Instruments

Metrology is where a trainee moves from 'making something' to 'making something to specification'. The CTS learning outcomes state tolerances explicitly — basic fitting to ± 0.25 mm in the first year, fits and interference work to ± 0.04 mm and 30 minutes of arc, precision fitting to ± 0.02 mm at CITS level — and the CITS outcomes add comparators, sine bar, dial test indicator, digital instruments and the coordinate measuring machine. Instruments in this group are precision items: they must be calibrated, handled as instruments and never used as tools.

34 entries in this group.

M-001 Vernier caliper (analogue)

Also called: Vernier 150/200/300 mm, least count 0.02 mm

Identify	A steel rule with a fixed jaw and a sliding jaw carrying a vernier scale, with inside jaws, outside jaws, a depth rod and a locking screw. Least count is normally 0.02 mm (or 0.05 mm on cheaper types). Shop issue in Fitter, Turner, Machinist, Motor Vehicle and Welding trades.
Operate	Close the jaws and check the zero reading. Open the jaws slightly wider than the work, then close them with light contact, lock, and read the main scale before the vernier zero plus the vernier division that coincides, multiplied by the least count. Read from directly in front of the scale. Use the inside jaws for bores and the depth rod for depths.
Safety / precautions	Do not use the caliper as a pair of pliers, a scribing tool or a clamp; the jaws are ground and easily sprung. Do not measure a moving or hot workpiece.
Care / calibration	Wipe the jaws and beam before and after use; keep the slide very lightly oiled; store in the case, jaws slightly open; check zero at every use and have the instrument calibrated annually.
Trades	Fitter, Turner, Machinist, Tool & Die Maker, Mechanic Motor Vehicle, Welder, Millwright, Instrument Mechanic
PPE	None beyond normal shop PPE

M-002 Digital vernier caliper

Also called: Digital caliper 150/200 mm, resolution 0.01 mm

Identify	Same layout as the analogue caliper, with a battery-powered LCD readout, an on/zero/off key and a mm/inch key; some have a data output port.
Operate	Switch on, close the jaws and zero the display, then measure as for the analogue caliper. Use the hold function for reading in a confined position. Re-zero after changing the battery.
Safety / precautions	The electronics and the glass scale are fragile and sensitive to coolant and magnetic fields — do not immerse, and keep away from welding current and magnet chucks.
Care / calibration	Remove the battery for long storage; wipe the beam free of ferrous dust; store in the case; recalibrate against a gauge block.
Trades	Fitter, Turner, Machinist, Motor Vehicle, Sheet Metal, CITS Fitter
PPE	None

M-003 Vernier height gauge

Also called: Height gauge 300 mm with scriber

Identify	A heavy base with a vertical beam and a sliding measuring head carrying a scriber or a dial/digital indicator, used on a surface plate. Vernier or dial type, least count 0.02 mm.

M-003 Vernier height gauge

Operate	Place on a clean surface plate, zero the scriber on the plate, set the height required with the fine adjustment, lock, then scribe or check a height. Read against the vernier as for a caliper.
Safety / precautions	The scriber is sharp and the base is heavy — carry with two hands and never let the gauge topple. Do not use the scriber to cut deeply.
Care / calibration	Clean the plate and the base before use; oil the beam lightly; store upright in the case.
Trades	Fitter, Tool & Die Maker, Machinist, CITS Fitter
PPE	Safety glasses, safety shoes

M-004 Vernier depth gauge and depth micrometer

Also called: Depth gauge 0–200 mm; depth micrometer 0–100 mm

Identify	The depth gauge has a flat base bar and a sliding rule or rod; the depth micrometer has a series of interchangeable rods and a micrometer head with a flat base.
Operate	Seat the base firmly on the reference surface, extend the rod until it touches the bottom, and read. Take three readings around the feature and use the largest for a blind hole depth.
Safety / precautions	Do not force the rod against a rough or tapered bottom; the reading will be wrong and the rod may bend.
Care / calibration	Wipe the rods and the base; oil; store in the case with the rods in their slots.
Trades	Fitter, Tool & Die Maker, Machinist, Mechanic Machine Tools Maintenance
PPE	None

M-005 Outside micrometer

Also called: Micrometer 0–25, 25–50, 50–75 mm with setting standards

Identify	A C-shaped frame with a fixed anvil, a spindle driven by a precision screw, a barrel with a datum line, and a thimble graduated to 0.01 mm with a ratchet or friction stop. Shop issue in sets of 0–25, 25–50 and 50–75 mm with test pieces.
Operate	Clean the anvils, check zero with the anvils closed (or with the setting standard for larger sizes), open beyond the work and close with the ratchet until it slips three clicks, then read the barrel plus the thimble. Take several readings along and around the work to detect taper and ovality.
Safety / precautions	Never close the micrometer hard on the work or use the thimble as a vice — the screw is precision and does not tolerate force. Do not drop it; a fall of a few centimetres can put it out of adjustment.
Care / calibration	Wipe the anvils with a clean cloth and light oil before storage, keep the anvils very slightly open, store in the case, and have it checked with gauge blocks annually.
Trades	Fitter, Turner, Machinist, Tool & Die Maker, Motor Vehicle, Millwright
PPE	None

M-006 Inside micrometer and telescopic (T-) gauge

Also called: Inside micrometer 25–50 mm; telescopic gauge up to 150 mm

Identify	The inside micrometer uses extension rods and a micrometer head with spherical anvils; the telescopic gauge has a spring-loaded plunger that locks and is then measured with an outside micrometer.

M-006 Inside micrometer and telescopic (T-) gauge

Operate	For the inside micrometer, set the size approximately on the head, insert and rock gently to find the true diameter, then read. For the telescopic gauge, collapse the plunger, insert, expand until it touches both walls, lock, withdraw and measure across the plunger ends with an outside micrometer. Rock the gauge in two planes to find the maximum.
Safety / precautions	Insert and withdraw with care in a blind bore; a jammed gauge can be damaged or can damage the bore.
Care / calibration	Oil the plunger and screw; keep the anvils clean; store in the case.
Trades	Fitter, Turner, Machinist, Motor Vehicle, Millwright, Refrigeration & AC (compressor bores)
PPE	None

M-007 Digital micrometer and digital dial indicator

Also called: Digital outside micrometer 0–25 mm; digital lever-type dial test indicator

Identify	Micrometer or indicator with an LCD display, zero-set and mm/inch keys, and a data output on better types. CITS Fitter requires digital micrometer, digital caliper, digital height gauge and digital dial test indicator.
Operate	Zero the instrument on its standard before use, take the reading on the display and use the hold key where the display cannot be seen. Treat the measurement exactly as you would an analogue one — verify with a second instrument when the tolerance is tight.
Safety / precautions	Batteries and electronics are sensitive to coolant and impact; do not use near welding or induction heating.
Care / calibration	Remove the battery for long storage; keep clean and dry; recalibrate with gauge blocks.
Trades	Fitter, Tool & Die Maker, Machinist, CITS Fitter
PPE	None

M-008 Screw thread micrometer and three-wire method

Also called: Thread micrometer with V-anvil and cone; three-wire set

Identify	A micrometer with a 60 degree V-anvil and a matching cone for measuring the pitch diameter of a thread, or three hardened wires of known diameter used with a standard micrometer.
Operate	For the thread micrometer, seat the V on one flank and the cone in the opposite groove and read the pitch diameter directly. For the three-wire method, place one wire in one groove and two in the opposite grooves, measure over the wires and calculate the pitch diameter from the wire diameter, thread angle and pitch.
Safety / precautions	Wires are tiny and easily lost or bent; keep them in the holder and do not force them into a damaged thread.
Care / calibration	Keep the V-anvil free of swarf; oil; replace bent wires; store in the case.
Trades	Fitter, Turner, Machinist, Tool & Die Maker, Instrument Mechanic
PPE	None

M-009 Screw pitch gauge

Also called: Thread pitch gauge, metric and BSW set

M-009 Screw pitch gauge

Identify	A folding set of thin hardened leaves, each cut with a known thread form and pitch, marked in millimetres or threads per inch.
Operate	Offer leaves to the thread until one seats fully across the crests with no light gap; read the pitch from the leaf marking. Use it with a thread table or a standard to identify the thread series.
Safety / precautions	The thin leaves bend and cut; open only the leaf in use.
Care / calibration	Wipe off oil and swarf; close the set; do not force a leaf into a dirty thread.
Trades	Fitter, Turner, Machinist, Motor Vehicle, Plumber, Millwright
PPE	None

M-010 Thread plug gauge and thread ring gauge (Go / No-Go)

Also called: Go and No-Go plug gauge; ring gauge; caliper gauge

Identify	Hardened, lapped gauges made to the limits of the thread. The Go plug must enter the full thread by hand; the No-Go plug must not enter more than two turns. Ring gauges do the same for external threads.
Operate	Clean the thread and the gauge, apply light oil, and screw the Go gauge in by hand — it must pass. Then try the No-Go, which must not pass. Never use a wrench.
Safety / precautions	Gauges are expensive reference standards; a dropped or forced gauge is scrap. Never use a thread gauge to clean a thread.
Care / calibration	Oil and case after use; keep separate from production tooling; recertify on schedule; never regrind.
Trades	Fitter, Turner, Machinist, Tool & Die Maker
PPE	None

M-011 Limit gauges — plug, snap, ring and caliper

Also called: Plug gauge single-ended 5–55 mm; snap gauge Go/No-Go 25–50 mm

Identify	Plain (non-threaded) hardened gauges: plug gauges for holes, snap or gap gauges for shafts, ring gauges for cylinders. The Go end is made to the maximum material limit, the No-Go to the minimum.
Operate	Insert or apply without force — the Go must pass freely and the No-Go must not. The judgement is by feel and by weight of the gauge, not by pressure.
Safety / precautions	Do not force or hammer a gauge into a hole; it will seize and both parts will be damaged.
Care / calibration	Clean, oil and store in a wooden case; handle by the insulated or knurled part to avoid warming the gauge.
Trades	Fitter, Turner, Machinist, Tool & Die Maker, Motor Vehicle
PPE	None

M-012 Dial test indicator with magnetic base

Also called: DTI 0.01 mm; plunger or lever type; magnetic stand

Identify	A small precision instrument with a spring-loaded plunger or lever that drives a needle over a graduated dial, mounted on an articulated arm and a magnetic or mechanical base. Least count 0.01 mm, travel typically 0–10 mm.

M-012 Dial test indicator with magnetic base

Operate	Mount rigidly on the base, set the stylus approximately perpendicular to the surface, pre-load the needle by about half a revolution, zero the bezel, and rotate or move the work to read runout, squareness or alignment. For centring work in a four-jaw chuck, adjust jaws until the needle deflection is halved.
Safety / precautions	Never let the plunger snap back; the mechanism is delicate. Keep the magnetic base away from watches, cards and electronic instruments.
Care / calibration	Release the stylus gently, store in the case, avoid coolant ingress, and have it calibrated. Check repeatability on a known surface.
Trades	Fitter, Turner, Machinist, Tool & Die Maker, Millwright, Mechanic Machine Tools Maintenance
PPE	None

M-013 Comparators — mechanical, electrical, pneumatic and stand mounted

Also called: Dial comparator stand; Sigma comparators; air gauge

Identify	A comparator does not measure an absolute size; it measures the difference between the work and a setting master. Mechanical types use a lever and gear train, electrical types a transducer, pneumatic types an air jet and back pressure.
Operate	Set the comparator to zero using a gauge block or setting master of nominal size, then present the work and read the deviation. Magnification can be high, so a small deviation reads clearly.
Safety / precautions	Setting masters and gauge blocks are reference standards; handle with tongs or gloves and never leave them on a bench.
Care / calibration	Keep the measuring head and stand clean; protect the air supply from moisture on pneumatic types; calibrate against a master at each shift.
Trades	Fitter, Tool & Die Maker, Machinist, Mechanic Machine Tools Maintenance, Instrument Mechanic
PPE	None

M-014 Vernier bevel protractor

Also called: Universal bevel protractor, least count 5 minutes

Identify	A protractor head with a vernier scale and an adjustable blade, used with a stock and a clamp for measuring and setting angles.
Operate	Set the blade against one face of the angle and the stock against the other, lock, and read the degree scale plus the vernier minutes. Use the acute angle attachment for small angles.
Safety / precautions	The blade is sharp; do not force it into a rough corner — the reading will be wrong.
Care / calibration	Oil the slide; keep the scale legible; store in the case.
Trades	Fitter, Tool & Die Maker, Machinist, Millwright
PPE	None

M-015 Sine bar and sine centre

Also called: Sine bar 100/200 mm with rollers; sine centre

Identify	A hardened steel bar with two accurately ground rollers at a known centre distance; the angle is set by raising one roller on slip gauges. CITS Fitter requires sine bar and its safety precautions explicitly.

M-015 Sine bar and sine centre

Operate	Place the bar on a surface plate with one roller on the plate and the other on a slip gauge stack of height h; the angle is the arcsine of h divided by the roller centre distance L. Check the setting with a dial indicator along the work.
Safety / precautions	The sine bar is a reference standard and must not be clamped in a vice or used as a support; a bent bar is useless. Keep slip gauges wringing-free of grit.
Care / calibration	Handle by the ends, oil, store in the case; keep the roller centres clean; protect from rust.
Trades	Tool & Die Maker, Fitter (advanced), Machinist, Instrument Mechanic
PPE	None

M-016 Slip gauge (gauge block) set

Also called: Johansson metric set, grade 0/1/2

Identify	A set of hardened, lapped rectangular blocks of precise thickness supplied in a wooden case; blocks wring together to build a dimension. CITS Fitter lists a Johnson metric set.
Operate	Wring blocks together by sliding one over the other with light pressure after cleaning, build the required dimension from the largest blocks downward, and use the stack to set a height gauge, sine bar or comparator.
Safety / precautions	The lapped faces are easily scratched; a single particle of grit ruins the wring. Never leave a stack assembled for long, as it can seize.
Care / calibration	Wipe with a soft cloth and a volatile solvent, apply the supplied anti-corrosion oil, and return each block to its slot; keep the case closed; recertify periodically.
Trades	Tool & Die Maker, Fitter (advanced), Machinist, Instrument Mechanic, Metrology lab
PPE	Handle with cotton gloves where specified

M-017 Feeler (thickness) gauge

Also called: Feeler gauge 10 or 13 blades; spark plug gauge

Identify	A set of thin, hardened, marked leaves of known thickness on a pivot.
Operate	Insert the leaf between the surfaces and note whether it enters with a light drag; combine leaves to build a thickness. Used for tappet clearances, bearing clearances and contact breaker gaps.
Safety / precautions	The leaves kink permanently if forced; a kinked leaf gives a false reading. Do not use a feeler gauge to clean a gap.
Care / calibration	Wipe off oil and grit, keep the leaves straight, and close the set.
Trades	Mechanic Motor Vehicle, Diesel, Two Wheeler, Refrigeration & AC, Millwright
PPE	None

M-018 Radius gauge and fillet gauge

Also called: Radius gauge set 1–25 mm; fillet or screw thread radius gauge

Identify	A folding set of leaves with convex and concave radiused ends, marked in millimetres.
Operate	Match the leaf to the radius until no light shows through; use the convex leaf for an internal fillet and the concave leaf for an external radius.
Safety / precautions	Thin leaves bend easily; use the correct leaf and do not force it around a burr.

M-018 Radius gauge and fillet gauge

Care / calibration	Wipe and oil; close the set; replace bent leaves.
Trades	Fitter, Tool & Die Maker, Machinist, Sheet Metal
PPE	None

M-019 Wire gauge and sheet metal gauge

Also called: Standard wire gauge (SWG); Birmingham gauge; metric sheet gauge

Identify	A circular steel plate with slots of graded width, marked with gauge numbers or millimetres, used to size wire and sheet.
Operate	Insert the wire or sheet edge into the slot that it fits without slack; read the gauge number and convert to millimetres using the standard table.
Safety / precautions	The sheet edge is sharp; wear gloves when gauging a cut edge.
Care / calibration	Keep the slots free of burrs and rust; check against a vernier for worn slots.
Trades	Electrician, Electronics Mechanic, Sheet Metal Worker, Welder
PPE	Gloves for sheet edges

M-020 Surface roughness comparator plates

Also called: Roughness standard N1–N12; Rubert gauge

Identify	Plates or blocks carrying machined specimens of known roughness, marked in Ra or N grades, used visually and by touch.
Operate	Compare the work surface with the standard under good light, by eye and with a fingernail, and match the closest grade. CITS Fitter lists the N1–N12 set.
Safety / precautions	Finger contact with a rough surface can cut; use a light touch or a fingernail.
Care / calibration	Keep clean and oiled to prevent corrosion of the reference surfaces.
Trades	Fitter, Machinist, Tool & Die Maker, CITS Fitter
PPE	None

M-021 Spirit level and precision level

Also called: Level 150/300/600 mm; engineer's precision level 0.02 mm/m

Identify	A sealed glass vial with an air bubble in a metal or wooden frame; the bubble centres when the base is horizontal. Precision types are graduated and read 0.02 mm per metre.
Operate	Place on the surface, read the bubble position, then reverse the level end-for-end and re-read to cancel any error in the level itself. Use for machine installation, levelling and checking twist.
Safety / precautions	Do not drop or heat the level; the vial is fragile and the frame distorts if clamped.
Care / calibration	Wipe the base free of grit; store in the case; check the vial is not loose.
Trades	Millwright, Mechanic Machine Tools Maintenance, Fitter, Lift & Escalator, Refrigeration & AC, Masonry

M-021 Spirit level and precision level

PPE	None

M-022 Straight edge and feeler for flatness

Also called: Steel straight edge 450/1000 mm; knife edge

Identify	A hardened, ground straight bar with a bevelled or knife edge for checking flatness against a light source.
Operate	Place the edge on the surface, view against a light source, and estimate the gap or measure it with a feeler gauge. Check in several directions.
Safety / precautions	The knife edge is sharp and the bar can spring; handle with both hands.
Care / calibration	Oil and store hanging or flat on a padded support; never use as a lever.
Trades	Fitter, Millwright, Machinist, Sheet Metal
PPE	None

M-023 Bore gauge (dial and telescopic)

Also called: Dial bore gauge 18–150 mm; cylinder bore gauge

Identify	A dial indicator mounted on a body with a measuring plunger and a centralising bridge, set to a known size with a micrometer or a setting ring.
Operate	Zero the gauge in the setting master, insert into the bore, rock gently to find the minimum, and read the deviation. Measure at several depths and in two directions at 90 degrees to detect taper and ovality.
Safety / precautions	Insert with the plunger compressed; forcing it into a rough or stepped bore damages the anvil.
Care / calibration	Oil the plunger; keep the setting ring clean; store in the case; recalibrate.
Trades	Mechanic Motor Vehicle, Diesel, Fitter, Millwright, Refrigeration & AC
PPE	None

M-024 Coordinate measuring machine (CMM)

Also called: CMM with touch probe; bridge or cantilever type

Identify	A granite table carrying a moving bridge or column with a touch-trigger probe, driven by software that records X, Y, Z coordinates and computes size, form and position. CITS Fitter requires demonstration of CMM operation and its safety precautions.
Operate	Clean the table and the work, establish the part datum, qualify the probe with a reference sphere, and then run the measuring programme or take manual points. Interpret the report — size, position, concentricity, flatness — against the drawing.
Safety / precautions	Keep hands clear of the moving bridge, use the speed override and the emergency stop, never lean on the granite table, and control the environment so that temperature variation does not invalidate the measurement. Do not touch the probe stylus.
Care / calibration	Keep the machine clean and covered, the air supply dry and filtered, and the probe calibrated; a CMM room is normally temperature controlled around 20 ± 2 degrees C.
Trades	Tool & Die Maker, Fitter (advanced), Machinist, CITS Fitter, Quality inspection
PPE	None; avoid loose clothing near moving axes

M-025 Optical profile projector and toolmaker's microscope

Also called: Profile projector 300/500 mm screen; toolmaker's microscope

Identify	An optical instrument that projects a magnified silhouette of a small part onto a screen with measuring graticules, or a microscope with a measuring stage.
Operate	Focus the profile, align the feature with the graticule or the stage coordinates, and measure lengths and angles by moving the stage. Use for small profiles, threads, form tools and inserts.
Safety / precautions	Do not look at the light source directly; keep fingers out of the stage mechanism; handle the part with tweezers as it is hot after machining.
Care / calibration	Keep the optics covered and dust-free; clean lenses only with lens tissue; protect the stage from swarf.
Trades	Tool & Die Maker, Machinist (advanced), Fitter, Quality inspection
PPE	None

M-026 Brinell / Rockwell hardness tester

Also called: Brinell hardness tester with ball indenter; Rockwell tester

Identify	A machine that applies a known load through a ball or diamond indenter and measures the indentation, either optically (Brinell) or by depth (Rockwell). CITS Fitter lists a Brinell hardness tester.
Operate	Prepare a flat, clean surface, select the load and indenter for the material and thickness, raise the specimen to the anvil, apply the load, and read the indentation. Take three readings and average.
Safety / precautions	The load mechanism can crush fingers — keep hands clear of the anvil during application, and support thin specimens so that the anvil does not punch through.
Care / calibration	Keep the indenter and anvil clean; verify with a certified test block; service the hydraulic or load cell system.
Trades	Fitter, Forger & Heat Treater, Tool & Die Maker, Metallurgy lab
PPE	None; keep hands clear

M-027 Dumpy level, auto level and levelling staff

Also called: Surveying level with tripod; telescopic staff 3 m

Identify	A telescope with a spirit level and a levelling head on a tripod; the auto level uses a compensator. The staff is graduated in metres and millimetres and is read through the telescope.
Operate	Set the tripod firm, level the instrument with the foot screws (or rely on the compensator), take a backsight on a known reduced level, then foresights, and reduce the levels by height of instrument or rise and fall method.
Safety / precautions	Do not look at the sun through the telescope; carry the tripod over the shoulder and tighten the head; keep the staff clear of overhead power lines.
Care / calibration	Wipe the optics, keep the tripod screws oiled, store the instrument in its case with desiccant.
Trades	Surveyor, Draughtsman (Civil), Mason (Building Constructor), Architectural Draughtsman
PPE	Safety shoes, high-visibility vest

M-028 Theodolite and total station

Also called: Electronic theodolite; total station with EDM and data logger

M-028 Theodolite and total station

Identify	A precision angle-measuring instrument on a tribrach, with horizontal and vertical circles; a total station adds an electronic distance meter, an on-board computer and a data store.
Operate	Set up over the station point, centre and level with the optical plummet and foot screws, orient to a reference bearing, then observe angles and distances. Use the instrument's programmes for setting out, traversing and coordinate measurement.
Safety / precautions	Never look at the sun or a laser through the telescope; keep the instrument away from traffic and always use a warning triangle; secure it against wind and theft.
Care / calibration	Keep dry, charge the battery properly, clean the optics, and have it calibrated. Store in the case, not on the tripod.
Trades	Surveyor, Draughtsman (Civil), Architectural Draughtsman, Mining
PPE	High-visibility vest, safety shoes, sun protection

M-029 Laser level, line laser and rotary laser

Also called: Cross-line laser; self-levelling rotary laser with detector

Identify	A battery-powered diode unit on a tripod or bracket that projects a levelled line, cross or rotating plane; some have a receiver for outdoor use.
Operate	Set on a firm base, allow the unit to self-level, and use the projected line for setting heights, partitions, suspended ceilings, tiling and machine alignment.
Safety / precautions	Never stare into the beam or point it at anyone's eyes. Most ITI units are class 2 or class 3R; treat any laser as an eye hazard, do not use it on scaffolding in wind, and never look through binoculars at the beam.
Care / calibration	Remove the battery, store in the case, protect from shock, and check calibration against a known level periodically.
Trades	Mason, Carpenter, Plumber, Electrician, Interior Design, Refrigeration & AC
PPE	Laser safety glasses where specified

M-030 Plumb bob, square, bevel and marking gauge (building and wood)

Also called: Plumb bob 250/500 g; try square; sliding bevel; mortise gauge

Identify	The plumb bob is a pointed weight on a cord; the sliding bevel has an adjustable blade for transferring angles; the mortise gauge has two adjustable spurs for marking parallel lines on timber.
Operate	Suspend the plumb bob from a line to establish verticality; set the bevel from a protractor or the work and transfer angles; set the gauge to the timber thickness and mark with the head against the face side.
Safety / precautions	The plumb bob point is sharp and the weight can swing into the face; keep the cord clear of people.
Care / calibration	Oil the gauge screw and the bevel lock; keep the spur sharp; store the bob in its case.
Trades	Carpenter, Mason, Interior Design, Building Maintenance
PPE	Safety glasses, safety shoes

M-031 Tachometer and stroboscope

Also called: Contact / optical tachometer; stroboscope

Identify	The tachometer measures rotational speed by contact wheel, optical reflective mark or magnetic pickup; the stroboscope flashes a xenon or LED lamp at an adjustable rate to freeze a rotating part.

M-031 Tachometer and stroboscope

Operate	For contact types apply the tip to the shaft centre and hold for a steady reading; for optical types place a reflective mark and aim. For the stroboscope, adjust the flash rate until the mark appears stationary.
Safety / precautions	Contact tachometers put a hand near rotating shafts — keep sleeves clear and use an optical type where possible. Stroboscopes can trigger photosensitive epilepsy, and the apparent frozen motion is an illusion: never touch a part believing it is stopped.
Care / calibration	Keep the optical window clean; check the reflective tape; store dry.
Trades	Mechanic Motor Vehicle, Millwright, Refrigeration & AC, Mechanic Machine Tools Maintenance
PPE	Safety glasses, hearing protection

M-032 Vibration meter, sound level meter and lux meter

Also called: Vibration pen; sound level meter dB(A); lux meter

Identify	Battery-powered hand-held instruments with a sensor — an accelerometer, a microphone or a photodiode — and a digital display of the measured quantity.
Operate	Hold or mount the instrument at the measuring point, select the correct range and weighting (for example dB(A) for hearing risk), and record the steady value. Use for condition monitoring, noise surveys and lighting checks.
Safety / precautions	Measurement is not a substitute for control — if noise exceeds about 85 dB(A), hearing protection is required whatever the meter reads.
Care / calibration	Calibrate with the reference source; protect the microphone from wind and dust; remove the battery for storage.
Trades	Health Safety & Environment, Millwright, Mechanic Machine Tools Maintenance, Electrical maintenance
PPE	Hearing protection while measuring in noise

M-033 Handheld thermometer, infrared thermometer and pyrometer

Also called: Dial thermometer; IR gun; thermocouple pyrometer

Identify	A temperature sensor — bimetal dial, thermocouple, resistance thermometer or infrared detector — with a display, sometimes with a laser sighting dot.
Operate	For contact types place the probe in the medium or against the surface with good contact; for infrared types set the emissivity, hold at the correct distance to fill the field of view, and scan. Use for bearing temperatures, oven and furnace checks, motor and panel temperatures, and refrigerant line temperatures.
Safety / precautions	Infrared thermometers cannot measure through glass or shiny metal reliably, and the laser sighting dot is an eye hazard. Do not touch a hot surface to 'confirm' a reading.
Care / calibration	Protect the lens; keep the probe clean; check against a known reference (ice point or boiling water) periodically.
Trades	Refrigeration & AC, Mechanic Motor Vehicle, Welder, Forger & Heat Treater, Electrical maintenance, Food trades
PPE	Heat-resistant gloves near hot surfaces

M-034 Trammel and large-diameter measuring tools

Also called: Trammel bar; inside/outside caliper for large work; pi tape

Identify	A long bar with two adjustable, hardened points for scribing large circles and transferring large distances; a pi tape measures circumference and reads diameter directly.

M-034	Trammel and large-diameter measuring tools
Operate	Set the points to the radius with a rule, lock, and scribe by rotating about a centre; for pi tape wrap squarely around the diameter and read the diameter scale.
Safety / precautions	Long bars need two people or a support; a swinging point can injure the hand.
Care / calibration	Oil the points and the lock screw; keep the tape clean and free of kinks.
Trades	Sheet Metal, Welder/Fabricator, Fitter, Millwright, Plumber (large pipe)
PPE	Gloves, safety glasses

Group 3 — Cutting, Drilling, Sawing, Shearing and Grinding

This group is where the hand tool becomes a powered tool and where the injury severity rises sharply: rotating spindles, abrasive wheels that burst, blades that grab. The CTS curriculum places drilling, reaming, tapping and dieing within the first year of every engineering trade and introduces power saws, pedestal grinders and lathes under specific learning outcomes on machine operation and overhauling. Everything here requires the power-tool and machine-tool rules of Chapter 5 in addition to the item-specific notes.

25 entries in this group.

C-001 Twist drill (straight and taper shank)

Also called: HSS drill 1.5–25 mm straight shank; Morse taper shank 6–25 mm

Identify	A fluted cylindrical cutter with two helical flutes, a 118 degree point angle (135 degrees for harder materials), a chisel edge and either a straight shank for a chuck or a tapered shank to fit a spindle sleeve.
Operate	Select the drill for the material and the tapping size, centre-punch and start with a pilot drill for large sizes, set the correct spindle speed — slower for larger diameters and harder materials — and use coolant. Clear the flutes by withdrawing the drill (peck drilling) in deep holes. Sharpen on a bench grinder at the correct point angle and with equal lip lengths.
Safety / precautions	Never hold the work by hand; use a machine vice or clamp. Wear glasses and keep sleeves and hair clear of the rotating drill. Do not use a drill with a bent shank or a chipped point. Ease the pressure as the drill breaks through so that it does not screw into the work.
Care / calibration	Wipe off coolant and swarf, oil lightly, store in a drill index so that the edges are protected, and re-point when the margins show wear.
Trades	All engineering trades
PPE	Safety glasses, no gloves near the spindle, hearing protection for high-speed work

C-002 Drill chuck, key and Morse taper sleeve

Also called: Key type and keyless chuck 0–13 mm; drill sleeve MT1–MT2–MT3; drift

Identify	A three-jaw scroll chuck with a taper shank; the key or a knurled collar tightens the jaws. Sleeves adapt a smaller taper shank to a larger spindle socket; a drift removes taper-shank tools.
Operate	Insert the drill fully into the jaws and tighten all three holes of a key chuck in turn. For taper shanks, clean both tapers, seat firmly with a light tap, and remove with a drift through the slot.
Safety / precautions	Remove the chuck key immediately after tightening and never leave it in the chuck — this is the single most common drill-press injury. Use the correct sleeve; a mismatched taper spins and scores the spindle.
Care / calibration	Keep the jaws and tapers clean and dry (never oil a Morse taper); replace a chuck with worn jaws; store keys on a chain at the machine.
Trades	All engineering trades
PPE	Safety glasses

C-003 Portable electric drill

Also called: Hand drill 6/10/13 mm; hammer drill; impact driver

C-003 Portable electric drill

Identify	A pistol or body-grip power tool with a chuck, a trigger switch and often a variable speed control and a reversing switch. Hammer drills add a percussion action; impact drivers use a rotary impact mechanism with a hex bit holder.
Operate	Select the correct bit and speed, mark and centre-punch the hole, hold the drill square with both hands, start slowly to establish the hole, then increase pressure. Use the hammer action only for masonry. Clear the flutes periodically.
Safety / precautions	Use an RCD-protected supply, check the cable and plug, and never drill into a wall without checking for concealed cables and pipes. Keep the bit sharp; a blunt bit needs force and can snap. Watch for reaction torque when a large bit grabs, and never wear loose gloves or sleeves. Wear hearing and eye protection.
Care / calibration	Clean the chuck and vents of dust; do not use the cable to carry the tool; have brushes and the cable replaced by a competent person.
Trades	Electrician, Plumber, Carpenter, Fitter, Fabricator, Refrigeration & AC
PPE	Safety glasses, ear plugs, dust mask, no loose clothing

C-004 Pillar (bench) drilling machine

Also called: Pillar drill 13/20 mm capacity; radial drill

Identify	A vertical column carrying a table and a head with a spindle, driven by a belt or gearbox giving several speeds, with a feed handle. A radial drill has a rotating arm for large work.
Operate	Clamp the work to the table or in a machine vice, set the spindle speed from the chart for the material and drill size, set the depth stop, align the drill with the punch mark, and feed steadily with coolant. Retract to clear chips on deep holes.
Safety / precautions	Never hold work by hand — it will spin. Guard the belt and pulley, keep the chuck key on its chain, and never brake the spindle with the hand. Clamp the table and the head before drilling, and make sure the drill is clear of the table before starting.
Care / calibration	Lubricate the spindle and the column, keep the table T-slots clean, change the coolant, and check the belt tension.
Trades	Fitter, Turner, Machinist, Fabricator, Tool & Die Maker
PPE	Safety glasses, hearing protection, no gloves

C-005 Radial arm drill and gang drill

Also called: Radial drilling machine; multi-spindle gang drill

Identify	A heavy drill with a horizontal arm that swings and slides on a column, allowing the spindle to be positioned over large work; the gang drill mounts several spindles on one table for sequential operations.
Operate	Clamp the work to the base or a table, release the arm clamps, position the spindle, re-clamp the arm and head before drilling, then drill with the correct speed and feed.
Safety / precautions	Always lock the arm and the head before the drill engages — an unlocked arm will swing under load. Large work needs a crane or two people; never lift it by hand.
Care / calibration	Lubricate the arm ways and the column; keep the locks adjusted; check the coolant system.
Trades	Fabrication, Heavy engineering, Fitter, Machinist
PPE	Safety glasses, hearing protection, steel-toe boots

C-006 Hole saw, annular cutter and step drill

Also called: Bi-metal hole saw 14–150 mm; annular cutter; cone/step drill

Identify	The hole saw is a toothed cup on an arbor with a pilot drill; the annular cutter is a hollow, multi-tooth cutter for thick plate; the step drill is a conical fluted drill that enlarges a hole in thin sheet in steps.
Operate	Use a low speed and steady pressure with cutting fluid, and start with the pilot drill. For an annular cutter use a magnetic base drill with the safety chain attached. Support thin sheet with a backing board.
Safety / precautions	Hole saws grab badly at breakthrough — use a support and keep both hands on the tool. Annular cutters on a magnetic drill must always be used with the safety chain and with the magnet confirmed on clean steel.
Care / calibration	Remove the swarf from the teeth, oil, replace the pilot drill when worn, and do not overheat the teeth.
Trades	Electrician, Plumber, Fabricator, Refrigeration & AC, Carpenter
PPE	Safety glasses, face shield, hearing protection, gloves away from the spindle

C-007 Magnetic base drilling machine

Also called: Mag drill with annular cutter; electromagnetic drill press

Identify	A portable drill with an electromagnetic base that clamps to steel plate when energised, on a slide with a feed lever and a coolant bottle.
Operate	Clean the steel surface, position the magnet, energise and confirm the holding force, attach the safety chain, then drill with an annular cutter at the recommended speed and feed.
Safety / precautions	If the power fails the magnet releases — always fit the safety chain. Do not use on non-magnetic steel or on a painted or rusty surface. Keep the cable clear of the cutter and do not drill overhead without a secondary restraint.
Care / calibration	Clean the magnet face; check the chain; oil the slide; replace cutters when dull.
Trades	Fabrication, Welder, Electrician (busbar and panel work), Millwright
PPE	Safety glasses, face shield, hearing protection

C-008 Abrasive cut-off saw (chop saw)

Also called: Cut-off machine 355 mm with abrasive or carbide blade

Identify	A bench or floor machine with a hinged head carrying a large abrasive or toothed blade in a guard, with a vice for clamping the section.
Operate	Clamp the work firmly, set the mitre if needed, allow the blade to reach full speed before contact, and cut with a light, steady pressure, letting the blade do the work. Do not cut thin material without support.
Safety / precautions	The blade must be rated for the machine speed and be undamaged. Always use the guard and the vice; never cut freehand. Sparks and hot burrs fly far — clear the area of flammables and wear a face shield and hearing protection. A cut-off blade can shatter if it is side-loaded.
Care / calibration	Inspect the blade for cracks and missing segments, check the guard and the vice, clean the spark deflector, and earth the machine.
Trades	Welder/Fabricator, Fitter, Plumber, Electrician, Refrigeration & AC
PPE	Face shield, hearing protection, gloves, apron

C-009 Power hacksaw and band saw

Also called: Hydraulic power hacksaw; horizontal band saw; vertical band saw

C-009 Power hacksaw and band saw

Identify	A machine that feeds a toothed blade through clamped work under hydraulic or gravity pressure; the horizontal band saw uses a continuous blade loop and is faster and quieter; vertical types are used for contour cutting.
Operate	Clamp the work, select the blade pitch and the cutting speed for the material, set the feed and coolant, position the blade clear of the vice, and start. Allow the machine to finish the cut and lift the head before unloading.
Safety / precautions	Keep hands clear of the blade path and never reach into the cut to clear chips while the blade runs. Keep the blade guard and the wheel covers fitted. A blade that is too coarse for thin work will strip teeth.
Care / calibration	Check the blade tension and guide alignment, brush the chips from the machine, top up the coolant, and change the blade when it is dull or has missing teeth.
Trades	Fitter, Turner, Fabricator, Tool & Die Maker, Mechanic Machine Tools Maintenance
PPE	Safety glasses, hearing protection, no gloves near the blade

C-010 Bench grinder and pedestal grinder

Also called: Double-ended grinder 150/200 mm; tool grinder

Identify	One or two abrasive wheels on a common spindle driven by a motor, with adjustable tool rests, spark deflectors and eye shields, mounted on a bench or pedestal.
Operate	Stand to one side while starting, check the wheel runs true, set the rest to within about 2 mm of the wheel and the tongue guard within about 3 mm, hold the work against the rest and apply light pressure, quenching carbon-steel tools frequently in water. Use the correct wheel grade for the material.
Safety / precautions	Ring-test bonded wheels and match the wheel's rated speed to the machine. Wear goggles or a face shield, never wear gloves or loose sleeves, and grind on the face of the wheel only. Do not grind aluminium or wood on a standard wheel — it loads the wheel, which then bursts. Never stand in line with the wheel when starting.
Care / calibration	Dress the wheel with a wheel dresser to keep it true and free-cutting, adjust the rests as the wheel wears, keep the coolant clean, and replace a wheel that is cracked, out of balance or worn below the marked minimum.
Trades	All engineering, fabrication and maintenance trades
PPE	Goggles or face shield, dust mask, hearing protection, no gloves

C-011 Portable angle grinder

Also called: 4 inch / 5 inch / 7 inch angle grinder; die grinder

Identify	A high-speed right-angle tool with a side handle, a wheel guard and a spindle lock, taking depressed-centre grinding discs, cut-off discs, flap discs or wire brushes. The die grinder is a smaller, higher-speed straight tool for burrs.
Operate	Select the disc for the material and the task, fit it with the correct flanges, set the guard to protect the operator, hold with both hands, let the disc reach full speed before contact, and grind at a shallow angle. Use a cutting disc for cutting only, without side pressure.
Safety / precautions	This is one of the highest-risk tools in the shop. Always use the guard. Check the disc speed rating against the tool, inspect for damage and never fit a disc larger than the guard allows. Wear a face shield, hearing protection, dust mask and apron; keep sparks away from the body and from flammables. A disc that breaks ejects fragments at high speed, so never remove the guard and never grind without the side handle.
Care / calibration	Clean the air vents, check the cable and the switch, replace flanges if damaged, and have the tool serviced when the bearings sound rough.
Trades	Welder/Fabricator, Fitter, Plumber, Masonry, Mechanic Motor Vehicle, Refrigeration & AC

C-011 Portable angle grinder

PPE	Face shield, hearing protection, dust mask or respirator, apron and gloves

C-012 Shearing machine, guillotine and nibbler

Also called: Hand lever shear; treadle guillotine; power shears; nibbler

Identify	The guillotine holds sheet between a fixed bed and a moving blade driven by a lever, treadle or motor; the nibbler punches small bites with a reciprocating punch; hand shears are bench-mounted lever blades.
Operate	Set the back gauge for the cut width, hold the sheet flat against the bed and the gauge, keep fingers well behind the hold-down, and make the cut with a firm stroke. For the nibbler, start from an edge or a drilled hole and follow the marked line.
Safety / precautions	Guillotine blades are extremely sharp and the hold-down can crush fingers — use a hold-down and keep hands at least a hand's width from the blade line. Fit the finger guard and never remove it. Nibblers throw small sharp slugs; wear a face shield.
Care / calibration	Oil the blade guides, check the blade clearance for the sheet thickness, keep the hold-down working, and sharpen or reverse the blades when dull.
Trades	Sheet Metal Worker, Fabricator, Refrigeration & AC, Electrician (panel work)
PPE	Cut-resistant gloves, safety glasses or face shield, hearing protection

C-013 Bandsaw blade, hacksaw blade and jigsaw blade

Also called: Bimetal band 6–32 TPI; HSS hacksaw blade; jigsaw blade T-shank

Identify	Toothed cutting blades supplied by pitch and tooth material: bimetal for general cutting, carbide for hard materials, HSS for hacksaws. Pitch is chosen by section thickness.
Operate	Fit with the teeth pointing in the cutting direction (forward on a hacksaw, towards the motor on a band saw) and tension correctly. Use a coarser pitch for thick sections, a finer pitch for thin sections and tube.
Safety / precautions	Damaged or blunt blades need force and break; replace rather than persist. Never touch a running blade, and never reach into the cut.
Care / calibration	Keep blades dry and oiled before storage; store in a rack so that the teeth are not blunted against other tools.
Trades	All metal, wood and fabrication trades
PPE	Safety glasses, hearing protection

C-014 Files, rasps and rotary burrs (power)

Also called: File 250/300 mm; wood rasp; carbide rotary burr (rotary file)

Identify	Files are covered in Group 1; here the reference is to powered versions — rotary burrs in a die grinder or drill, and rasps for wood. A burr is a carbide, double-cut or single-cut head in various shapes (ball, cylinder, tree, cone).
Operate	Run the burr at the speed recommended for the material, keep it moving so that it does not groove the work, and use light pressure. For wood rasps, work along the grain.
Safety / precautions	Rotary burrs grab and kick on edges; use both hands, a face shield, and no gloves on a fast spindle. Carbide burrs shatter if dropped.
Care / calibration	Clean the teeth with a file card or a wire brush; oil lightly; discard chipped burrs.
Trades	Tool & Die Maker, Fitter, Carpenter, Fabricator

C-014 Files, rasps and rotary burrs (power)

PPE	Face shield, dust mask, hearing protection

C-015 Reamer, counterbore and countersink

Also called: Machine reamer; counterbore with pilot; countersink 60/82/90 degree

Identify	The machine reamer is a fluted cutter with a chamfered lead for sizing a drilled hole; the counterbore has a pilot that follows the drilled hole to cut a flat seat; the countersink cuts a conical seat for a screw head.
Operate	Drill undersize, then ream at about two-thirds of the drilling speed with twice the feed and plenty of cutting fluid, feeding in one direction only. Use the correct pilot and a slow speed for counterboring and countersinking.
Safety / precautions	Reamers must never be turned backwards; doing so dulls them and can seize. Keep hands clear of the flutes.
Care / calibration	Clean and oil after use, store in the case, and protect the cutting edges.
Trades	Fitter, Turner, Machinist, Tool & Die Maker, Fabricator
PPE	Safety glasses

C-016 Tap, die and thread rolling head

Also called: Tap set M3–M20; split die; tap wrench; thread rolling head

Identify	Taps are threaded, fluted cutters with taper, second and bottom leads; dies are split rings in a die stock. A thread rolling head forms threads by displacement rather than cutting.
Operate	Use the correct tapping drill size, keep the tap square with a tap guide, apply cutting fluid, and back off a quarter turn every half turn. Chase threads by hand, not under power, unless using a tapping attachment with a clutch.
Safety / precautions	Taps are brittle; a broken tap in a hole is a costly defect. Use a tap wrench of the right size, not pliers, and never power-tap by hand-holding the work.
Care / calibration	Clean and oil the flutes, store in the box, and use a tap extractor for broken taps.
Trades	Fitter, Turner, Machinist, Plumber, Millwright, Tool & Die Maker
PPE	Safety glasses

C-017 Belt sander, disc sander and linisher

Also called: Belt and disc sander; linisher; orbital sander

Identify	An abrasive belt or disc driven over rollers or a platen, with a work table and dust extraction; portable orbital sanders use a pad with a clamped or hook-and-loop abrasive sheet.
Operate	Set the table close to the belt, hold the work firmly, let the abrasive cut, and move the work to avoid burning or grooving. Select the grit for the stock removal required.
Safety / precautions	Dust from wood, paint and metal is a fire and health hazard — use extraction and a dust mask. Never wear loose gloves or sleeves near a belt, and do not sand small parts that can be pulled into the machine. Sparks from metal sanding can ignite wood dust, so keep the two operations separate.
Care / calibration	Replace worn belts, clean the platen, empty the extraction bags, and check the tracking of the belt.
Trades	Carpenter, Painter, Fabricator, Fitter, Foundry

C-017 Belt sander, disc sander and linisher

PPE	Safety glasses, dust mask or respirator, hearing protection

C-018 Woodworking circular saw and table saw

Also called: Table saw with fence and blade guard; mitre saw; circular saw

Identify	A toothed blade projecting through a table or a base plate, with a riving knife, an upper and lower guard, a rip fence and a mitre gauge.
Operate	Set the blade height just above the stock, set the fence parallel to the blade, stand to one side of the blade line, feed with push sticks, and use a feather board for narrow rips. Support long stock at both ends.
Safety / precautions	Never remove the blade guard or the riving knife. Never reach over or behind the blade. Kickback is the main hazard — stand out of the line of the blade and use the fence correctly. No gloves near the blade, and check for nails.
Care / calibration	Clean the table and the rise-and-fall mechanism, check the guard return, sharpen or replace the blade, and check the alignment of the fence and the blade.
Trades	Carpenter, Interior Design, Building Maintenance, Furniture
PPE	Safety glasses, hearing protection, dust mask, no gloves

C-019 Woodworking planer, thicknesser and jointer

Also called: Surface planer; thickness planer; hand plane

Identify	A rotating cutter block with knives or disposable inserts, set in a table with an infeed and outfeed, and a fence on the jointer. The hand plane is a body with a sole and an adjustable iron.
Operate	On a jointer, feed with the grain, keep hands behind the cutter block and use push blocks. On a thicknesser, set the depth of cut to about 1–2 mm, feed one end, and let the machine take the stock. On a hand plane, set the iron for a fine cut and plane along the grain.
Safety / precautions	The cutter block is unguarded for the width of the cut — keep hands well away and use push blocks. Never plane stock shorter than about 300 mm on a machine, and check for nails and grit. Hearing protection is essential.
Care / calibration	Check knife condition and tightness, wax the tables, check the belt and bearing condition, and align the fence.
Trades	Carpenter, Interior Design, Furniture, Building Maintenance
PPE	Safety glasses, hearing protection, dust mask, no gloves

C-020 Woodworking router and spindle moulder

Also called: Plunge router with cutters; spindle moulder; trimmer

Identify	A high-speed vertical spindle carrying a profiled cutter, adjustable in depth, with a fence or a template guide.
Operate	Select the cutter, set the depth in small increments, feed against the rotation (left to right along a fence), and use a template or a jig for curves. Take shallow cuts and make several passes.
Safety / precautions	Routers and moulders are unforgiving of small or unsupported work — use jigs, push sticks and feather boards. Check the cutter is tight before starting and never reach under the work. Feed direction matters: climb cutting will throw the work.
Care / calibration	Clean the collet and the cutter shanks, check the spindle brake, and keep cutters sharp.
Trades	Carpenter, Interior Design, Furniture

C-020 Woodworking router and spindle moulder

PPE	Safety glasses, hearing protection, dust mask

C-021 Metal cutting chisels, trepanning tool and tube cutter

Also called: Hole cutter for sheet; chassis punch; tube cutter

Identify	A hole cutter is a toothed or fluted cylinder with a pilot; a chassis punch is a draw-bolt punch and die set; a tube cutter has a cutting wheel and rollers.
Operate	For a chassis punch, drill a pilot hole, assemble the punch and die through the hole, and tighten the draw bolt steadily. For a tube cutter, rotate the tool around the tube, tightening the wheel slightly each revolution, then ream.
Safety / precautions	Chassis punches have high stored force; keep fingers out of the punch area and check the dies are matched. Tube cutters leave a sharp internal burr — ream it, especially on refrigeration lines.
Care / calibration	Oil the threads and the cutting wheel; replace a chipped wheel; keep the punch and die pairs together.
Trades	Electrician, Plumber, Refrigeration & AC, Fabricator
PPE	Safety glasses, gloves

C-022 Hand snips, notcher and pneumatic shear

Also called: Power notcher; pneumatic shears; hand nibbler

Identify	Bench or portable machines that remove a notch or shear sheet; pneumatic shears use a compressed-air reciprocating blade.
Operate	Mark the notch, align with the stops, and make the cut in one clean stroke. For pneumatic shears, keep the tool square to the sheet and let the blade cut rather than forcing it.
Safety / precautions	Air-operated tools need a whip check and the correct pressure; never exceed the rated pressure. Sheet edges are sharp and the cut-off piece can fall on a foot.
Care / calibration	Lubricate the air motor, drain the air line, and sharpen or replace the blades.
Trades	Sheet Metal Worker, Fabricator, Refrigeration & AC
PPE	Cut-resistant gloves, face shield, hearing protection

C-023 Hydraulic press and arbor press

Also called: Hydraulic workshop press 10–100 tonne; arbor press; screw press

Identify	A heavy frame with a hydraulic ram or a mechanical rack and pinion, a pressure gauge and a work table, used for pressing bearings, straightening, broaching and assembly.
Operate	Centre the work under the ram, use a suitable pressing sleeve or drift that bears on the inner ring of a bearing, apply pressure slowly while watching the gauge and the work, and use a shield or blanket when pressing brittle parts.
Safety / precautions	The stored energy in a press is enormous — never place hands between the ram and the work, never exceed the rated pressure, and always use a guard or a polycarbonate shield. A part that shatters under pressure ejects fragments. Check the hoses and the relief valve.
Care / calibration	Check the hydraulic oil level and hoses, keep the ram and table clean, and never leave the press under load.
Trades	Mechanic Motor Vehicle, Millwright, Fitter, Tool & Die Maker, Bearings and assembly shops
PPE	Safety glasses or face shield, gloves, safety shoes

C-024 Sharpening equipment — oilstone, diamond hone, grinder jig

Also called: Oilstone; diamond plate; wet stone; sharpening jig

Identify	Abrasive blocks or plates in graded grits, used to restore a cutting edge by hand, with an angle jig to hold a constant bevel.
Operate	Flood the stone with the recommended lubricant, hold the tool at the correct angle, and work in smooth strokes over the full length of the stone, finishing on a finer grit and removing the burr on the flat face.
Safety / precautions	A sharp edge that slips off a stone can cut deeply — cut away from the body and keep the stone on a non-slip base.
Care / calibration	Flatten the stone periodically on a diamond plate or emery on glass; keep it covered; do not drop it.
Trades	Carpenter, Fitter, Painter, Butcher/Food trades, Tool & Die Maker
PPE	Cut-resistant gloves

C-025 Ultrasonic and abrasive jet cutting (advanced)

Also called: Abrasive water jet; ultrasonic cutter; wire EDM

Identify	Computer-controlled machines that cut by abrasive jet, high-frequency vibration or spark erosion rather than by a toothed edge. Found in advanced CITS trade technology and in industry attachment.
Operate	Under supervision: set the work on the table, load the programme, verify the offset and the cutting path, and start with the safety enclosure closed.
Safety / precautions	Never open an enclosure while a jet, wire or spark is active; high-pressure water can inject through skin, and EDM dielectric and debris are hazardous. Follow the manufacturer's procedure exactly.
Care / calibration	Per the manufacturer's schedule — filters, nozzle wear, dielectric quality and wire tension.
Trades	Tool & Die Maker (advanced), Additive Manufacturing, industry OJT
PPE	Per machine specification; safety glasses and hearing protection at minimum

Group 4 — Machine Tools and Machine-shop Accessories

The machine tool turns a cutting edge into a controlled process, and it is where the largest share of serious injuries in an ITI occurs. The CTS Fitter outcome for the first year requires plain turning, facing, step turning, parting, chamfering, grooving, knurling, boring, taper turning and external V threading to specification; the CITS outcome adds CNC turning, part programming, DNC and machine erection and alignment. Each machine below assumes the machine-tool rules of Chapter 5 as a baseline, and the item notes only the specific hazards.

15 entries in this group.

T-001 Centre lathe (all-geared headstock)

Also called: Lathe 1000 mm between centres, 150 mm centre height; SS and SC lathe

Identify	A bed with V and flat ways carrying a headstock with a geared spindle, a tailstock, a carriage with apron, cross slide and compound rest, a tool post, a lead screw and a feed shaft. The CITS Fitter Annexure specifies an all-geared centre lathe with 150 mm centre height, 1000 mm centre distance, three- and four-jaw chucks, auto feed, safety guard, motorised coolant and lighting.
Operate	Check the oil level and the chuck guard, select the spindle speed and feed from the chart, mount the work in the chuck or between centres, set the tool height on centre, and take a trial cut. Use the correct cutting speed for the material and the tool, apply coolant, and measure with the machine stopped. Perform facing, plain and step turning, parting, chamfering, grooving, knurling, boring, taper turning and threading with the appropriate tool and feed.
Safety / precautions	Remove the chuck key before starting — every time. Never leave a chuck key, spanner or file on the machine. No gloves, no loose sleeves, no rings or watches; tie back hair. Never reach across a rotating workpiece, and never measure, polish or clear chips while the spindle turns. Guard the chuck, keep the workholding tight, and clamp the toolbit with minimum overhang. Beware of long stringy swarf — use a hook, not hands. Never brake the chuck by hand.
Care / calibration	Oil the ways and the apron daily, top up the headstock oil, clean the coolant tank and filter, brush the chips from the ways and the tray, check the belt tension and the gib adjustment, and level the machine periodically.
Trades	Turner, Fitter, Machinist, Tool & Die Maker, Mechanic Machine Tools Maintenance
PPE	Safety glasses, close-fitting coverall, safety shoes, no gloves

T-002 Lathe chucks — three-jaw self-centring and four-jaw independent

Also called: 3-jaw scroll chuck 125/160 mm; 4-jaw independent chuck; collet chuck

Identify	The three-jaw chuck has a scroll plate that moves all jaws together for round and hexagonal work; the four-jaw has independent screws for irregular work and for accurate setting with a dial indicator. Collets grip small round work accurately.
Operate	Mount on a clean spindle nose, tighten all jaws in sequence, and check that the work runs true. For the four-jaw, set each jaw with the chuck key and clock the work with a dial indicator, halving the error at each adjustment. Support long work with the tailstock or a steady.
Safety / precautions	A chuck is heavy — use a chuck cradle or a second person and never lift it by the jaws. Ensure the jaws and the work do not project beyond the safe margin. Never start the spindle with the chuck key in place. Do not grip work on a rough cast surface without soft jaws or packing.
Care / calibration	Strip, clean and grease the scroll and jaws periodically; keep the jaw threads free of swarf; check for bell-mouthed jaws; store chucks on a wooden rack.
Trades	Turner, Fitter, Machinist, Tool & Die Maker
PPE	Safety glasses, safety shoes, gloves only for handling the chuck when the machine is isolated

T-003 Lathe cutting tools — HSS and carbide inserts

Also called: HSS bit 6/8/10 x 100 mm; brazed carbide tool; indexable insert holder

Identify	A single-point tool with a ground cutting edge: HSS bits ground to the correct rake, clearance and nose radius, brazed carbide tips, or indexable inserts in a holder with the manufacturer's grade and geometry. The CITS Annexure lists HSS lathe tool sets and bits.
Operate	Select the tool geometry for the material and operation, set the tool exactly on centre with a gauge, use minimum overhang, and choose the speed, feed and depth of cut. Grind HSS tools on a bench grinder with a slight nose radius and hone the edge on an oilstone.
Safety / precautions	A tool that is below centre or has excessive overhang will dig in and can break or throw the work. Keep hands clear when setting the tool and never touch a tool or chip straight after cutting — it is hot.
Care / calibration	Hone HSS edges frequently, replace a chipped insert and the seat, keep the tool post and shims clean, and store tools in a rack with the edges protected.
Trades	Turner, Fitter, Machinist, Tool & Die Maker
PPE	Safety glasses; heat-resistant gloves when handling hot tools

T-004 Tailstock, live centre, dead centre and steady rest

Also called: Revolving centre; dead centre; fixed steady; travelling steady

Identify	The tailstock slides on the bed and carries a quill with a Morse taper socket; the live centre rotates with the work, the dead centre is fixed and needs lubrication; the fixed steady supports long work from the bed and the travelling steady follows the carriage.
Operate	Drill and countersink the work end, lubricate a dead centre, bring the tailstock up and lock it, and apply light pressure with the quill. Set the steady fingers to a running fit with oil.
Safety / precautions	A loose tailstock or a dry dead centre will seize and burn the centre hole or throw the work. Do not over-tighten the quill, which bows the work.
Care / calibration	Keep the tapers clean and dry, oil the quill, check the centre point for damage, and adjust the steady fingers for the work diameter.
Trades	Turner, Fitter, Machinist
PPE	Safety glasses

T-005 Vertical milling machine and horizontal milling machine

Also called: Turret / knee-type vertical mill; horizontal mill; universal mill

Identify	A machine with a vertically or horizontally oriented rotating spindle carrying a milling cutter, on a table with X, Y and Z movement, with a quill, a drawbar and a spindle taper (R8, ISO 30/40).
Operate	Clamp the work in a machine vice or on the table with clamps and a dial indicator, set the spindle speed and feed for the cutter and material, select up-cut or down-cut milling direction, and use coolant. Use an edge finder or a dial test indicator for setting the datum, and take a trial cut with the cutter clear.
Safety / precautions	Milling cutters are multi-tooth and grab; keep hands well away, use a brush to clear chips and never reach across the table while the spindle turns. Ensure the drawbar is tight and the cutter rotates in the correct direction. Wear glasses or a face shield and never use gloves near the cutter.
Care / calibration	Oil the ways and the lead screws, keep the table and T-slots clean, check the gib adjustment, and clean the coolant system.
Trades	Machinist, Tool & Die Maker, Fitter (advanced), Millwright
PPE	Safety glasses or face shield, hearing protection, no gloves

T-006 Milling cutters and arbors

Also called: End mill; slot drill; face mill; side and face cutter; slitting saw; boring head

Identify	Multi-tooth cutters in HSS, cobalt or carbide: end mills cut on the end and the periphery, slot drills can plunge, face mills plane a surface, side and face cutters cut slots, slitting saws part off, and the boring head fits the spindle for accurate hole sizing.
Operate	Select the cutter for the feature and the material, mount it with minimum overhang in a clean arbor or collet, and set the speed for the cutter diameter and material. Climb mill only with backlash eliminators, otherwise conventional mill.
Safety / precautions	Cutters are sharp and brittle — handle with a cloth, and never with the machine running. A cutter that is too large for the arbor or too long in overhang will vibrate and shatter.
Care / calibration	Clean the flutes, protect the edges in a rack, and have cutters reground when the edges dull; keep the arbors and spacers clean.
Trades	Machinist, Tool & Die Maker, Fitter
PPE	Safety glasses, no gloves, hearing protection

T-007 Shaping machine and planing machine

Also called: Shaper with clapper box; planer; slotter

Identify	The shaper drives a single-point tool in a reciprocating ram over a stationary workpiece on a cross-sliding table; the planer moves the work under a stationary tool; the slotter cuts internal keyways with a vertical ram.
Operate	Clamp the work, set the stroke length and position, set the table feed, and use the clapper box so that the tool lifts on the return stroke. Take light finishing cuts with a round-nosed tool.
Safety / precautions	The ram and the table move under power — keep clear of the full stroke, guard the drive and check the tool is clear before starting. Chips are hot and sharp; a shaper can throw them some distance.
Care / calibration	Oil the ram ways and the lead screws, check the stroke adjustment and the clapper spring, and clean the swarf from the table.
Trades	Machinist, Fitter, Tool & Die Maker, Millwright
PPE	Safety glasses or face shield, no gloves

T-008 Grinding machines — surface, cylindrical and tool and cutter

Also called: Surface grinder with magnetic chuck; cylindrical grinder; tool and cutter grinder

Identify	A precision machine with a high-speed grinding wheel spindle, a work table and, on cylindrical types, a workhead and tailstock. The surface grinder usually has a magnetic chuck.
Operate	Dress the wheel, set the depth of cut at a few hundredths of a millimetre, balance the wheel, and use copious coolant. For the surface grinder, check the magnet is energised, place the work with a stop block for small parts, and use down-feed in small increments.
Safety / precautions	Grinding dust is a respiratory hazard and wheel burst is the critical hazard. Check the wheel speed rating, ring-test, use the guard and the eye shield, stand to one side when starting, and never grind on the side of a straight wheel. Beware of residual magnetism and of the work being thrown if the magnet fails.
Care / calibration	Balance and dress wheels, change the coolant and clean the settling tank, check the magnetic chuck for holding force, and keep the ways oiled.
Trades	Machinist (Grinder), Tool & Die Maker, Fitter (advanced), Mechanic Machine Tools Maintenance
PPE	Safety glasses or face shield, dust mask or LEV, hearing protection, no gloves

T-009 CNC lathe (turning centre)

Also called: CNC lathe with Fanuc/Siemens control; teach lathe

Identify	A lathe with a servo-driven carriage and cross slide, a turret with multiple tools, an enclosure with a sliding door and interlock, a control panel with a screen, and an RS-232, USB or network interface. CITS Fitter requires CNC turning practice, classification of CNC systems, part programming and DNC.
Operate	Power up and reference the axes, load the programme, set the tool offsets and the work offset (G54), run a dry run with rapid override reduced and single block, then machine. Inspect the first component and use tool-wear offsets to correct size. Write simple G-code programmes with G00, G01, G02/G03, canned cycles and tool changes.
Safety / precautions	Never open the enclosure door while the spindle turns — the interlock should prevent it, and it must never be bypassed. Verify tool offsets before the first cut because a wrong offset drives the tool into the chuck. Keep the door closed, the panel clean and hands clear of the turret. Know the feed hold, single block and emergency stop before running any programme.
Care / calibration	Change the way-lube oil and hydraulic oil, clean the coolant and the chip conveyor, back up programmes and parameters, and preserve the batteries that hold the parameters.
Trades	Turner, Fitter (advanced), Machinist, Operator Advanced Machine Tools
PPE	Safety glasses, safety shoes, no loose clothing; follow the enclosure

T-010 CNC machining centre (vertical and horizontal)

Also called: VMC with automatic tool changer; HMC; turn-mill centre

Identify	A CNC mill in an enclosure with a tool magazine and automatic tool changer, a spindle with a taper, coolant through nozzles, a chip conveyor and a control panel.
Operate	Reference the axes, load the programme and tool table, set the work offset with an edge finder or probe, simulate, run single block with reduced feed and rapid, then run. Always verify the tool length and diameter offsets.
Safety / precautions	The tool changer moves fast and without warning — stay out of the enclosure. Never bypass the door interlock. Verify the programme and the offsets before the first cut; most collisions and injuries come from a wrong offset or a wrong tool.
Care / calibration	Clean the taper, check the coolant concentration, clean the chip conveyor, and back up the parameters.
Trades	Machinist, Tool & Die Maker, Operator Advanced Machine Tools, Additive Manufacturing
PPE	Safety glasses, safety shoes

T-011 Machine vice, angle plate and clamping kit

Also called: Machine vice 100/150 mm; step clamps and T-nuts; angle plate; parallel blocks

Identify	A precision vice with a screw and a fixed jaw, bolted to the T-slots; a clamping kit with step clamps, T-nuts, studs and a clamp bar; hardened parallel blocks to raise the work.
Operate	Clean the table and the T-slots, align the vice with a dial test indicator, seat the work on parallels so it is not rocked by chips, and tighten the clamp nearest the cut first. Tap the work down with a soft mallet while tightening.
Safety / precautions	Work that is not seated or clamped can be thrown from the machine — always check that the jaws bear on the full face and that the clamps are clear of the cutter path. Do not over-tighten and distort thin work.
Care / calibration	Stone any burrs from the table and the vice base, oil the screw and the T-nuts, and store clamps in a tray.
Trades	Machinist, Fitter, Tool & Die Maker, Millwright
PPE	Gloves when handling heavy vices, safety shoes

T-012 Arbor press and broaching press

Also called: Hand arbor press; broaching machine

Identify	A press with a rack and pinion or a hydraulic ram, used with broaches, bearing installation tools and presses.
Operate	Centre the work and the tool under the ram, apply pressure slowly and straight, and use the correct broach and shims.
Safety / precautions	Broaches are hard and brittle; a misaligned broach shatters. Use a guard and keep hands clear.
Care / calibration	Oil the rack and the ram, keep the broach guides clean, and store broaches in their holders.
Trades	Fitter, Tool & Die Maker, Mechanic Machine Tools Maintenance
PPE	Safety glasses, gloves

T-013 Machine tool alignment tools

Also called: Test mandrel; dial test indicator; precision level; laser alignment kit

Identify	Precision instruments used to check that a machine is level, that the spindle is true, that the tailstock is aligned with the headstock, and that the ways are straight.
Operate	Level the bed with a precision level in both directions, sweep the spindle taper with a dial indicator on a test mandrel to check runout, and turn a test bar between centres to check tailstock alignment.
Safety / precautions	Isolate and lock off the machine before making alignment checks, and remove all instruments before restarting.
Care / calibration	Calibrate the instruments, keep them in cases, and clean the mounting surfaces.
Trades	Mechanic Machine Tools Maintenance, Millwright, Fitter, Lift & Escalator
PPE	None beyond normal shop PPE

T-014 Coolant system, chip conveyor and swarf handling

Also called: Coolant pump and tank; suds; chip conveyor; swarf bin

Identify	A pump, tank and nozzle system delivering cutting fluid to the cutting zone; a conveyor or auger to remove chips from the machine; a swarf bin for collection.
Operate	Check the level and concentration, direct the nozzle at the cutting zone, and clean the tank and strainer on schedule. Never use compressed air to clear chips.
Safety / precautions	Coolant mist is a respiratory irritant and can cause dermatitis — keep the concentration correct, use local exhaust or a mask, and wear gloves. Wet floors around machines are a serious slip hazard; mop up and use drip trays.
Care / calibration	Skim tramp oil, change the coolant periodically, clean the tank, and dispose of used coolant and swarf as hazardous or recyclable waste per the local rules.
Trades	All machine shop trades
PPE	Gloves, apron, safety shoes, mask where mist forms

T-015 Additive manufacturing (3D printer) and accessory tools

Also called: FDM printer; resin (SLA) printer; filament dryer; deburring and support-removal tools

T-015	Additive manufacturing (3D printer) and accessory tools
Identify	A computer-controlled machine that builds parts layer by layer from filament (FDM) or resin (SLA); accessories include nozzles, build plates, scrapers, flush cutters, tweezers, IPA for resin cleaning and a post-cure unit. The Additive Manufacturing trade and the CITS emerging-technology outcome both require it.
Operate	Prepare the model in the slicer, check the bed levelling and the filament, start the print, and monitor the first layer. Remove supports with flush cutters and deburring tools and post-cure resin parts as specified.
Safety / precautions	The nozzle and the heated bed reach 200–300 degrees C and cause serious burns. Resin is a skin sensitiser and an eye irritant — wear nitrile gloves and goggles, keep the printer ventilated, and never pour resin down a drain. Enclosure fumes from ABS and ASA need extraction. Keep fingers out of moving axes and never reach into a printer that is running.
Care / calibration	Clean the nozzle, level the bed, store filament dry, filter the resin, clean the screen, and dispose of waste per the resin manufacturer's instructions.
Trades	Additive Manufacturing, Tool & Die Maker (prototyping), Draughtsman, CITS emerging technology
PPE	Nitrile gloves, goggles, ventilation; heat-resistant gloves for hot parts

Group 5 — Joining, Welding, Thermal Cutting and Fastening

Joining covers the widest range of hazards in one group: high current, ultraviolet radiation, fuel gas at pressure, hot metal, fume and stored energy. The CTS and CITS outcomes require the trainee not only to weld but to set up the equipment, adjust the parameters, inspect the joint and observe the safety norms for arc, TIG and gas welding, soldering, brazing and riveting. Chapter 3.8 on gas cylinders and 3.9 on fire apply to everything below.

14 entries in this group.

W-001 Arc welding machine (transformer, inverter and rectifier)

Also called: SMAW set 200–400 A; AC transformer; DC inverter; welding generator

Identify	A power source with an on/off switch, a current range selector or digital display, a polarity switch on DC sets, two output terminals (electrode holder and work return clamp) and a cooling fan. Inverter sets are light, transformer sets heavy and robust, and engine-driven generators are used on site.
Operate	Connect the electrode holder to the correct polarity for the electrode, connect the work return clamp to bare clean metal close to the joint, set the current for the electrode diameter and the material thickness, strike and maintain the arc, and inspect the bead. Adjust current as the work heats or the position changes.
Safety / precautions	The open-circuit voltage is lethal — never touch the holder jaws or bare metal with the machine on. Isolate before changing cables or electrodes. The arc emits intense ultraviolet that causes 'arc eye' within minutes: use a proper welding helmet of the correct shade, keep the screen and the flash guards in place, and warn others. Keep cables and the return clamp in good condition; a poor return causes stray currents that can damage bearings and shock the operator. Earth the casing, keep the work area dry and never weld in a wet or confined space without specific precautions.
Care / calibration	Blow out dust, check the fan, inspect the cable insulation and the holder, keep the terminals tight, and store the set dry.
Trades	Welder (Gas & Electric), Welder (Pipe), Fabricator, Fitter, Mechanic Motor Vehicle
PPE	Welding helmet, gauntlets, leather apron, safety boots, fume extraction

W-002 Electrode holder and welding cables

Also called: Insulated stinger 300/400 A; welding cable 25–70 mm2; earth clamp

Identify	The holder is an insulated tong assembly with a spring or screw grip for the electrode; the cable is a heavy flexible, finely stranded copper cable with rubber insulation; the earth clamp is a spring or screw clamp with copper jaws.
Operate	Insert the electrode at about 90 degrees to the jaws for flat work and adjust for position work. Keep the cable run short and untangled, and fix the earth clamp to clean bare metal on the workpiece itself.
Safety / precautions	A damaged holder, exposed jaw or cracked insulation is a direct electrocution hazard — withdraw the tool. Never coil a cable around the body, never run a cable over sharp edges or a hot area, and never repair a cable with tape. Disconnect before changing electrodes on high-current sets.
Care / calibration	Inspect the insulation before each use, keep the jaws clean and replace worn ones, and coil the cable loosely when storing.
Trades	All welding and fabrication trades
PPE	Dry insulating gloves, welding gauntlets, rubber-soled boots

W-003 Welding helmet, hand shield and filter glass

Also called: Auto-darkening helmet; passive helmet shade 10–13; hand shield

Identify	A helmet or hand-held shield with a filter plate of a specified shade and a clear cover lens. Auto-darkening types have a liquid crystal filter, a solar or battery supply and adjustable shade and sensitivity.
Operate	Set the shade for the process and current — around shade 10 for low-current TIG and up to shade 13 for heavy arc gouging — adjust the headband, and ensure the clear cover lens is in good condition. Test the auto-darkening filter before each session.
Safety / precautions	Never weld without the helmet down, never look at an arc with the naked eye, and never use a helmet with a cracked or missing filter — the ultraviolet passes straight through. Do not rely on a solar auto-darkening helmet in poor light without checking that it darkens.
Care / calibration	Replace scratched cover lenses frequently (they are cheap), clean with a soft cloth, store the helmet where it will not be scratched, and remove the battery for long storage.
Trades	All welding trades
PPE	The helmet is itself PPE; add gauntlets, apron, ear protection when chipping

W-004 Welding table, positioner, clamps and fixtures

Also called: Arc welding table with positioner 100 x 75 x 70 cm; magnetic clamps; earth boss

Identify	A steel table, often slotted or with a fixture grid, on which work is tacked and clamped; a positioner rotates or tilts heavy work; magnetic and toggle clamps hold alignment.
Operate	Set the earth connection to the table for small work, tack the joint, check alignment with a square, and complete the weld in the flat position where possible by using the positioner.
Safety / precautions	Never weld with the earth current passing through a bearing or a hydraulic ram — use a proper return path. Hot offcuts fall through slots: keep the floor clear and wear spats. Clamps and fixtures must be rated for the current and must not be made of unknown alloys that fume.
Care / calibration	Keep the table free of spatter with a light anti-spatter spray, check the earth boss and clamps, oil the positioner bearings.
Trades	Welder, Fabricator, Fitter
PPE	Helmet, gauntlets, apron, safety boots, spats

W-005 Gas welding and cutting set (oxy-acetylene / LPG)

Also called: Oxy-acetylene welding torch; cutting torch with nozzles; regulators; hoses

Identify	Two cylinders with regulators (oxygen right-hand thread, acetylene left-hand thread), coloured hoses (oxygen black or green, acetylene red), flashback arresters, a torch body with a mixing head, welding nozzles and cutting nozzles.
Operate	Open the cylinder valves slowly, set the regulator pressures for the nozzle size, purge the hoses, light with a spark lighter using acetylene first then oxygen, and adjust to a neutral flame. For cutting, preheat the edge, open the oxygen jet and move steadily. Shut down in reverse order — oxygen first, then acetylene — and drain the hoses.
Safety / precautions	Never use oil or grease on any oxygen fitting. Never leave the torch burning unattended or hanging on a cylinder. Keep cylinders upright and chained, at least about 3 m from the flame and clear of slag. Fit flashback arresters at both ends. Test joints with soap solution, never a flame. Do not use a worn or greasy hose. Beware of acetylene at pressures above about 1 bar (15 psi) — it can decompose explosively. After work, close the valves and bleed the regulators to zero.
Care / calibration	Check the hoses for cracks, replace perished hose, clean the nozzle tips with the correct tip cleaner (never a drill), leak-test the regulators periodically, and store the torch with the valves closed.
Trades	Welder (Gas & Electric), Fabricator, Plumber, Refrigeration & AC, Sheet Metal

W-005 Gas welding and cutting set (oxy-acetylene / LPG)

PPE	Goggles shade 4–5, gauntlets, leather apron, spark-proof clothing

W-006 Gas cylinders and regulators

Also called: Oxygen cylinder; dissolved acetylene cylinder; LPG cylinder; argon and CO2 cylinders

Identify	Seamless steel cylinders in standard colours with a valve and a valve cap, identified by shoulder colour and marking. Regulators are two-gauge (cylinder contents and delivery pressure) or single-gauge with a diaphragm.
Operate	Identify the cylinder by marking, not colour alone. Crack the valve briefly to clear dust, fit the correct regulator, set the delivery pressure with the adjusting screw while watching the delivery gauge, and open the valve slowly.
Safety / precautions	Acetylene cylinders must never be laid on their side or used within about 30 minutes of being upright; the acetone and porous mass will be disturbed. Do not use a cylinder showing impact damage or a leaking valve. Keep full and empty cylinders segregated and marked, and never drop or roll one with the valve uncapped. A regulator with a damaged diaphragm will pass full cylinder pressure to the torch.
Care / calibration	Store under cover and ventilated, chained, with caps on; replace damaged gauges; have regulators tested; record cylinder receipt and return.
Trades	All welding, plumbing, refrigeration and fabrication trades
PPE	Gloves, goggles, safety boots; face shield when connecting

W-007 TIG welding set (GTAW)

Also called: TIG inverter with HF start; argon regulator and flowmeter; torch with tungsten electrode

Identify	A DC or AC/DC constant-current power source with a gas valve or solenoid, a torch with a ceramic cup and a collet holding a tungsten electrode, a foot pedal or torch amperage control, and an argon cylinder with a regulator and flowmeter.
Operate	Grind the tungsten to a point (longitudinal, not radial), set the current and the gas flow (typically 8–12 litres per minute), start the arc with the HF or lift-arc, form a weld pool and add filler rod at a shallow angle. Control the heat with the pedal. The CITS Fitter outcome requires identification of TIG welding tools.
Safety / precautions	TIG produces ozone and metal fume, and the arc is very bright — use extraction and a shade 10–11 lens. Tungsten electrodes can become radioactive with thoriated types; grind with dust extraction and store separately. The HF start can interfere with pacemakers and electronics. Hot tungsten and hot filler ends are easily touched — cut the filler end rather than breaking it.
Care / calibration	Clean the collet and the cup, replace a contaminated tungsten, check the gas hose for leaks, and keep the torch leads dry.
Trades	Welder (advanced), Fabricator (stainless and aluminium), Refrigeration & AC (pipe joints), Fitter
PPE	Auto-darkening helmet shade 10–11, gauntlets, fume extraction, safety glasses under the helmet

W-008 MIG / MAG welding set (GMAW)

Also called: MIG welder with wire feeder; spool gun for aluminium; CO2/Argon shielding gas

Identify	A power source with a wire feed unit, a spool of filler wire, a torch with a contact tip and a gas nozzle, a regulator and flowmeter, and an earth clamp. Parameter controls are voltage, wire feed speed and gas flow.
Operate	Fit the correct liner, drive rolls and contact tip for the wire, set the voltage and wire speed, check the gas flow with the trigger, and weld with the torch pushed or dragged as the process requires. Maintain a stick-out of about 10–15 mm and a consistent travel speed.

W-008 MIG / MAG welding set (GMAW)

Safety / precautions	The wire is fed under power and can pierce the hand — never point the torch at yourself and always keep the nozzle away from the body. Watch for burn-back and for the gas flow being lost. Ultraviolet and fume hazards are the same as for arc welding, with additional metal fume from the filler. Do not use a wire spool that has absorbed moisture.
Care / calibration	Replace the contact tip and liner when worn, blow out the wire path, drain the gas line, and store wire in a dry place.
Trades	Welder (Gas & Electric), Fabricator, Mechanic Motor Vehicle (body repair)
PPE	Helmet shade 10–12, gauntlets, apron, fume extraction

W-009 Soldering irons, soldering station and solder

Also called: Soldering iron 25–100 W; temperature-controlled station; solder wire with flux; solder pot

Identify	A heated bit or element with a copper or plated tip; the station has a temperature control and a stand with a sponge; solder is supplied as wire with a flux core, as paste or as bar for a pot.
Operate	Set the temperature for the solder and the joint (about 300–350 degrees C for leaded, slightly higher for lead-free), tin the tip, heat the joint and apply solder to the far side so that it flows into the joint, and remove the solder then the iron. Use the minimum time to avoid damaging components.
Safety / precautions	The iron reaches 300–400 degrees C and looks cool — always return it to the stand, never lay it on the bench. Solder fume from flux is a respiratory irritant; use extraction or a fan. Lead solder is toxic: no eating, drinking or smoking at the bench, wash hands after work, and keep solder away from children. Molten solder spits; wear glasses.
Care / calibration	Clean and re-tin the tip frequently, replace a corroded tip, keep the sponge damp, and store the iron in the stand.
Trades	Electronics Mechanic, Electrician, ICT, Mechanic Consumer Electronics, Refrigeration & AC (electrical joints)
PPE	Safety glasses, fume extraction, ESD-safe practices

W-010 Brazing and silver soldering equipment

Also called: Brazing torch; silver solder and flux; brazing hearth with firebricks

Identify	A gas torch with a brazing nozzle, filler rods (bronze or silver-based) and a flux; a hearth lined with firebricks to contain the heat.
Operate	Clean the joint, apply flux, heat evenly to the brazing temperature (the flux becomes clear), and feed the filler into the joint by capillary action without playing the flame on the filler.
Safety / precautions	Brazing fumes from flux and from cadmium-bearing filler are toxic — use extraction and avoid cadmium fillers in training. The work stays hot and looks cold; quench or handle with tongs. Fluoride fluxes burn skin.
Care / calibration	Clean the torch tip, store flux in a closed container, and keep firebricks dry.
Trades	Refrigeration & AC, Plumber, Sheet Metal, Welder, Electrician
PPE	Goggles, gauntlets, apron, fume extraction

W-011 Riveting tools — hand, pneumatic and pop riveter

Also called: Rivet set and snap; pneumatic rivet hammer with bucking bar; lazy tongs; pop riveter

Identify	Solid riveting uses a rivet set in a hammer and a bucking bar behind the work; pop or blind riveting uses a mandrel pulled by a hand or power tool.

W-011 Riveting tools — hand, pneumatic and pop riveter

Operate	Drill the hole to the rivet size, insert the rivet, hold the bucking bar against the tail, and upset the head with the rivet gun or hammer in short bursts. For blind rivets, insert and squeeze until the mandrel snaps.
Safety / precautions	Pneumatic rivet hammers cause vibration injury and hearing damage — use anti-vibration gloves, ear protection and a trigger with a dead-man control. The bucking bar can slip and crush a finger. Blind rivet mandrels eject at speed; wear glasses.
Care / calibration	Keep the rivet sets and snap dressed, oil the air motor, and check the hose connection with a whip check.
Trades	Sheet Metal Worker, Fabricator, Mechanic Motor Vehicle (body), Aircraft maintenance
PPE	Safety glasses, hearing protection, anti-vibration gloves

W-012 Brazing and welding inspection tools

Also called: Welding gauge (fillet and butt); weld size gauge; magnifying glass; dye penetrant kit

Identify	A folding gauge with fingers for measuring fillet leg length, throat thickness, undercut, misalignment and root gap; a magnifier; a dye penetrant or magnetic particle test kit for surface defects.
Operate	Clean the weld, place the gauge fingers against the profile, and read the dimension. For dye penetrant, clean, apply penetrant, dwell, remove the excess, apply developer and inspect under white light.
Safety / precautions	Penetrant chemicals are irritants and are often flammable — gloves, glasses and ventilation. Welds and the surrounding metal are hot; allow cooling before inspection.
Care / calibration	Clean the gauge after use, keep the penetrant kit closed and within shelf life, and calibrate the gauge.
Trades	Welder, Fabricator, Quality inspection, Mechanic Motor Vehicle
PPE	Gloves, safety glasses, ventilation for chemicals

W-013 Adhesive, sealant and mechanical fastening tools

Also called: Caulking gun; sealant applicator; nut runner; torque multiplier; pop riveter; thread-locking compound

Identify	Hand or air-powered tools for dispensing adhesives and sealants, and powered wrenches and multipliers for tightening.
Operate	Prepare the surface, apply the sealant in a continuous bead, and tool it before it skins. For threaded fasteners, clean the threads, apply the specified torque, and use thread-locking compound where specified.
Safety / precautions	Many adhesives and sealants are flammable, sensitising or corrosive in the uncured state — read the safety data sheet, wear gloves and goggles, and ventilate. Air tools need a whip check.
Care / calibration	Seal the cartridge after use, clean the nozzle, and store the compound within its shelf life.
Trades	Mechanic Motor Vehicle, Carpenter, Refrigeration & AC, Fabricator, Building Maintenance
PPE	Gloves, goggles, ventilation

W-014 Oxy-fuel cutting torch with nozzles and tip cleaner

Also called: Cutting torch; cutting nozzle size 0–6; tip cleaner set

Identify	A torch with a cutting oxygen lever and a mixing head for two preheat flames, fitted with a cutting nozzle numbered by plate thickness.

W-014 Oxy-fuel cutting torch with nozzles and tip cleaner

Operate	Select the nozzle for the thickness, set the pressures, light and adjust to a neutral flame, preheat the edge to bright orange, then depress the cutting lever and move steadily. Clean the nozzle orifices with the correct tip cleaner.
Safety / precautions	Never use a drill or a wire to clean a nozzle. Slag sprays downwards and bounces off the floor — protect feet and shins, clear the area below, and keep combustibles away. Never cut a sealed container or one that held fuel.
Care / calibration	Clean the nozzles, inspect the torch body for leaks, and store with the valves closed.
Trades	Welder, Fabricator, Sheet Metal, Mechanic Motor Vehicle (scrap cutting)
PPE	Goggles, gauntlets, apron, spats, hearing protection

Group 6 — Electrical Tools, Wiring Accessories and Test Instruments

The Electrician and Wireman trades use the largest single Annexure in the CTS system because the trade spans hand tools, wiring accessories, rotating machines, control gear and a full bench of test instruments. The CITS Electrician and Electronics Mechanic curricula add the demonstration and calibration duties. Every instrument in this group is a measuring device that can also be a hazard: a megger generates high voltage, a clamp meter is used on live conductors, and a multimeter is only as safe as its category rating and its leads.

20 entries in this group.

E-001 Pliers set — combination, nose, side cutting and crimping

Also called: Combination plier 200 mm insulated; long-nose plier; diagonal cutter; crimping tool 1.5–16 mm2 and 16–95 mm2

Identify	Insulated handles marked with the voltage rating (typically 1000 V), serrated gripping jaws, a cutting edge near the pivot, and, on the crimper, a set of graduated nests marked with the conductor size.
Operate	Cut and strip small conductors with side cutters, grip and twist with combination pliers, and form terminations with the nose plier. For crimping, select the nest marked for the lug and conductor size, insert the lug fully and close the handles completely so that the ratchet releases.
Safety / precautions	Always isolate before working on a circuit; insulated tools reduce the risk of accidental contact and are not a licence for live work. Never cut a live conductor. Check the insulation for cuts and the cutting edge for nicks before use.
Care / calibration	Oil the pivot, keep the cutting edges dressed, and discard a tool with damaged insulation — the voltage rating depends on it.
Trades	Electrician, Wireman, Electronics Mechanic, Lift & Escalator, Refrigeration & AC
PPE	Insulating gloves, safety glasses, insulated footwear

E-002 Electrician's knife and cable stripping tool

Also called: Double-bladed electrician's knife 100 mm; automatic wire stripper; cable sheath stripper

Identify	A folding knife with one straight and one hooked blade, or a plier-like stripper with graduated cutting nests and an adjustable stop.
Operate	Score the sheath lightly around the cable, peel it back, and strip the insulation without nicking the conductor. Use the correct nest for the conductor size and keep the blade away from the copper.
Safety / precautions	Always cut away from the body and never towards the hand holding the cable; a nicked conductor is a weak point that will fail and heat. Never cut a live cable. Fold the blade closed when not in use.
Care / calibration	Keep the blade sharp, oil the hinge, and replace a worn stripper nest.
Trades	Electrician, Wireman, ICT, Electronics Mechanic
PPE	Cut-resistant gloves for stripping, safety glasses

E-003 Neon tester and voltage detector

Also called: Neon screwdriver tester 500 V; two-pole voltage tester; non-contact voltage pen

Identify	A single-pole tester has a neon lamp and a resistor in a transparent handle with a metal contact; a two-pole tester has two probes and a lamp or display; a non-contact pen senses the electric field and beeps or lights.

E-003 Neon tester and voltage detector

Operate	Test the tester on a known live source before and after the test. For the single-pole type, touch the probe to the conductor and the finger contact to the body; for the two-pole type, connect both probes across the conductors.
Safety / precautions	A neon tester proves presence, never absence of voltage with confidence, and it cannot indicate the voltage level. Use a two-pole tester or a proven multimeter for isolation verification. A non-contact pen gives false readings near other conductors. Never rely on a single tester; test, then verify with a second instrument and prove the tester on a known source.
Care / calibration	Check the lamp and resistor periodically, replace a cracked body, and keep the probes clean.
Trades	Electrician, Wireman, Lift & Escalator
PPE	Insulating gloves, safety glasses, insulated footwear

E-004 Multimeter (analogue and digital)

Also called: DMM with CAT III/IV rating; analogue multimeter; clamp-on ammeter

Identify	A meter with a rotary range switch, a display, a common and a voltage/ohms input socket, sometimes a separate 10 A socket, and test leads with probe tips. A digital meter displays values; an analogue meter has a moving pointer.
Operate	Select the function and range before connecting the leads, connect in parallel for voltage and in series for current, and select the highest range when the value is unknown. For resistance or continuity, isolate and de-energise the circuit and check for parallel paths. Use the continuity buzzer for fault finding and check diodes on the diode range.
Safety / precautions	Check the meter's safety category against the circuit — CAT II for appliances, CAT III for distribution boards, CAT IV for service entrance. Use fused, shroud-protected leads in good condition and never use a cracked probe. Never measure resistance or continuity on a live circuit. On an analogue meter, wrong range selection can destroy the movement; on any meter, a wrong function on a live circuit can arc and injure.
Care / calibration	Check the battery, replace blown fuses with the correct rating, keep the probes clean, and have the meter calibrated for training and assessment use.
Trades	Electrician, Wireman, Electronics Mechanic, Refrigeration & AC, Mechanic Motor Vehicle, Lift & Escalator, Solar technician
PPE	Insulating gloves, safety glasses

E-005 Clamp meter (tong tester)

Also called: Clamp-on ammeter 0–100 A digital; leakage clamp meter

Identify	A meter with a hinged jaw that encircles a single conductor and measures current by magnetic induction, with a rotary switch for AC current, voltage and resistance. A leakage clamp has higher resolution for small currents.
Operate	Set the range, open the jaws, encircle one conductor only, close the jaws fully, and read the current. For a three-phase circuit, clamp each phase separately.
Safety / precautions	Clamp round a single conductor — clamping a cable containing both live and neutral reads zero. Keep fingers behind the guard, never clamp a bare conductor, and use the CAT-rated instrument for the panel. A clamp meter is not an isolation device.
Care / calibration	Check the jaw faces are clean and close fully, replace the battery, and verify the reading against a known load.
Trades	Electrician, Wireman, Lift & Escalator, Refrigeration & AC
PPE	Insulating gloves, arc-rated clothing for panel work

E-006 Megger (insulation resistance tester)

Also called: Megger 500 V and 1000 V; insulation tester with hand crank or electronic source

Identify	An instrument that applies a high DC voltage (commonly 500 V or 1000 V) and measures resistance in megohms, with two (or three, with a guard) test terminals.
Operate	Isolate and discharge the circuit, disconnect sensitive equipment, connect the leads across the insulation, apply the test voltage, and read the resistance after the reading stabilises (typically 60 seconds for polarisation index). Test between phase and earth, phase and phase, and across the winding.
Safety / precautions	The megger generates a high voltage that can give a painful shock, and it stores charge in the instrument and in the circuit under test — discharge to earth afterwards and never touch the leads during or immediately after the test. Never test a live circuit or an electronic device. Do not use on a capacitor bank without discharging it.
Care / calibration	Check the battery or the crank drive, keep the leads in good condition, and have the instrument calibrated.
Trades	Electrician, Wireman, Lift & Escalator, Mechanic Motor Vehicle, Millwright
PPE	Insulating gloves, safety glasses, insulated footwear; second person present

E-007 Earth tester and earth resistance measurement

Also called: Three- and four-terminal earth tester; clamp-on earth tester

Identify	An instrument with spikes and leads for the fall-of-potential method, or a clamp-on type that measures the loop resistance without disconnecting.
Operate	Drive the potential and current spikes at the specified spacing in a straight line from the electrode, connect in the fall-of-potential configuration, and read the resistance. Take readings at several spacings to confirm a plateau.
Safety / precautions	Isolate the installation before testing, drive spikes clear of buried services, and do not test during a thunderstorm.
Care / calibration	Clean the spikes, check the leads, and calibrate the instrument.
Trades	Electrician, Wireman, Electrical maintenance
PPE	Gloves, safety boots, hi-vis vest

E-008 Wiring accessories — switches, sockets, holders and MCBs

Also called: Modular switch and socket; batten holder; ceiling rose; MCB 6–63 A; RCCB/ELCB; distribution board

Identify	Insulated moulded devices rated for voltage and current: switches break the phase, sockets receive the plug, holders carry the lamp, and protection devices (MCB for overcurrent, RCCB or ELCB for earth leakage, MCCB for larger currents) sit in a distribution board.
Operate	Select the correct rating, wire the phase to the switch and the neutral to the holder or socket as per the standard, tighten terminals to the specified torque, and test the polarity and the earth continuity. Fit the correct size of protective device for the cable and the load.
Safety / precautions	Isolate and lock off the supply, verify dead, then work. Never bridge a protective device or fit a fuse or MCB of a higher rating than the cable can carry. Test the RCCB with its test button after installation and periodically. Do not mix polarity — a switch in the neutral leaves the holder live even when switched off.
Care / calibration	Keep terminals tight, clean the contacts, replace a device that shows arcing or heat discolouration, and label the circuits in the board.
Trades	Electrician, Wireman, Lift & Escalator, Building Maintenance
PPE	Insulating gloves, safety glasses, insulated tools

E-009 Conduit, bending and threading tools

Also called: Conduit bender (hand and hydraulic); PVC bending spring; conduit die and stock; fish tape

Identify	The bender is a shoe and handle that bends conduit to a radius; a bending spring supports PVC conduit internally; a die and stock cut threads on mild steel conduit; the fish tape is a flexible steel or nylon tape for pulling cables through conduit.
Operate	Measure and mark, place the conduit in the bender with the mark on the arrow, and bend with steady pressure, checking the angle with a square. For PVC, insert the spring, heat gently and bend, then cool before removing the spring. Thread steel conduit with cutting oil, reversing to break the chip.
Safety / precautions	A hydraulic bender develops high force — keep fingers out of the shoe. Heat PVC evenly; overheating releases hydrogen chloride fumes. Debris in conduit can cut the hand when fishing; wear gloves.
Care / calibration	Oil the bender pins and the die, keep the spring clean, and store the fish tape coiled without kinks.
Trades	Electrician, Wireman, ICT cabling
PPE	Gloves, safety glasses, fume extraction when heating PVC

E-010 Cable pulling, jointing and termination tools

Also called: Cable roller; cable jack; crimping tool; heat shrink gun; cable jointing kit; binding wire

Identify	Rollers and jacks support heavy cable; crimping tools make compression joints; a heat gun shrinks the insulation sleeve; jointing kits contain tape, mastic and ferrules.
Operate	Set rollers at intervals, pull with steady tension and no kinks, and maintain the minimum bend radius. For a termination, strip the correct length, fit the ferrule or lug, crimp in the marked nest, and insulate with heat-shrink.
Safety / precautions	Cable pulling involves high tension — never stand inside a bight or allow anyone to stand where a rope or cable could snap. Heat guns reach about 600 degrees C. Crimping produces a sharp burr.
Care / calibration	Keep crimp nests clean, check the ratchet release, and store the heat gun on its stand.
Trades	Electrician, Wireman, Lift & Escalator, ICT
PPE	Gloves, safety glasses, safety boots

E-011 Electric motors and starters (demonstration set)

Also called: Three-phase induction motor; single-phase motor; DOL starter; star-delta starter; 3-point and 4-point DC starter

Identify	A motor with a terminal box and a nameplate giving voltage, current, speed, frame and insulation class; starters with contactors, overload relays and push buttons. The Electrician Annexure lists 3-point and 4-point DC starters for 2.5 kW motors.
Operate	Read the nameplate, connect the winding for the supply voltage (star or delta), check the direction of rotation, set the overload relay to the motor full-load current, and start and stop with the correct starter. Measure the winding resistance and insulation resistance before energising.
Safety / precautions	Isolate and lock off before opening the terminal box. Beware of the residual charge in capacitors on single-phase motors and of the high inrush current. Couple the motor to a load only with the guard fitted. Rotating shafts and couplings must be guarded.
Care / calibration	Lubricate the bearings, keep the cooling fan and fins clean, check the terminal tightness, and test the overload relay.
Trades	Electrician, Wireman, Lift & Escalator, Refrigeration & AC, Millwright
PPE	Insulating gloves, safety glasses, insulated tools

E-012 Transformer and winding equipment

Also called: Small transformer kit; winding machine; coil winder; varnish and baking oven; laminations

Identify	A winding machine with a counter and a spindle; bobbins and formers; enamelled copper wire; laminations; insulation paper; a varnish dip and a bake oven.
Operate	Wind turns evenly in layers with insulation between layers, terminate the leads securely, assemble the core, dip in insulating varnish and bake, then test the turns ratio, insulation resistance and no-load current.
Safety / precautions	The oven is hot and the varnish is flammable and gives off solvent vapour — use extraction and keep ignition sources away. The winding machine can trap fingers and hair.
Care / calibration	Keep the wire spools clean, check the counter, and store varnish closed.
Trades	Electrician, Electronics Mechanic, Mechanic Transformer
PPE	Gloves, safety glasses, fume extraction, heat-resistant gloves at the oven

E-013 Battery, charger and DC supply

Also called: Lead-acid battery; lithium-ion pack; battery charger; SMPS; UPS

Identify	Cells with terminals marked positive and negative, an electrolyte level on flooded types, a vent, and a charger with voltage and current settings or an automatic profile. Lithium packs carry a battery management board.
Operate	Connect positive to positive and negative to negative, charge in a ventilated space, and monitor the temperature and the gassing. Equalise and top up flooded cells with distilled water only, and test the specific gravity with a hydrometer.
Safety / precautions	Lead-acid batteries release hydrogen, which is explosive: no sparks or flames, and ventilate. The electrolyte is sulphuric acid — goggles, gloves and an eye wash are mandatory, and acid is added to water, never the reverse. Lithium cells can go into thermal runaway if shorted, overcharged or punctured: never charge unattended, never use a damaged pack, and use the correct charger. Remove metal jewellery — a shorted ring causes deep burns.
Care / calibration	Keep the terminals clean and greased, store charged, and dispose of batteries through the authorised recycler.
Trades	Electrician, Mechanic Motor Vehicle, Solar technician, UPS maintenance, Electronics Mechanic
PPE	Chemical goggles, acid-resistant gloves and apron, face shield, eye wash nearby

E-014 Oscilloscope and signal generator

Also called: Digital storage oscilloscope; function generator; logic analyser

Identify	An oscilloscope displays voltage against time on a screen, with vertical (volts per division) and horizontal (time per division) controls and one or more channels; the signal generator produces sine, square and triangle waveforms of set amplitude and frequency.
Operate	Set the probe attenuation and compensate the probe, connect the ground clip to circuit ground, set the vertical and timebase controls, and trigger on the signal. Use the generator to inject a known signal for tracing.
Safety / precautions	An oscilloscope ground clip is connected to the protective earth — never connect it to a live conductor or to a point above earth potential. Use an isolating transformer or a differential probe for mains-referenced measurements. Do not exceed the probe's voltage rating.
Care / calibration	Store the probe carefully, keep the instrument cooled and dry, and check the calibration output.
Trades	Electronics Mechanic, Mechanic Industrial Electronics, ICT, Mechanic Medical Electronics, CITS Electronics

E-014 Oscilloscope and signal generator

PPE	Safety glasses, no live contact

E-015 LCR meter, component tester and logic probe

Also called: LCR bridge; transistor tester; logic probe; ESR meter

Identify	Bench instruments with test clips that measure inductance, capacitance and resistance, identify semiconductor junctions, or indicate logic high, low and pulse states.
Operate	Discharge capacitors, isolate the component, connect the clips and read the value; compare with the marked value and tolerance.
Safety / precautions	Capacitors hold a charge after the power is removed — discharge through a resistor and verify with a meter before touching. Solder joints and hot components are a burn hazard.
Care / calibration	Calibrate, keep the clips clean, and store in a dry place.
Trades	Electronics Mechanic, Mechanic Consumer Electronics, Mechanic Computer Hardware, ICT
PPE	Safety glasses, ESD precautions

E-016 Soldering and desoldering station with ESD protection

Also called: Temperature-controlled soldering station; desoldering pump and wick; hot air rework station

Identify	A station with a temperature-controlled iron, a hot-air handpiece on rework types, and an ESD-safe grounded tip; a desoldering pump is a spring-loaded suction tool and desoldering wick is braided copper.
Operate	Set the temperature, tin the tip, heat the joint and apply solder, then remove the solder and the iron. For desoldering, heat the joint and use the pump or wick to remove the solder.
Safety / precautions	The tip and the hot air nozzle reach 300–450 degrees C — burns are instant and the iron must always be returned to the stand. Flux fume causes asthma; use extraction. The pump can eject molten solder.
Care / calibration	Re-tin the tip, clean the heating element, and replace the tip when the plating is gone.
Trades	Electronics Mechanic, Mechanic Computer Hardware, ICT, Mechanic Consumer Electronics
PPE	Safety glasses, fume extraction, ESD wrist strap

E-017 Insulated hand tools with voltage rating

Also called: VDE 1000 V pliers, screwdrivers, cutters and spanners

Identify	Tools with a two-colour insulating layer and a VDE or IEC 60900 marking with the rated voltage (1000 V AC). Insulation covers the handle and most of the shaft.
Operate	Use the correct type and size for the fastener and apply pressure in line with the screw or nut. Keep the insulated shaft between the hand and the live part.
Safety / precautions	The insulation is rated for 1000 V AC and must be inspected before every use — a cut, a burn or a puncture voids the rating. Insulated tools do not make live work safe; they reduce the consequence of an accidental contact. Never use them above the rated voltage.
Care / calibration	Clean with a damp cloth and mild detergent only, store in a dry tool roll, and discard a tool with any damage to the insulation.
Trades	Electrician, Wireman, Lift & Escalator
PPE	Insulating gloves, arc-rated clothing, face shield

E-018 Electrical safety equipment — LOTO, RCD and insulated mat

Also called: Lockout padlock and tags; residual current device; insulating mat and gloves; voltage detector; arc flash shield

Identify	A lockout hasp and padlock with a tag for isolation; an RCD (30 mA for personnel protection) in the supply; rubber insulating mats and gloves rated for the voltage; a voltage detector as the first line of proof.
Operate	Identify the isolation point, open it, lock and tag, prove dead with a detector of the correct voltage, and discharge stored energy. Only then work. Restore by removing personnel, checking the work, and removing the lock with the key holder present.
Safety / precautions	Never defeat an RCD; test it with the test button monthly. Insulating gloves must be air-tested and within their test date. An insulating mat with oil or a hole in it does not insulate. A tag without a lock is advisory, not a control.
Care / calibration	Record and date-test the protective equipment, store gloves in a bag away from sunlight, and replace damaged items.
Trades	All electrical trades
PPE	These are the PPE for the trade

E-019 Solar PV tools and test gear

Also called: DC clamp meter; irradiance meter; MC4 crimper and spanner; module tester

Identify	A DC-rated clamp meter for strings, an irradiance meter for performance ratios, MC4 crimping tools and spanner sets for module connectors, and an insulation and continuity tester for the array.
Operate	Isolate the array at the DC isolator, use the DC clamp to measure string current, and crimp MC4 connectors with the matched crimper and check the pull test.
Safety / precautions	A PV array generates voltage whenever there is light — it cannot be switched off, so cover or isolate at the string level and treat DC conductors as live. DC arcs do not self-extinguish, which is why DC-rated devices and tools are required. Work with insulated tools and gloves, and never use an AC-rated meter on a DC string.
Care / calibration	Keep connectors clean and dry, inspect for hot spots with a thermal camera, and record the test results.
Trades	Solar technician, Electrician (renewables)
PPE	Insulating gloves rated for DC, arc-rated clothing, safety glasses

E-020 Ladder, platform and insulated access equipment

Also called: Aluminium and fibreglass ladder; step ladder; mobile platform; insulated platform

Identify	A ladder with side stiles and rungs, a class or duty rating and a non-slip foot. The fibreglass type is non-conductive and mandatory near live electrical work.
Operate	Place at the correct angle (about 1 in 4), secure the top and the foot, maintain three points of contact, and never overreach. Use a permit and a harness for work at height.
Safety / precautions	Aluminium ladders conduct — never use one for electrical work. Inspect the ladder for cracks and missing rungs, never stand on the top two rungs, and never place a ladder on an unstable or slippery surface. Do not work alone at height.
Care / calibration	Store horizontally on racks, keep the feet clean, and withdraw a damaged ladder.
Trades	Electrician, Wireman, Lift & Escalator, ICT cabling, Building Maintenance
PPE	Safety harness for work at height, helmet, non-slip footwear

Group 7 — Electronics, Computer Hardware and ICT Tools

Electronics and ICT trades share a bench, a set of fine hand tools and a strict electrostatic-discharge discipline. The distinguishing hazards are small and easy to underestimate: a soldering tip that looks cold, a capacitor that holds a charge, a laser in a DVD drive, a lithium cell in a laptop, a CRT with stored high voltage. The CITS Electronics Mechanic curriculum also expects the instructor to demonstrate the operation and care of every instrument below.

12 entries in this group.

X-001 Anti-static wrist strap, mat and ESD station

Also called: ESD wrist strap; dissipative bench mat; grounding cord; ioniser

Identify	A fabric or elastic wrist band with a conductive inner surface and a coiled cord ending in a 1 megohm resistor and a clip; the mat is a dissipative rubber or vinyl sheet with a snap for the grounding cord.
Operate	Wear the strap on bare skin, clip the cord to a verified earth point through the resistor, and place the mat on the bench. Handle boards by the edges only, and keep components in conductive bags or tubes until needed.
Safety / precautions	The 1 megohm resistor limits current if the wearer contacts a live conductor — never substitute a plain wire. Do not wear a strap while working on a live mains circuit.
Care / calibration	Test the strap and cord periodically, clean the mat with an approved cleaner, and replace a broken cord.
Trades	Electronics Mechanic, Mechanic Computer Hardware, ICT, COPA, Mechanic Industrial Electronics
PPE	The ESD station is itself protective equipment

X-002 Precision screwdriver and bit set

Also called: Tweaker set; Torx, hex and Phillips precision drivers; nut driver set

Identify	Small screwdrivers with swivel tops and hardened tips in sizes from 0 to 3 mm, and a bit set on a magnetic holder.
Operate	Match the tip exactly to the screw head, apply axial pressure, and turn. Use a magnetic retriever for screws in tight places. Keep a screw tray so that nothing is lost inside equipment.
Safety / precautions	Small blades slip easily into the palm; hold the work in a clamp or on the bench, not in the hand. Do not force a worn screw. Avoid touching components on the board with the shaft.
Care / calibration	Keep the tips dressed, oil the swivel cap, and store the set in its case.
Trades	Mechanic Computer Hardware, Electronics Mechanic, ICT, Mobile repair
PPE	Safety glasses, ESD strap

X-003 Long-nose plier, flush cutter and tweezers

Also called: Snipe-nose plier 120 mm; side cutter; anti-magnetic tweezers

Identify	Small pliers with fine serrated jaws, a flush cutter with a hardened edge that cuts component leads flush with the board, and pointed or curved tweezers for placing surface-mount parts.
Operate	Bend and form leads with the plier, cut leads with the flush side against the board, and place components with tweezers while holding the iron in the other hand.
Safety / precautions	Cut leads can fly into the eye — wear safety glasses and cut away from the face. Tweezers are sharp; never use them to probe a live circuit.

X-003 Long-nose plier, flush cutter and tweezers

Care / calibration	Oil the pivot, keep the cutter edges dressed, and store tweezers with tips protected.
Trades	Electronics Mechanic, ICT, Mechanic Computer Hardware
PPE	Safety glasses, ESD strap

X-004 Desoldering pump, wick and hot air rework station

Also called: Solder sucker; desoldering braid; hot air gun with nozzles; preheater

Identify	A spring-loaded suction pump, a braided copper wick, and a hot-air station with a temperature and airflow control and interchangeable nozzles.
Operate	Heat the joint until the solder melts, then release the pump or apply the wick with the iron. For hot-air rework, select the nozzle, set the temperature and airflow, preheat the board, and lift the component with tweezers when the solder flows.
Safety / precautions	The hot-air nozzle reaches 400 degrees C and can ignite the board or the bench — keep it on its stand and away from solvents, and let it cool before changing the nozzle. Solder splash from the pump can burn. Use extraction for flux fume.
Care / calibration	Clean the pump barrel and nozzle, replace the tip, and check the airflow.
Trades	Electronics Mechanic, Mechanic Computer Hardware, ICT
PPE	Safety glasses, fume extraction, heat-resistant gloves for hot parts

X-005 Multimeter, logic probe and battery tester

Also called: Bench DMM; logic probe; battery load tester; PSU tester

Identify	As in Group 6, plus a logic probe for digital levels and a PSU tester with connectors for the standard power supply rails.
Operate	Read ATX rail voltages under load with the PSU tester, use the logic probe to trace a digital signal, and use the DMM for continuity and voltage checks.
Safety / precautions	The primary side of a switch-mode power supply holds about 340 V DC on the reservoir capacitor after the supply is unplugged — discharge it and wait before touching. Never probe a live primary side.
Care / calibration	Keep the probes clean, replace the battery, and check the calibration.
Trades	Mechanic Computer Hardware, Electronics Mechanic, ICT
PPE	Safety glasses, one hand in the pocket rule on live boards

X-006 Computer assembly and service tools

Also called: Anti-static bag; cable ties; crimping tool for RJ45; punch-down tool; cable tester; Torx and hex driver sets

Identify	RJ45 crimpers with a ratchet and a cutting and stripping function; a punch-down tool with 66 and 110 blades for patch panels; a cable tester with a remote terminator; cable ties and a tie gun.
Operate	Assemble the PC on the ESD mat, fit the CPU with the correct orientation, apply thermal paste, seat the RAM fully, and connect the power leads. For networking, untwist the minimum length, crimp the RJ45 with the correct colour code, and test the patch lead and the run.
Safety / precautions	Never force a connector or a CPU into a socket. The power supply holds a charge — work with it unplugged. Do not use a vacuum cleaner on a PC; static and dust ignite.

X-006 Computer assembly and service tools

Care / calibration	Clean the case fans and filters, replace the thermal paste periodically, keep the crimper clean, and store the tester with the battery removed.
Trades	Mechanic Computer Hardware, ICT System Maintenance, COPA, Network Technician
PPE	ESD strap, safety glasses when cutting cable ties

X-007 Network tools — crimper, punch-down, tester and tone generator

Also called: RJ45 crimper; 110 punch-down tool; LAN cable tester; tone and probe kit; fibre cleaver and connector kit

Identify	Hand tools for terminating twisted-pair and fibre cable, and test sets that verify continuity, pinout and (on better testers) length, attenuation and crosstalk.
Operate	Terminate to T568A or T568B consistently at both ends, test with the remote unit, and label both ends. For fibre, strip, cleave, and terminate with the specified connector and polish if required.
Safety / precautions	Fibre offcuts are microscopic glass splinters that cause serious eye injury and cannot be seen on skin — use a black mat, keep the waste pot labelled, never use bare hands, and wear eye protection. Laser light in a fibre system is invisible and can damage the retina — never look into a fibre end or a connector.
Care / calibration	Clean fibre connectors with lint-free swabs and IPA, keep the cleaver blade clean, and store the tester charged.
Trades	Network Technician, ICT System Maintenance, COPA
PPE	Safety glasses, gloves for fibre waste handling

X-008 Signal generators, power supply (bench) and frequency counter

Also called: Regulated DC bench supply 0–30 V / 0–5 A; function generator; frequency counter; isolation transformer

Identify	Bench instruments with digital displays, output terminals, current limiting and, on the supply, constant-voltage and constant-current modes.
Operate	Set the voltage and current limit before connecting the load, connect with the polarity correct, and monitor the current. Use the isolation transformer when servicing equipment with a live chassis.
Safety / precautions	Verify the output before connecting a component — an accidental 30 V destroys a 3.3 V board and can cause a fire. Do not exceed the current rating. Keep the benches dry.
Care / calibration	Keep the vents clear, check the calibration, and replace the fuse with the correct type.
Trades	Electronics Mechanic, Mechanic Industrial Electronics, ICT
PPE	Safety glasses

X-009 Electronic trainer kits and project boards

Also called: Analogue and digital trainer; breadboard; microcontroller board; sensor modules

Identify	A self-contained training panel with a breadboard area, a power supply, potentiometers, lamps, switches and built-in test points; microcontroller boards with USB and GPIO headers.
Operate	Build the circuit on the breadboard with the power off, verify the connections against the schematic, then apply the supply. For microcontroller work, upload the programme and use the serial monitor to debug.
Safety / precautions	The trainer has exposed conductors on the breadboard — keep metal objects and liquids away. Electrolytic capacitors and the regulator can become hot. Do not exceed the board supply rating and never short the rails.

X-009 Electronic trainer kits and project boards

Care / calibration	Keep the trainer covered, clean the breadboard contacts, and store components in labelled drawers.
Trades	Electronics Mechanic, ICT, COPA, Artificial Intelligence Programming Assistant, CITS Electronics
PPE	Safety glasses, ESD precautions

X-010 Display, printer and scanner service tools

Also called: Suction cup for screens; plastic opening tools and spudger; screwdriver set; USB cable; cleaning kit

Identify	Non-marring plastic pry tools, suction cups for lifting glass panels, a magnetic mat with screw positions, and lint-free cleaning cloths and fluid.
Operate	Power down and disconnect, discharge, use the plastic tools to release clips without forcing, record the screw positions, and reassemble in reverse.
Safety / precautions	A laptop or monitor may retain charge in the inverter or power board; wait and discharge. The display panel is fragile and cuts when broken; wear gloves. Broken backlight tubes in older LCDs contain mercury — handle as hazardous waste.
Care / calibration	Clean the screen only with the specified fluid, avoid pressure on the panel, and store the tool kit organised.
Trades	Mechanic Computer Hardware, Electronics Mechanic, ICT
PPE	Cut-resistant gloves, safety glasses, ESD strap

X-011 Laser and optical tools (printer, DVD, photo equipment)

Also called: Laser printer unit; DVD drive laser; digital camera and lens; optical power meter

Identify	Laser printer assemblies with a laser scanner unit and a fuser; DVD and Blu-ray drives with a laser diode; cameras with a lens and sensor; an optical power meter for fibre work.
Operate	Follow the manufacturer's service procedure, replace toner and drum units as instructed, and never open the laser scanner unit in training.
Safety / precautions	Laser radiation in a printer or drive is enclosed — never defeat the interlock. The fuser reaches about 200 degrees C. Toner is an eye and respiratory irritant; vacuum it with the specified toner-safe vacuum, never with a domestic cleaner. In cameras, never look at the sun through the lens.
Care / calibration	Follow the maintenance kit schedule, clean the optics with lint-free tissue only, and store the lens capped.
Trades	Mechanic Computer Hardware, Digital Photographer, Printing trades, ICT
PPE	Safety glasses, mask when handling toner

X-012 Digital camera, camcorder and tripod (training aid)

Also called: Digital video camera with hard disk or SD card; tripod; light panel

Identify	A camera with a lens, a sensor, a recording medium and controls for aperture, shutter speed and ISO; a tripod with a levelling head and a quick-release plate.
Operate	Set the exposure for the subject, brace the camera on the tripod, frame the shot, record, and transfer the file. The CITS Training Methodology curriculum specifically requires the application and care of the digital camera for recording demonstrations.
Safety / precautions	Do not point a camera at the sun or a welding arc — the sensor can be destroyed and the eye injured. Avoid tripod legs in walkways, and obtain consent before filming trainees.

X-012	Digital camera, camcorder and tripod (training aid)
Care / calibration	Charge and remove the battery, clean the lens with lens tissue, store the card separately, and keep the camera in a dry box in humid conditions.
Trades	CITS all trades (Training Methodology), Digital Photographer, Multimedia Animation & Special Effects
PPE	None beyond normal workshop PPE

Group 8 — Automotive and Vehicle Service Tools

Automotive trades add three hazards that the bench trades do not have: a heavy vehicle above the mechanic, a fuel system under pressure, and a high-current electrical system with an explosive battery. The Mechanic Motor Vehicle, Two Wheeler, Diesel, Auto Electrical and Agricultural Machinery trades all share the lifting, measuring, diagnostic and special-service tools below. Vehicle lifting and support is the single most important control in this group.

14 entries in this group.

A-001 Trolley jack and axle stands

Also called: Hydraulic floor jack 2–3 tonne; axle stands or jack stands; wheel chocks

Identify	A wheeled hydraulic jack with a lifting saddle and a release valve; axle stands are rated, toothed or pinned supports with a broad base; chocks are wedge blocks.
Operate	Chock the wheels, place the jack under the specified jacking point, lift to the required height, and immediately place axle stands under the chassis or axle at rated points before going under. Lower the jack so that the vehicle rests on the stands, not the jack.
Safety / precautions	Never work under a vehicle supported only by a jack. Never exceed the rated capacity and never place a stand on a soft or sloping surface. Bleed and check the jack for leaks before use. Keep the jack handle clear of the vehicle when lowering.
Care / calibration	Check the hydraulic oil level, bleed air from the cylinder, inspect the seals and the castors, and store the stands with the pins in place.
Trades	Mechanic Motor Vehicle, Diesel, Light & Heavy Vehicle, Agricultural Machinery, Two Wheeler (ramp)
PPE	Safety glasses, safety boots, gloves

A-002 Vehicle hoist and service pit

Also called: Two-post and four-post hoist; scissor lift; inspection pit

Identify	A hydraulic lift with arms or platforms, a control panel, safety locks and, on two-post types, adjustable arm pads; a pit is a below-floor trench with steps and lighting.
Operate	Position the vehicle so that the centre of gravity is between the posts, set the arms on the specified lifting points, raise to the working height, and engage the mechanical locks before going underneath.
Safety / precautions	Only trained and authorised staff operate a hoist. Confirm the locks are engaged, check the load is balanced, keep feet clear of the arms, and never stand under a raised vehicle that is not locked. Pits need guarding, drainage and ventilation for heavier-than-air fumes.
Care / calibration	Inspect the cables, chains, hoses and locks per the statutory schedule, keep the pads clean, and record the periodic inspection.
Trades	Mechanic Motor Vehicle, Diesel, Heavy Vehicle, Auto Electrical
PPE	Safety glasses, safety boots, helmet where specified

A-003 Torque wrench and angle gauge

Also called: Click and beam torque wrench 5–300 Nm; torque angle gauge; torque multiplier

Identify	A wrench with a calibrated scale or a preset click mechanism, a square drive, and a ratchet head; the angle gauge measures the additional rotation for torque-to-yield fasteners.
Operate	Look up the specified torque and tightening sequence, set the wrench, apply a steady pull to the click, and then the specified angle where required. Tighten in stages and in the specified pattern, for example spiral from the centre for a cylinder head.

A-003 Torque wrench and angle gauge

Safety / precautions	Reaction torque from a large fastener can injure the wrist and throw the operator off balance — brace yourself and use the correct drive size. Never use a torque wrench to undo a fastener or as a breaker bar.
Care / calibration	Return the wrench to its lowest setting after use, do not drop it, and have it calibrated annually. Store in the case.
Trades	Mechanic Motor Vehicle, Diesel, Two Wheeler, Agricultural Machinery, Refrigeration & AC (compressor assembly)
PPE	Gloves, safety glasses

A-004 Engine repair hand tools — ring spanner, socket and puller sets

Also called: Combination spanner set; deep socket set; bearing puller; gear puller; valve spring compressor; piston ring compressor; ridge reamer

Identify	A collection of forged and hardened tools shaped for engine work: flare-nut spanners for fuel and brake lines, split-ring and two- and three-leg pullers for bearings and gears, a valve spring compressor for removing valves, a ring compressor for fitting pistons, and a ridge reamer for worn bores.
Operate	Use the puller with the legs fully engaged and the centre screw lubricated; compress the valve spring squarely; compress the rings with the correct taper. Work to the workshop manual's sequence and torque specifications.
Safety / precautions	Valve springs store considerable energy — keep fingers clear and use the correct compressor. Pullers can slip off a bearing and the part can fly. Support the block securely on a stand.
Care / calibration	Oil the screw threads, keep the puller legs and jaws in good condition, and store the spring compressor in its case.
Trades	Mechanic Motor Vehicle, Diesel, Two Wheeler, Agricultural Machinery
PPE	Safety glasses or face shield, gloves, safety boots

A-005 Micrometer, bore gauge and Plastigauge for engine measurement

Also called: 0–100 mm outside micrometer; dial bore gauge; cylinder gauge; Plastigauge; vernier caliper

Identify	As described in Group 2, used for measuring piston diameter, bore wear, journal diameter, bearing clearance, valve stem and guide wear, and crankshaft end float.
Operate	Measure at the specified points — for a cylinder, at three levels and in two directions at 90 degrees. For bearing clearance, use Plastigauge: place a strand across the journal, assemble and torque the cap without rotating, then measure the flattened width against the scale.
Safety / precautions	Keep instruments clean of oil and grit — a false reading here leads to an engine failure. Never measure a hot component.
Care / calibration	Wipe and oil after use, store in the case, and calibrate.
Trades	Mechanic Motor Vehicle, Diesel, Two Wheeler, Agricultural Machinery
PPE	Gloves; no other special PPE

A-006 Compression tester and cylinder leakage tester

Also called: Petrol compression gauge; diesel compression gauge; leak-down tester

Identify	A pressure gauge with a hose and an adapter that screws into the injector or spark plug hole; the leak-down tester supplies regulated air and measures the percentage lost.

A-006 Compression tester and cylinder leakage tester

Operate	Warm the engine, disable the ignition and the fuel supply, remove all plugs or injectors, fit the gauge, crank for a fixed number of revolutions, and record the reading for each cylinder. For leak-down, set the cylinder to top dead centre on compression and apply the specified pressure.
Safety / precautions	Disable the ignition before cranking — a spark near an open plug hole and fuel vapour is a fire. Diesel compression can exceed 30 bar; use a diesel-rated gauge. Keep clear of the fan and belts during cranking.
Care / calibration	Check the hose and O-rings, zero the gauge, and store in the case.
Trades	Mechanic Motor Vehicle, Diesel, Two Wheeler
PPE	Safety glasses, gloves

A-007 Fuel injection and common-rail service tools

Also called: Injector puller; nozzle tester; common-rail pressure gauge; diagnostic scanner; injection timing light

Identify	Special tools for removing and testing injectors, a high-pressure gauge and hose for rail pressure, an OBD scanner with a vehicle interface, and a timing light for ignition timing.
Operate	Depressurise the system per the manual, use the specified puller, test the injector on the nozzle tester, and read live data with the scanner. Never crack a high-pressure union to bleed it.
Safety / precautions	Common-rail pressure reaches 2,000 bar or more — a jet of diesel at that pressure penetrates skin and causes injection injury leading to amputation. Depressurise and wait the specified time before opening any union, never put a hand near a suspected leak, and use cardboard to check for a leak. Diesel is a fire hazard.
Care / calibration	Keep the test oil clean, store the scanner battery charged, and protect the high-pressure gauge from impact.
Trades	Mechanic Motor Vehicle, Diesel, Agricultural Machinery
PPE	Safety glasses or face shield, diesel-resistant gloves, no hand near leaks

A-008 Brake service tools

Also called: Brake bleeding kit; brake spring pliers; disc brake piston retractor; shoe adjuster; flare nut spanner; brake lathe

Identify	A pressure or vacuum bleeder with a reservoir; pliers with shaped jaws for shoe return springs; a piston retractor for caliper pistons; a flaring tool and tube bender for brake lines; a brake lathe for disc or drum machining.
Operate	Bleed in the specified order (usually furthest from the master cylinder first), keep the reservoir topped up, retract the caliper piston squarely, and torque the caliper and wheel fasteners.
Safety / precautions	Brake fluid damages paint and is toxic — wear gloves and goggles and wash off spills. Brake dust from older linings may contain asbestos: never blow it with compressed air; use a wet method or a dust-rated vacuum and a respirator.
Care / calibration	Clean the bleeder and the lathe, store brake fluid sealed, and check the lathe tooling.
Trades	Mechanic Motor Vehicle, Light & Heavy Vehicle, Two Wheeler
PPE	Chemical gloves and goggles, respirator when handling brake dust

A-009 Suspension and steering tools

Also called: Coil spring compressor; ball joint separator; tie rod puller; wheel alignment gauge; camber and caster gauge; steering wheel puller

A-009 Suspension and steering tools

Identify	Heavy screwed compressors that grip the coils, fork or scissor separators for tapered joints, and alignment gauges or laser/computer alignment equipment.
Operate	Compress the spring evenly on both sides, keeping the compressor square; separate the joint with the correct tool rather than a hammer; set the alignment to the manufacturer's specification with the vehicle at kerb weight.
Safety / precautions	A compressed coil spring stores enormous energy and can kill if the compressor slips — use the correct spring compressor with the safety pins fitted, stand to the side, and never use a makeshift compressor. Never hammer a ball joint taper — it deforms the housing.
Care / calibration	Check the compressor threads for wear, oil them, and inspect the alignment equipment calibration.
Trades	Mechanic Motor Vehicle, Light & Heavy Vehicle, Two Wheeler
PPE	Face shield, gloves, safety boots

A-010 Tyre service equipment

Also called: Tyre changer; wheel balancer; tyre pressure gauge; tread depth gauge; bead breaker; wheel brace; impact wrench

Identify	A tyre changer with a turntable and a mount/demount head; a balancer with a spindle and a weight station; a pressure gauge with a chuck; a tread depth gauge with a plunger.
Operate	Deflate fully before demounting, use the machine's bead breaker, lubricate the bead, mount with the correct head position, inflate with a clip-on chuck and a remote valve, and balance with the specified weights. Torque the wheel nuts in a star pattern.
Safety / precautions	An over-inflated or badly seated tyre can burst with lethal force — inflate in a safety cage or with the tyre on the machine and never stand over it or hold it by hand. Never exceed the marked pressure. Impact wrenches cause hearing damage and can shear studs.
Care / calibration	Keep the changer jaws and head clean, calibrate the balancer, and check the gauge against a master.
Trades	Mechanic Motor Vehicle, Light & Heavy Vehicle, Two Wheeler
PPE	Safety glasses or face shield, hearing protection, safety boots

A-011 Vehicle diagnostic scanner and oscilloscope

Also called: OBD-II scanner; manufacturer diagnostic tool; automotive scope; multimeter; CAN bus tester

Identify	A hand-held or PC-based tool with a vehicle connector (OBD-II or manufacturer specific), a display and a menu of systems; an automotive scope has multiple channels for injector, ignition and sensor signals.
Operate	Connect with the ignition off, switch on, read and record the fault codes and live data, and clear codes only after the repair. Use the scope to check a signal waveform against the expected pattern.
Safety / precautions	Never disconnect a battery or a module while the ignition is on, and never probe airbag circuits with a multimeter — the squib can fire. Keep the scanner cable clear of fans and belts.
Care / calibration	Update the software, charge the battery, and store the cable coiled.
Trades	Mechanic Motor Vehicle, Auto Electrical & Electronics, Diesel, Two Wheeler
PPE	Safety glasses

A-012 Auto electrical tools — alternator, starter and charging system

Also called: Battery load tester; charging system tester; growler; armature tester; soldering iron; heat gun

A-012 Auto electrical tools — alternator, starter and charging system

Identify	A load tester with a carbon pile or electronic load and a voltmeter; a growler for testing armatures; a charging system tester that reads voltage and current; soldering and heat-shrink tools for terminals.
Operate	Test the battery state of charge and then the load capacity, test the alternator output at the specified engine speed, and check the voltage drop across cables. Recondition or replace the starter and test the no-load current.
Safety / precautions	A lead-acid battery releases hydrogen — no sparks, and connect the negative last and disconnect it first. The load tester produces heat; keep it ventilated and never touch the carbon pile. Remove rings and watches.
Care / calibration	Keep the tester clean and calibrated, store the soldering iron in its stand, and keep the battery terminals clean.
Trades	Mechanic Auto Electrical & Electronics, Mechanic Motor Vehicle, Two Wheeler
PPE	Goggles, acid-resistant gloves, face shield near batteries

A-013 Lubrication and fluid service equipment

Also called: Oil drain pan; oil filter wrench; grease gun; oil pump; coolant tester; air line and blow gun

Identify	A drain pan with a spout, a strap or cup filter wrench, a lever or pneumatic grease gun, a hand or air-operated oil pump, a hydrometer or refractometer for coolant, and a blow gun with a safety nozzle.
Operate	Drain the oil warm, replace the filter with a lightly oiled gasket, fill to the specified level and check with the dipstick, grease every nipple to the specified schedule, and test the coolant concentration.
Safety / precautions	Waste oil is a skin sensitiser and a serious environmental hazard — collect it in a closed container for authorised recycling and never pour it down a drain. Hot oil causes burns. Compressed air must never be used on skin or clothing; use a safety nozzle limited to about 2 bar.
Care / calibration	Clean the grease gun, keep the oil containers labelled, and maintain the air line filter and lubricator.
Trades	Mechanic Motor Vehicle, Diesel, Agricultural Machinery, Refrigeration & AC (compressors)
PPE	Nitrile gloves, safety glasses, apron

A-014 Vehicle body repair tools

Also called: Panel beating hammers and dollies; slide hammer; stud welder; spray gun; masking materials; MIG welder for body panels

Identify	Shrinking and planishing hammers with matching dollies, a slide hammer with a claw or a hook, a stud welder that spot-welds pins for pulling, and a gravity or HVLP spray gun.
Operate	Assess the damage, pull the panel with the slide hammer or stud welder, planish with the dolly, fill and shape, then prime, sand and paint in a booth.
Safety / precautions	Body repair involves solvents, isocyanates in two-pack paint, dust and welding fume — use a supplied-air or appropriate cartridge respirator in the booth, eye protection and gloves. Do not weld a galvanised or coated panel without extraction. The slide hammer can slip and strike the operator.
Care / calibration	Clean the spray gun after every use, store solvents closed, and maintain the booth filters.
Trades	Mechanic Motor Vehicle (body repair), Painter (Industrial/Automotive)
PPE	Respirator, goggles, gloves, coverall

Group 9 — Refrigeration, Air-Conditioning, Plumbing and Fluid Services

This group works with pressure, vacuum and refrigerant. The hazards are largely invisible: a refrigerant leak that displaces oxygen in a basement, a refrigerant jet at minus 40 degrees C, a brazing flame next to an oil charge, a water hammer or a pipe under pressure. Plumbing adds hot water and confined-space work; chemical plant trades add corrosive and toxic fluids and permit-to-work systems.

14 entries in this group.

R-001 Manifold gauge set and hoses

Also called: Two-valve and four-valve refrigerant manifold; charging hoses; service couplers

Identify	Two or four pressure gauges (low side blue, high side red) on a body with hand valves and colour-coded hoses with shut-off or depressor fittings; the compound gauge reads both pressure and the corresponding saturation temperature.
Operate	Connect the low-side hose to the suction service port and the high side to the discharge port with the valves closed, purge the hoses, and read the operating pressures against the expected saturation temperature for the refrigerant.
Safety / precautions	Never open the high-side valve into a system that is not running, and never pressurise a low-side hose above its rating. Use only hoses rated for the refrigerant and never for a different refrigerant without flushing. Frostbite from a liquid refrigerant jet is instantaneous — wear gloves and goggles.
Care / calibration	Replace hoses that are cracked or have damaged depressor pins, keep the gauge faces clean, and calibrate against a known pressure.
Trades	Mechanic (Refrigeration & Air-Conditioner), HVAC service
PPE	Safety goggles or face shield, insulated and refrigerant-resistant gloves

R-002 Vacuum pump and micron gauge

Also called: Two-stage vacuum pump 3–6 CFM; vacuum gauge; evacuation manifold

Identify	A rotary-vane pump with an oil sight glass and an inlet port, and a micron or electronic vacuum gauge that reads absolute pressure in microns.
Operate	Connect through the manifold, open all valves including the Schrader cores, run the pump until the micron level falls below the specified figure (commonly 500 microns) and holds with the pump isolated, then charge.
Safety / precautions	The pump oil becomes contaminated with refrigerant and moisture — change it and dispose of it properly. Do not run the pump without oil, and do not use the same pump and hoses across incompatible refrigerants.
Care / calibration	Change the oil at each evacuation of a contaminated system, cap the ports, and check the gauge calibration.
Trades	Mechanic (Refrigeration & Air-Conditioner)
PPE	Safety glasses, gloves

R-003 Refrigerant recovery and charging machine

Also called: Recovery unit; recovery cylinder; electronic scale; charging cylinder

Identify	A compressor-based recovery unit with inlet and outlet ports and an oil separator; a recovery cylinder that is clearly marked and never filled above 80 per cent; an electronic scale for weighing the charge.

R-003 Refrigerant recovery and charging machine

Operate	Recover the charge into a dedicated cylinder, weigh the recovered amount, evacuate, then charge by weight from the cylinder on the scale. Record the recovered and charged quantity.
Safety / precautions	Refrigerant must never be vented to the atmosphere — it is a controlled substance and, in a confined space, an asphyxiation hazard. Never fill a cylinder above 80 per cent liquid, never heat a cylinder with a flame, and never mix refrigerants in one cylinder. Use a recovery cylinder that has been pressure-tested and is in date.
Care / calibration	Change the filter-drier on the recovery unit, check the hoses and seals, and store the cylinders upright and secured.
Trades	Mechanic (Refrigeration & Air-Conditioner), Mechanic Motor Vehicle (car AC)
PPE	Safety goggles, gloves, ventilation, refrigerant monitor in plant rooms

R-004 Tube working tools — bender, flare, swage and cutter

Also called: Tube bender; flaring block and yoke; swaging punch set; tube cutter; deburring tool

Identify	A lever bender with a forming shoe, a flaring block with a yoke and cone, a swage set that expands the tube end, a tube cutter with a wheel and rollers, and a deburring pen.
Operate	Cut square and deburr, then bend to the required radius without flattening the tube; make a flare with the correct depth of tube above the block, or swage for a solder or braze joint. Always purge with nitrogen when brazing.
Safety / precautions	A refrigerant line must never be brazed while it contains refrigerant or oil without purging — refrigerant decomposes into phosgene-like compounds and the oil can ignite. Keep the tube ends capped until assembly to prevent moisture ingress.
Care / calibration	Oil the bender and the flaring cone, keep the cutter wheel sharp, and store the tools dry.
Trades	Mechanic (Refrigeration & Air-Conditioner), Plumber, Instrument Mechanic
PPE	Safety glasses, gloves

R-005 Brazing and soldering set for refrigeration

Also called: Oxy-acetylene or air-acetylene brazing torch; nitrogen purging kit; heat shield

Identify	A small torch with a brazing tip, a nitrogen regulator and purging hose, and a heat shield blanket to protect adjacent surfaces.
Operate	Purge the system with a slow nitrogen flow during brazing to prevent oxidation inside the pipe, heat the joint evenly, and feed the filler as the joint reaches temperature. Cool and pressure-test afterwards.
Safety / precautions	Nitrogen must be used with a regulator and a relief path — never cap a pipe that is being purged, or the pressure will build and burst it. The torch is an open flame near oil and refrigerant. Wear goggles, gloves and a heat-resistant apron.
Care / calibration	Leak-test the torch and hoses, keep the nitrogen regulator in good order, and store cylinders upright and chained.
Trades	Mechanic (Refrigeration & Air-Conditioner), Plumber, Sheet Metal
PPE	Goggles shade 4–5, gauntlets, apron

R-006 Leak detection equipment

Also called: Electronic refrigerant leak detector; UV dye and lamp; soap solution; nitrogen pressure test kit

Identify	A hand-held electronic detector with a heated sensor and a flexible probe; UV dye with a UV lamp; a soap or bubble solution; a nitrogen regulator and gauge set for pressure testing.

R-006 Leak detection equipment

Operate	Move the probe slowly below and around each joint (some refrigerants are heavier than air), or pressurise the system with dry nitrogen and a trace of refrigerant and test with soap solution, or use dye and the UV lamp.
Safety / precautions	Never use oxygen or compressed air to pressure-test a refrigeration system — oxygen plus oil can explode. Use dry nitrogen only, with a regulator and a relief valve. Do not use a flame as a leak detector.
Care / calibration	Replace the sensor in the detector per the manufacturer, store the dye sealed, and keep the test kit complete.
Trades	Mechanic (Refrigeration & Air-Conditioner), Plumber, Chemical plant
PPE	Safety glasses, gloves

R-007 Compressor service tools

Also called: Valve spring compressor; bearing puller; crankshaft gauge; oil pump; compressor lifting sling

Identify	Tools for opening reciprocating, scroll and rotary compressors: valve plate pullers, bearing pullers, feeler and micrometer sets, and a sling or lifting eye for the compressor body.
Operate	Recover the charge, isolate and remove the compressor, drain the oil, measure the wear, replace the valves and bearings and reassemble with the specified oil and torque.
Safety / precautions	Compressors are heavy and awkward — use a sling and two people. Residual refrigerant and oil can spray when a bolt is released. Do not mix POE oil with mineral oil, and never leave a system open to atmosphere.
Care / calibration	Cap all openings, keep the oil sealed, and clean the work area before assembly.
Trades	Mechanic (Refrigeration & Air-Conditioner)
PPE	Safety glasses, gloves, safety boots

R-008 Air handling, duct and airflow instruments

Also called: Anemometer; manometer; differential pressure gauge; psychrometer; duct leakage tester

Identify	A vane or hot-wire anemometer for air velocity; a water or digital manometer for static pressure; a sling or digital psychrometer for dry and wet bulb temperature and relative humidity.
Operate	Drill test ports, insert the probe to the recommended depth, and take readings across the section. Use the psychrometer readings to find the relative humidity and dew point from a chart.
Safety / precautions	Do not insert a probe into a moving fan or into a duct under high pressure without the correct gland; keep hands clear of belts and pulleys.
Care / calibration	Zero the manometer, protect the sensor, and check the calibration.
Trades	Mechanic (Refrigeration & Air-Conditioner), HVAC, Health Safety & Environment
PPE	Safety glasses, dust mask in ducts

R-009 Pipe and fitting tools — pipe vice, wrench, cutter and threading

Also called: Pipe vice 100 mm; pipe wrench 300/400 mm; wheel cutter; BSP die set; reamer

Identify	As described in Group 1 (H-031, H-032), used for GI and MS pipe work in plumbing and for service lines in refrigeration.

R-009 Pipe and fitting tools — pipe vice, wrench, cutter and threading

Operate	Cut square, ream the burr, thread with cutting oil, and assemble with the specified sealant or tape applied to the male thread only.
Safety / precautions	Support long pipe runs; a pipe that rolls in a vice can fall on a foot. Thread cutting oil irritates skin. Never use a pipe wrench on a chrome or plastic fitting.
Care / calibration	Oil the vice screw and the die head, clean the cutter wheel, and store the dies oiled.
Trades	Plumber, Mechanic (Refrigeration & Air-Conditioner), Fitter
PPE	Gloves, safety glasses, safety boots

R-010 Plumbing hand tools — adjustable wrench, basin wrench, plunger, auger and drain snake

Also called: Adjustable wrench; basin wrench; closet auger; drain snake; plunger; Teflon tape

Identify	A wrench with a swinging jaw for access behind a basin; a coiled spring auger with a crank for clearing a trap; a plunger with a rubber cup; and PTFE tape or thread sealant.
Operate	Clear a blockage with the plunger first, then the auger; remove and refit the trap with a new washer; apply tape clockwise to the thread. Tighten unions firmly but not to the point of cracking the fitting.
Safety / precautions	Waste water carries pathogens — wear gloves and eye protection, and disinfect the tools afterwards. A corroded fitting can shear and release hot water. Never use a drain chemical and then a plunger in the same drain without flushing.
Care / calibration	Clean and oil the auger cable, dry the tools after use, and store the snake coiled.
Trades	Plumber, Sanitary Hardware Fitter, Building Maintenance
PPE	Gloves, goggles, apron, mask where aerosols are possible

R-011 Water pressure test pump and leak test equipment

Also called: Hand test pump; hydrostatic test pump; pressure gauge; drain test plug

Identify	A hand-operated piston pump with a reservoir, a non-return valve and a gauge, connected to the pipework with a test plug.
Operate	Fill the system, bleed the air, pressurise to the specified test pressure (commonly 1.5 times the working pressure for pipework, or as the standard requires), and hold it for the specified period while watching the gauge.
Safety / precautions	Never exceed the specified test pressure and never test with a gas where a hydrostatic test is required. Air-pressurised plastic pipe stores energy and can burst violently. Stand clear of joints and caps during the test.
Care / calibration	Oil the pump piston, check the gauge, and drain and dry the pump after use.
Trades	Plumber, Fitter, Chemical plant, Fire protection
PPE	Safety glasses or face shield, gloves, safety boots

R-012 Pumps, valves and fluid machinery (demonstration)

Also called: Centrifugal pump; reciprocating pump; gate, globe, ball and butterfly valves; pressure relief valve; flow meter

R-012 Pumps, valves and fluid machinery (demonstration)

Identify	Pumps are identified by the casing, impeller and suction and discharge ports; valves by the body, the stem and the disc or ball; the relief valve is set to a specified pressure; the flow meter shows rate on a dial or a digital display.
Operate	Read the nameplate and the flow direction arrow, prime the pump, open the suction valve fully and the discharge valve partially, start and set the flow. Identify valve types by their construction and their function in a line.
Safety / precautions	Never run a centrifugal pump dry — it destroys the seal and can seize. Isolate and depressurise before opening a line, and never loosen a flange under pressure. Relief valves must never be plugged or blinded. Rotating couplings need guards.
Care / calibration	Check the gland or mechanical seal, lubricate the bearings, keep the strainer clean, and operate every valve to prevent seizing.
Trades	Pump Operator-cum-Mechanic, Attendant Operator (Chemical Plant), Instrument Mechanic, Plumber, Refrigeration & AC
PPE	Safety glasses, gloves, hearing protection, chemical PPE in plant

R-013 Instrumentation for fluid measurement

Also called: Pressure gauge; manometer; rotameter; orifice plate; flow transmitter; level gauge; thermowell

Identify	A pressure gauge with a Bourdon tube; a U-tube or digital manometer; a rotameter with a float in a tapered tube; an orifice plate with pressure tappings; a level gauge with a sight glass or a float.
Operate	Open the isolating valve slowly with the vent closed, read the gauge at eye level, and record. Use the correct range and the correct fluid-specific calibration.
Safety / precautions	A gauge can burst if over-ranged or if a diaphragm fails on a corrosive line — use the correct wetted material and a gauge isolator. Never look at a gauge face while opening a valve; stand to the side.
Care / calibration	Calibrate against a deadweight tester, keep the impulse lines clean, and replace cracked glass.
Trades	Instrument Mechanic, Instrument Mechanic (Chemical Plant), Maintenance Mechanic (Chemical Plant), Attendant Operator
PPE	Safety glasses or face shield, gloves, chemical PPE

R-014 Welding and fabrication for pipework

Also called: Pipe welding positioner; alignment clamps; purge equipment; beveling machine

Identify	A positioner that rotates the pipe for downhand welding, internal and external alignment clamps, purge dams, and a beveling machine or grinder for the weld preparation.
Operate	Bevel the pipe ends to the specified angle and land, align with the clamps, tack weld, and complete the root, fill and cap passes while purging the root with argon on stainless and alloy pipe.
Safety / precautions	Pipe welding is often done in a pit, a trench or a confined space — test the atmosphere first, use forced ventilation, and have a standby person outside. Grinding in a pipe is a confined-space fire hazard.
Care / calibration	Inspect the positioner and clamps, keep the purge equipment sealed, and maintain the beveling tools.
Trades	Welder (Pipe), Fabricator, Plumber (industrial), Chemical plant maintenance
PPE	Helmet, gauntlets, apron, respirator; confined-space harness and gas detector

Group 10 — Building, Construction, Woodworking and Painting

Construction trades work with heavy materials, dust, silica and heights, and their tools are deceptively simple. A trowel, a chisel or a masonry drill each carry a specific hazard that is easy to overlook in a syllabus that covers plastering or bricklaying. Carpentry adds sharp edges and dust; painting adds solvents, isocyanates and work at height. The Silica hazard from concrete cutting and chasing deserves particular attention in an Indian ITI setting.

16 entries in this group.

B-001 Trowel, float and hawk

Also called: Steel trowel; wooden float; sponge float; hawk board; notched trowel

Identify	A flat steel or wooden plate with a handle: the steel trowel finishes, the wooden float is for rendering, the sponge float textures, and the notched trowel spreads tile adhesive in regular ribs.
Operate	Load mortar onto the hawk, pick up a workable amount on the trowel, spread and level, then finish with the appropriate float. Keep the blade at a shallow angle and use the whole edge.
Safety / precautions	Wet mortar is alkaline and dries the skin; wear gloves and wash hands before eating. A dropped trowel from height is lethal — tether tools when working above ground level.
Care / calibration	Clean the blade immediately after use, oil the metal edge lightly, and do not let mortar set on the trowel.
Trades	Mason (Building Constructor), Painter, Building Maintenance, Interior Design
PPE	Gloves, safety boots, goggles when mixing

B-002 Masonry drill, hammer and chisel

Also called: Masonry drill bit; SDS hammer drill; brick chisel; mortar chisel; club hammer

Identify	A drill bit with a carbide insert and a flute for masonry; a hammer drill with a percussion action; chisels with a wide or pointed forged edge; a club or lump hammer, about 1–2 kg.
Operate	Mark the position, use a short pilot, apply light pressure with the hammer action on masonry, and clear the dust. Chisel with short blows and a guard, working towards a free edge.
Safety / precautions	Cutting, chasing and drilling concrete or brick generates respirable crystalline silica, a serious lung hazard — use water suppression or dust extraction and an appropriate respirator. Goggles and a face shield are essential, and hearing protection for drilling. Never chisel with the tool pointing at anyone.
Care / calibration	Replace a blunt masonry bit, dress a mushroomed chisel head, and clean and oil the hammer handle.
Trades	Mason, Plumber, Electrician (wall chasing), Building Maintenance, Painter
PPE	Respirator (silica), goggles, face shield, hearing protection, gloves

B-003 Mixing tools — trough, mixer and vibrator

Also called: Mortar trough; concrete mixer; hand paddle; immersion vibrator; wheelbarrow; gauge boxes and screed

Identify	A trough for hand mixing, a drum mixer with a tilting wheel, a gauge box for measuring aggregate, and a flexible-shaft vibrator with a needle for compacting concrete.
Operate	Measure by gauge box or weight, dry-mix in the ratio specified, add water gradually to the correct workability, then place in layers and compact.

B-003 Mixing tools — trough, mixer and vibrator

Safety / precautions	Cement is a strong alkali and can cause chemical burns, particularly with prolonged contact — gloves, boots and eye protection are essential. Never put a hand or a shovel into a running mixer. A vibrator can electrocute in wet conditions, so it must be RCD-protected.
Care / calibration	Wash the mixer out after use, oil the drum bearings, and store the vibrator needle clean.
Trades	Mason, Building Constructor, Plumber, Civil works
PPE	Gloves, face shield, boots, respiratory protection for cement dust

B-004 Measuring, levelling and setting-out tools for building

Also called: Water level; spirit level 600–1800 mm; plumb bob; square; line and pins; folding rule; marking gauge

Identify	A transparent water-filled hose shows equal height at both ends; a laser or spirit level shows level and plumb; a line and pins set a straight axis; a folding rule or tape measures.
Operate	Establish a datum line, transfer levels with the water level or the laser, set out right angles with the 3-4-5 method or a square, and string a line between pins for a wall.
Safety / precautions	Do not use a laser level while looking into the beam; keep the work area clear of materials that could trip; props and formwork must not be left unbraced.
Care / calibration	Calibrate the level, keep the water level free of airlocks, store the folding rule dry, and protect the laser from impact.
Trades	Mason, Carpenter, Draughtsman (Civil), Surveyor, Interior Design
PPE	Safety shoes, helmet on site, laser safety glasses where specified

B-005 Carpentry hand tools — saw, plane, chisel, mallet, square

Also called: Handsaw; tenon saw; coping saw; jack plane; wood chisel set; wooden mallet; try square

Identify	A saw with a hardened, set-tooth blade; a plane with a body, an iron, a cap iron and an adjustable mouth; a bevel-edged chisel with a socket or tang handle; a mallet with a beech or rubber head.
Operate	Mark and knife the line, saw on the waste side using long strokes at about 45 degrees, pare with the chisel and strike with the mallet, and check with the square. Sharpen the plane iron and the chisel bevel on the oilstone.
Safety / precautions	Wood chisels are the most common cause of serious hand cuts in a workshop. Always chisel with both hands behind the edge, clamp the work, and never hold a piece in the hand. Saws cut on the forward stroke and can jump; keep the other hand clear.
Care / calibration	Hone the edges, oil the plane screws, keep the saw teeth protected in a rack, and store the tools dry and off the floor.
Trades	Carpenter, Interior Design & Decoration, Building Maintenance, Furniture
PPE	Safety glasses, dust mask when sanding

B-006 Portable power woodworking tools — circular saw, jigsaw, router, planer

Also called: Circular saw 7 inch; jigsaw; plunge router; power planer; reciprocating saw

Identify	Hand-held power tools with a guard system and a base plate; the circular saw has a spring-loaded lower guard, the jigsaw has an orbital action switch, and the router has a plunge base and a depth stop.
Operate	Clamp the work, set the depth so the blade projects a few millimetres below the stock, connect to an extractor, start the tool before contact, and feed steadily with both hands.

B-006 Portable power woodworking tools — circular saw, jigsaw, router, planer

Safety / precautions	Never remove or tie back the lower blade guard. Keep the cable behind the tool, never in the cut line. Kickback when ripping is violent — use a rip guide and stand to one side. Eye and hearing protection are essential, and a dust mask for MDF and hardwood dust.
Care / calibration	Clean the guards and vents, brush out the motor, check the brushes and the cable, and keep the blades sharp.
Trades	Carpenter, Interior Design, Building Maintenance, Furniture, Painter (boarding)
PPE	Goggles, hearing protection, dust mask, no gloves

B-007 Woodworking machine tools — circular saw bench, planer and spindle moulder

Also called: Table saw; surface planer; thicknesser; spindle moulder; mortiser

Identify	Stationary machines with a cast-iron or steel table, a fence, guards and a dust extraction hood: the table saw has a rising blade with a riving knife, the planer has a cutter block, the moulder has a spindle with cutters, and the mortiser has a hollow chisel.
Operate	Set the fence and the depth, fit and adjust the guards, feed with push sticks and feather boards against the rotation of the cutter, and never reach over the cutter head.
Safety / precautions	These are the highest-risk machines in a carpentry shop: the cutter block is unguarded for the width of cut and a kickback can break a wrist. Guards, push sticks and a riving knife are mandatory, short stock must not be machined, and hearing protection and extraction are essential.
Care / calibration	Check the knife or cutter tightness, wax the tables, check the belts and bearings, and empty the extraction.
Trades	Carpenter, Furniture, Interior Design, Building Maintenance
PPE	Goggles or face shield, hearing protection, dust mask, no gloves

B-008 Painting and decorating tools

Also called: Brush set; roller and tray; spray gun; putty knife; sanding block; paint kettle and strainer; ladder and platform

Identify	Brushes are natural bristle for solvent paint and synthetic for water-based; rollers have a sleeve and a frame; the spray gun has a nozzle, a needle and an air cap; the putty knife is a flexible blade for filling.
Operate	Prepare and sand the surface, prime, fill and sand again, then apply the finish in thin coats. For spraying, set the fluid and air, keep the gun square and at a constant distance, and use the correct thinner.
Safety / precautions	Paints, thinners, two-pack hardeners and adhesives give off flammable and toxic vapour. Use a cartridge or supplied-air respirator, ventilate, and eliminate ignition sources. Isocyanate hardeners sensitise the lungs and can cause asthma. Beware of lead paint in old buildings, and of working on a ladder with wet paint.
Care / calibration	Clean brushes and rollers after each use, strain and store paint, clean the gun with the correct solvent, and store solvents in a fireproof cabinet.
Trades	Painter General, Domestic Painter, Industrial Painter, Building Maintenance, Interior Design
PPE	Respirator, goggles, gloves, coverall, ventilation

B-009 Scaffolding, ladder and working platform

Also called: Scaffold tower; trestle; mobile scaffold; ladder; safety harness

Identify	A modular tube-and-fitting or system scaffold with standards, ledgers, transoms, boards and guard rails; a ladder with a rated duty; a full-body harness with a lanyard and an anchor point.

B-009 Scaffolding, ladder and working platform

Operate	Erect to the manufacturer's or the standard's configuration, brace and tie it, provide toe boards and guard rails at all working levels, and access by an internal ladder. Use a harness anchored above the working position.
Safety / precautions	Falls are the leading cause of construction deaths. Never work from a scaffold without guard rails, never climb the outside of a scaffold, and never overload a platform. Inspect the scaffold after alteration or high wind. A ladder must be at the correct angle, secured and extending above the landing.
Care / calibration	Inspect the components for damage and corrosion, store under cover, and tag and quarantine defective items. Record the scaffold inspection.
Trades	Mason, Painter, Carpenter, Electrician, Building Maintenance, Fabricator
PPE	Full-body harness, helmet, safety boots, high-visibility vest

B-010 Concrete cutting and surface preparation tools

Also called: Disc cutter (cut-off saw); core drill; scarifier; floor grinder; needle scaler; sandblasting equipment

Identify	Petrol, electric or hydraulic disc cutters with a water or dust suppression system; a core drill with a diamond core and a water feed; a floor grinder with diamond segments; a sandblasting pot with a nozzle and a supplied-air helmet.
Operate	Mark the line, use water suppression or extraction, start the cut with the blade clear, and cut in stages. For core drilling, anchor the rig, start the water, and let the core bit cut without forcing.
Safety / precautions	Silica dust from concrete cutting is a confirmed carcinogen — water suppression plus a properly fitted respirator is mandatory, and dry cutting indoors should be prohibited. Blade guards must stay on, and the operator needs hearing and eye protection. Sandblasting requires a supplied-air helmet and full protective clothing.
Care / calibration	Check the blade and the water supply, flush the core drill, and maintain the dust extraction filters.
Trades	Mason, Building Maintenance, Stone Processing, Industrial Painter
PPE	Respirator P2/P3, face shield, hearing protection, gloves, waterproofs

B-011 Bar bending, cutting and steel fixing tools

Also called: Bar bender; bar cutting machine; rebar cutter and bender; tying hook; bolt cutter; hacksaw

Identify	A lever or hydraulic bar bender with a former and a roller; a shear-type bar cutter; a hook for tying binding wire; a long-handled bolt cutter.
Operate	Set the former for the required bend radius, place the bar against the stop, and bend in one steady stroke. Cut with the correct machine capacity and tie the cage with the specified wire and pattern.
Safety / precautions	Rebar ends are sharp and a bar under bending stores energy — keep clear of the bar while it bends and never stand in the path of a cut bar. Bars can fall from a stack; wear gloves and boots.
Care / calibration	Oil the machine, check the formers and the shear blades, and store bars on dunnage.
Trades	Mason, Building Constructor, Civil works
PPE	Heavy gloves, safety glasses, safety boots, helmet

B-012 Formwork, shuttering and propping tools

Also called: Plywood shuttering; props and jacks; clamps; release agent; nail gun; crow bar

Identify	Plywood or steel panels with timber or steel bearers, adjustable props with pins, clamps for the joints, and a release agent for the face.

B-012 Formwork, shuttering and propping tools

Operate	Assemble to the drawing, prop and brace, check the level and the line, apply the release agent, and strike only when the concrete has reached the specified strength.
Safety / precautions	Formwork failure during a pour is a collapse hazard — props must be on a sound base and correctly braced, and the load must not exceed the design. Nail guns cause penetrating injuries; never point one at a person, and treat every gun as loaded.
Care / calibration	Clean and oil the panels, store the props on racks, and inspect the props for buckled pins.
Trades	Mason, Building Constructor, Carpenter
PPE	Gloves, helmet, safety boots, safety glasses

B-013 Demolition hand tools

Also called: Crowbar; wrecking bar; sledge hammer; pickaxe; bolt cutter; dust suppression spray

Identify	Heavy forged bars with a claw, chisel or point; a pick with a pointed and a chisel end; a heavy sledge.
Operate	Assess the structure, isolate services, work from the top down and from the farthest point towards the exit, keep a clear escape route, and suppress dust with water.
Safety / precautions	Demolition is a high-hazard activity: uncontrolled collapse, buried services, asbestos and silica. It requires a survey, isolation of electricity, gas and water, and a permit. Never demolish a load-bearing element without an engineer's instruction.
Care / calibration	Inspect the bars and the handles, and remove burrs.
Trades	Building Maintenance, Mason, General construction
PPE	Helmet, respirator, gloves, boots, high-visibility vest

B-014 Tile cutting and fixing tools

Also called: Manual tile cutter; wet tile saw; angle grinder with diamond blade; notched trowel; spacers; tile levelling system

Identify	A scoring wheel and breaker on a rail; a wet saw with a diamond blade and a water tray; a notched trowel with square or round notches sized to the tile; plastic spacers and clips.
Operate	Mark and score, break with the breaker, or cut with the wet saw; apply adhesive with the notched trowel, back-butter large tiles, and set with spacers and a level.
Safety / precautions	Tile edges and cut slivers are razor sharp — gloves and goggles. Wet saws and grinders run in water and dust, so RCD protection and a respirator for silica are needed. Adhesive is alkaline and irritates the skin.
Care / calibration	Clean the wet saw tray, check the blade, and wash the trowel before the adhesive sets.
Trades	Mason, Interior Design, Painter, Building Maintenance
PPE	Goggles, gloves, respirator for silica, hearing protection

B-015 Glass and glazing tools

Also called: Glass cutter; running pliers; glazing shovel; suction lifter; putty knife

Identify	A small hardened wheel in a handle; pliers with a curved jaw for running a score; a suction lifter with a pump and a gauge; a flexible knife for putty.
Operate	Score once, firmly and in one pass, with the cutting fluid applied; run the score with the pliers or over a wire; and carry with suction lifters.

B-015 Glass and glazing tools

Safety / precautions	Glass cuts deeply and breaks without warning. Wear cut-resistant gloves, safety glasses, and covered forearms; never carry a pane against the body; keep a first aid kit with tweezers nearby. Dispose of broken glass in a rigid, labelled container.
Care / calibration	Oil the cutting wheel, store the cutter dry, and clean the suction cups.
Trades	Carpenter, Interior Design, Building Maintenance, Glazier
PPE	Cut-resistant gloves, safety glasses, forearm protection

B-016 Plastering and rendering machine

Also called: Mortar spray machine; plastering trowel machine; hopper and hose

Identify	A hopper-fed pump delivering mortar through a hose to a spray nozzle, or a rotating trowel machine with a float blade set.
Operate	Set the mix and the delivery rate, keep the nozzle square to the wall, and follow with a rule and a float. Clean the hose immediately after use.
Safety / precautions	The pump is under pressure and a blocked hose can burst or inject mortar into the skin — never look into or point a nozzle at a person, and bleed the pressure before opening a line. Cement dust and lime are corrosive.
Care / calibration	Flush the pump and hose with water after use, check the seals, and grease the bearings.
Trades	Mason, Painter, Building Maintenance
PPE	Face shield, gloves, boots, respirator

Group 11 — Textile, Printing, Media, Food, Beauty and Health Service Tools

The non-engineering trades make up more than half of the ITI trade list and use tools that are often excluded from generic workshop safety material: a sewing machine needle, a screen-printing squeegee, an oven, a pair of beauty scissors, an autoclave. The hazards are real — burns, needle punctures, chemical sensitisation, slips on wet floors and, in food and health trades, cross-contamination and infection. Each group below records the identification, operation and specific controls.

19 entries in this group.

S-001 Industrial sewing machine

Also called: Single needle lockstitch; overlock (serger); flatlock; buttonholer; bartack machine

Identify	By the stitch type, the number of needles and threads, the feed mechanism (drop feed, differential feed) and the attachments fitted. The overlock trims and overcasts an edge with three to five threads.
Operate	Thread in the correct sequence, set the stitch length and tension, test on a scrap of the same fabric, and guide the fabric without pulling. Keep fingers well away from the needle and use a finger guard where fitted. Change the needle with the machine off and the presser foot up.
Safety / precautions	The needle can pierce a finger through the nail in an instant, and the overlock knives are exposed. Never sew over pins, never disable the needle guard, and always switch off and unplug before threading, cleaning or changing the needle. A worn or bent needle breaks and can fly.
Care / calibration	Remove the throat plate and clean the lint after each shift, oil per the manual, replace a bent needle, and check the timing periodically.
Trades	Dress Making, Cutting & Sewing, Sewing Technology, Fashion Design & Technology, Footwear, Leather Goods
PPE	Safety glasses, no loose hair or jewellery, closed shoes

S-002 Cutting tools — shears, rotary cutter and straight knife

Also called: Dressmaker's shears 200–300 mm; pinking shears; rotary cutter with mat; straight-knife cutting machine; electric round knife

Identify	Shears have offset handles and one blunt blade for sliding along the table; the rotary cutter is a circular blade on a handle used with a self-healing mat; the straight knife is a vertical reciprocating blade on a base with a handle.
Operate	Lay out and pin or weight the pattern, cut with the blade low and steady, and keep the hand supporting the fabric behind the blade. Use the rotary cutter against a straight edge and always close the guard.
Safety / precautions	Rotary cutters cause severe lacerations because the blade stays sharp — always retract the guard between cuts, cut away from the body, and never cut towards the hand. The straight knife is heavy and the blade is exposed; it must be fitted with a hand guard and a blade guard.
Care / calibration	Sharpen or replace blades, oil the knife mechanism, keep the mat free of grooves in the working area, and store cutters with the guard closed.
Trades	Dress Making, Cutting & Sewing, Fashion Design, Tailoring, Leather Goods
PPE	Cut-resistant glove on the guiding hand, safety glasses

S-003 Measuring, marking and drafting tools for apparel

Also called: Measuring tape 150 cm; L-square; French curve; grading ruler; tailor's chalk; pattern notcher; dressmaker's wheel

S-003 Measuring, marking and drafting tools for apparel

Identify	A flexible fibre-glass tape; rigid acrylic or wooden squares and curves for pattern work; chalk or a tracing wheel for transferring markings.
Operate	Measure with the tape flat and not twisted, draft with the square and curves, and mark on the wrong side with chalk.
Safety / precautions	The tracing wheel has sharp points and can puncture the hand; work on a cutting mat.
Care / calibration	Keep the tape clean and un-stretched, store patterns flat, and sharpen the notcher.
Trades	Dress Making, Fashion Design, Cutting & Sewing, Interior Design
PPE	None beyond normal care

S-004 Iron, steam press and fusing machine

Also called: Steam iron; boiler iron; vacuum pressing table; fusing press

Identify	An iron with a thermostat and a steam function, or a press with a heated upper buck and a vacuum lower buck; a fusing press bonds interlining with heat and pressure.
Operate	Set the temperature for the fibre, test on a scrap, press with the grain, and use a pressing cloth. In fusing, follow the specified time, temperature and pressure.
Safety / precautions	Steam and the sole plate reach over 100 degrees C and cause serious scalding burns. Never fill a hot iron, never leave it face down, and beware of steam escaping when the press is closed. Keep the cord clear of the floor.
Care / calibration	Empty and rinse the water tank, clean the sole plate with the specified cleaner, descale the boiler, and inspect the cord.
Trades	Dress Making, Fashion Design, Tailoring, Laundry, Garment production
PPE	Heat-resistant gloves for the press, closed shoes

S-005 Embroidery and surface ornamentation tools

Also called: Computerised embroidery machine; hand embroidery frame; aari needle; sequin punch; fabric punch; zari and hook tools

Identify	A multi-head embroidery machine with hoops and a thread tension system; hand frames with a fabric tension; an aari hook needle mounted on a wooden handle.
Operate	Hoop the fabric evenly, digitise or select the design, set the tensions, run a test on a scrap, and monitor the first stitches. Work hand embroidery with the frame held firmly.
Safety / precautions	The machine head moves at high speed and the needle can pierce the hand; keep the guard closed and never reach under the head while it runs. The aari needle and the punch are sharp.
Care / calibration	Clean the lint from the bobbin area, oil per the manual, and store needles in a labelled box.
Trades	Surface Ornamentation Techniques, Computer Aided Embroidery & Designing, Fashion Design
PPE	Safety glasses, no loose clothing

S-006 Textile machinery — loom, knitting and spinning

Also called: Power loom; hand loom; circular knitting machine; ring frame; draw frame; carding machine

Identify	By the process: the loom interlaces warp and weft, the knitting machine forms loops, the ring frame drafts and twists yarn, and the carding machine opens and aligns fibre.

S-006 Textile machinery — loom, knitting and spinning

Operate	Learn the threading and the doffing procedure, keep hands clear of the nip points, and stop the machine before clearing a fault.
Safety / precautions	Textile machines have many in-running nip points, high-speed spindles and flying yarn or fibre. Guards must always be in place, hair must be fully covered, and loose sleeves are prohibited. Cotton dust is a respiratory hazard (byssinois) and requires extraction.
Care / calibration	Clean the lint daily, lubricate per the schedule, and check the guards and the emergency stops.
Trades	Weaving Technician, Spinning Technician, Textile Mechatronics, Textile Wet Processing
PPE	Dust mask, ear protection, no loose clothing, hair covering

S-007 Textile wet processing equipment

Also called: Dyeing vat; winch; jigger; padding mangle; stenter; washing and drying range

Identify	Open or closed vessels with heating, a liquor circulation system and a fabric transport; a mangle with rollers for squeezing; a stenter for drying and setting under tension.
Operate	Prepare the liquor to the recipe, control the temperature and the pH, run the fabric, and wash off. Handle dyes and chemicals with the correct measuring equipment.
Safety / precautions	Dye liquors are hot (often above 90 degrees C) and alkaline or acidic; dyes are sensitising and some are hazardous. Use chemical goggles, a face shield, an apron, rubber boots and gloves, and never add water to concentrated acid. The mangle nip is a severe crush hazard — never reach into it while running.
Care / calibration	Clean the vessels, check the temperature controllers and the nip guards, and store chemicals in a bunded area.
Trades	Textile Wet Processing Technician
PPE	Chemical goggles and face shield, apron, rubber boots, gloves, respirator

S-008 Printing press equipment — offset, digital and screen

Also called: Offset litho press; plate maker (CTP); screen printing table and squeegee; digital press; guillotine; creasing and folding machine; perfect binding machine

Identify	The offset press has plate, blanket and impression cylinders, ink and dampening rollers; the screen printing unit has a mesh stretched on a frame and a squeegee; the guillotine is a heavy blade with a clamp and a light curtain.
Operate	Make ready, set the ink and water balance, register the sheet, and check the first pull. On the guillotine, set the back gauge, use the clamp, keep hands behind the guard, and use the light curtain.
Safety / precautions	Printing machines have many in-running rollers and nip points that will draw in a hand or an arm. Guards and interlocks must never be defeated. The guillotine blade amputates; the light curtain and the two-hand control must be functional. Printing inks and cleaning solvents are flammable and dermatitic; the offset dampening solution contains isopropyl alcohol.
Care / calibration	Clean the rollers and blankets with the specified solvent, check the guards and interlocks, and maintain the extraction.
Trades	Litho Offset Machine Minder, Plate Maker-cum-Impositor, Process Cameraman, Desktop Publishing Operator
PPE	Safety glasses, gloves, apron, solvent-resistant gloves, extraction

S-009 Photography, image capture and processing equipment

Also called: Digital SLR or mirrorless camera; studio lights; flash meter; tripod; colour calibrator; large-format printer; film processing tank and enlarger

Identify	Cameras by the lens mount and sensor size; lights by their power and modifier; the colour calibrator with a sensor puck for the monitor and printer profiles; the enlarger with a lamphouse, a negative carrier and a lens.
Operate	Set the exposure using the meter, set the white balance, shoot tethered or to a card, then process and colour-manage the file before printing.
Safety / precautions	Studio flash and modelling lamps are hot and the flash tube can shatter; never touch a hot lamp or fire a flash close to the eyes. Darkroom chemistry is corrosive and sensitising — gloves, goggles and ventilation. Cables across a studio floor are a trip hazard.
Care / calibration	Clean the lens with lens tissue, store cards and batteries safely, and dispose of chemistry properly.
Trades	Photographer, Digital Photographer, Process Cameraman, Multimedia Animation & Special Effects
PPE	Safety glasses, gloves for chemistry, cable management

S-010 Commercial cooking equipment

Also called: Gas range; induction range; oven; combi steamer; deep fryer; griddle; tandoor; pressure cooker

Identify	By the heat source (gas, electric, induction) and the vessel: the deep fryer has a thermostat and a basket, the combi steamer has steam and convection modes, and the pressure cooker has a weighted valve and a safety plug.
Operate	Check the gas connections and the ventilation, preheat, cook to temperature, and use the correct utensils. Never fill a fryer above the marked line, and never immerse a wet basket in hot oil.
Safety / precautions	Hot oil at 180 degrees C causes the most severe burns in a kitchen and a water droplet makes it spatter violently. Gas leaks, carbon monoxide from a faulty flue, and steam scalds are the other major hazards. Never use water on an oil fire — smother it or use a wet chemical extinguisher. A pressure cooker must be cooled and depressurised before opening.
Care / calibration	Filter the fryer oil, clean the burners and flues, delime the steamer, and check the gas hoses and regulators.
Trades	Craftsman Food Production (General/Vegetarian), Baker & Confectioner, Catering & Hospitality Assistant
PPE	Heat-resistant gloves and apron, closed non-slip shoes, hair covering

S-011 Bakery equipment

Also called: Planetary mixer; dough sheeter; spiral mixer; deck oven; prover; pastry cutter; piping bag and nozzles

Identify	The planetary mixer has a bowl, a beater, a whisk and a dough hook; the sheeter has adjustable rollers; the deck oven has a stone sole and steam injection.
Operate	Weigh ingredients accurately, mix to the specified stage (short, developed or over-developed), prove at the controlled temperature and humidity, and bake at the specified temperature and time.
Safety / precautions	The mixer must be stopped before the bowl or the attachment is changed — a rotating dough hook breaks bones. The oven is hot and the steam from injection scalds; the deck is at or above 200 degrees C. Flour dust is explosive in high concentrations.
Care / calibration	Clean and grease the mixer per the manual, check the oven door seals and the thermostat, and keep the prover clean.
Trades	Baker & Confectioner, Craftsman Food Production, Catering & Hospitality
PPE	Heat-resistant gloves, apron, hair covering, closed shoes

S-012 Food preparation hand tools and machines

Also called: Chef's knife; boning knife; cleaver; mandoline; vegetable slicer; meat mincer; slicer; blender; can opener

Identify	Knives by blade shape and use; the mandoline has an adjustable slicing plate and a hand guard; the mincer has a hopper, a worm and a perforated plate; the slicer has a rotating blade and a carriage.
Operate	Keep knives sharp, use a stable cutting board, and cut with the claw grip (knuckles forward, fingertips tucked). Use the mandoline hand guard every time. Feed the mincer with the pusher, never the fingers.
Safety / precautions	A sharp knife is safer than a blunt one because it does not slip. Never put a knife in a sink of water, never catch a falling knife, and never leave a blade on the edge of a table. The slicer blade is exposed when the guard is removed and the mincer worm can amputate a finger. Cross-contamination between raw meat and cooked food causes illness — separate boards and wash hands.
Care / calibration	Hone the knife, wash and sanitise the boards, and dismantle and clean the machines after each use.
Trades	Craftsman Food Production, Catering & Hospitality, Food & Vegetable Processing
PPE	Cut-resistant glove, apron, non-slip shoes, hair covering

S-013 Food preservation and processing equipment

Also called: Dehydrator; canning and bottling unit; pulp extractor; pasteuriser; autoclave/retort; vacuum packer; chiller and freezer

Identify	A drying cabinet with a thermostat and a fan; a pasteuriser with a temperature controller and holding coil; a retort or pressure canner with a gauge and a safety valve; a vacuum packer with a chamber and a sealing bar.
Operate	Prepare and acidify as the recipe requires, fill and seal, process at the specified time and temperature, and cool rapidly. Record the batch with a code date.
Safety / precautions	A pressure canner or retort is a pressure vessel — its gauge and safety valve must work and it must never be opened under pressure. Hot product and steam cause burns. Botulism from inadequately processed low-acid canned food is a lethal risk, so process times are not negotiable.
Care / calibration	Calibrate the gauge and thermometer, test the seals, descale, and keep the records.
Trades	Food & Vegetable Processing, Preservation of Fruits and Vegetables, Dairying, Catering
PPE	Heat-resistant gloves, apron, face shield at the retort

S-014 Beauty and cosmetology tools

Also called: Barber and styling scissors; thinning shears; clipper and trimmer; hair dryer; straightener and curling tongs; make-up brushes; tweezers

Identify	Scissors by blade length and tension screw; the clipper by its cutting blade adjustment and attachments; thermal tools by their thermostat and coating; brushes by fibre and shape.
Operate	Section the hair, hold the scissors with the ring and the little finger, and cut with the tips. Adjust the thermal tool to the specified temperature for the hair type, and keep it moving.
Safety / precautions	Scissors and razors cut; a dropped blade is a penetrating hazard. Straighteners reach 200 degrees C and burn both the client and the operator — check the temperature and never leave one on the bench. Electrical tools near water are a shock hazard, so keep the station dry and use RCD protection.
Care / calibration	Oil the scissors and have them sharpened, clean the clipper blades and oil them, and clean the thermal plates.
Trades	Basic Cosmetology, Barber, Spa Therapy, Fashion Design (styling)
PPE	Gloves for chemical services, apron, closed shoes

S-015 Skin and nail care equipment

Also called: Facial steamer; magnifying lamp; extractor; nail drill; UV/LED lamp; pedicure unit; autoclave or UV steriliser

Identify	The steamer has a water tank and a nozzle; the extractor is a vacuum or a metal loop; the nail drill has a handpiece and interchangeable bits; the steriliser has a pressure or UV chamber.
Operate	Sanitise the hands and the tools, prepare the skin or nail, and use the tool at the correct angle and speed. Sterilise all metal implements between clients.
Safety / precautions	Infections including fungal and blood-borne disease transmit through unsterilised tools and through broken skin. Use single-use files and buffers, sterilise metal tools, and never use a sharp implement on a client's cuticle. UV lamps are an eye hazard. Nail dust is a respiratory irritant.
Care / calibration	Sterilise after every client, clean the steamer, change the drill bits, and keep the station disinfected.
Trades	Basic Cosmetology, Spa Therapy, Beauty & Wellness
PPE	Gloves, mask, eye protection, apron

S-016 Health care and hospital support equipment

Also called: Wheelchair; stretcher; hospital bed; nebuliser; suction machine; oxygen cylinder; BP apparatus; thermometer; autoclave; sharps container

Identify	By function and by the markings: an oxygen cylinder by its white shoulder and label, a sharps container by its yellow colour and biohazard symbol, an autoclave by its pressure gauge and safety valve.
Operate	Check the equipment before use, follow the safe patient-handling technique with two people or a hoist, and record the reading or the procedure.
Safety / precautions	Sharps injuries transmit blood-borne viruses — never recap a needle, never overfill a sharps container, and dispose of it per the protocol. Oxygen supports combustion: no oil, grease or flame near a cylinder. Manual handling of patients causes back injury; use the hoist or ask for help. An autoclave is a pressure vessel.
Care / calibration	Clean and disinfect between patients, calibrate the BP apparatus and the thermometer, and test the autoclave with a biological indicator.
Trades	Health Sanitary Inspector, Hospital House Keeping, Old Age Care Assistant, Physiotherapy Technician, Radiology Technician
PPE	Gloves, mask, apron, eye protection, sharps awareness

S-017 Housekeeping and cleaning equipment

Also called: Scrubber dryer; vacuum cleaner; wet and dry vacuum; mop and bucket; chemical dispenser; window cleaning kit; waste bins

Identify	Machines by their tank arrangement and brush deck; chemicals by their label and safety data sheet; colour-coded cloths and mops by their use area.
Operate	Follow the dilution ratio, use colour-coded cloths (for example red for sanitary areas, blue for general), clean from the cleanest area to the dirtiest, and display a wet floor sign.
Safety / precautions	Never mix cleaning chemicals — bleach with an acid or with ammonia releases chlorine or chloramine gas, which has killed people in bathrooms. Wear gloves and eye protection, ventilate, and keep chemicals in their original labelled containers. Wet floors and trailing cables cause falls and shocks.
Care / calibration	Rinse the tanks and filters, store chemicals closed and away from food, and check the cables and plugs.
Trades	Domestic House Keeping, Institution / Hospital House Keeping, Corporate House Keeping, Cabin or Room Attendant
PPE	Gloves, eye protection, non-slip shoes, apron

S-018 Fire fighting equipment (for all trades)

Also called: CO2 extinguisher; dry powder extinguisher; foam extinguisher; fire blanket; hose reel; sand bucket; smoke detector

Identify	Extinguishers by the colour code and the label — CO2 black, dry powder blue, foam cream, water red; the fire blanket is a folded wool or glass-fibre sheet in a quick-release pouch.
Operate	Raise the alarm first, then use the extinguisher by pulling the pin, aiming at the base of the fire, squeezing the lever and sweeping. Use PASS — Pull, Aim, Squeeze, Sweep. Approach with the wind behind you and always keep an exit behind.
Safety / precautions	Never fight a fire that is larger than a waste-paper basket, that is spreading, or that is between you and your exit. A CO2 horn becomes extremely cold — do not hold the horn. Never use water on an electrical or oil fire. Extinguishers are tested and refilled on schedule.
Care / calibration	Check the pressure gauge monthly, record the service date, and keep the access clear.
Trades	All trades (mandatory CTS safety content), Firemen, Fire Technology, HSE
PPE	The extinguisher protects the escape; do not rely on it as PPE

S-019 First aid kit and emergency equipment

Also called: First aid box; eye wash bottle or station; burn dressing; splint; stretcher; emergency shower; spill kit

Identify	A marked box with sterile dressings, adhesive plaster, bandages, scissors, disposable gloves, eye pads and, in a workshop, burn gel and an eye wash.
Operate	Assess the scene for safety, then treat in the order of catastrophic bleeding, airway, breathing and circulation. Cool burns with running water for at least ten to twenty minutes, irrigate eyes from the inner to the outer corner, and immobilise a suspected fracture before moving the casualty.
Safety / precautions	Do not move a casualty with a suspected spinal injury unless there is immediate danger. Do not apply ointment, ice or cotton wool to a burn. Never give anything by mouth to an unconscious person. Always call for qualified medical help and record the incident.
Care / calibration	Check and restock the kit monthly, replace expired items, and record the checks.
Trades	All trades
PPE	Disposable gloves, CPR barrier where available

PART C

Instructor and Reference Material

The CITS training-aid toolkit, the checklists used to assess tool competence, and the reference apparatus for the whole document.

- 8. The CITS instructor's toolkit (Training Methodology aids)
- 9. Assessing tool competence
- 10. Trade-to-tool group index
- 11. Glossary and bibliography
- 12. Limitations and maintenance of this document

The chapters that follow support the register: the instructor's training-aid toolkit for CITS Training Methodology, the checklists used to assess tool competence, a trade-to-tool index, a glossary and the source bibliography.

8. The CITS Instructor's Toolkit — Training Methodology Aids

Training Methodology is common to every CITS trade and is examined as a separate subject (40 per cent in Principles of Teaching in the NSTI scheme). Its tool list is short and mostly electronic, but it is assessed as rigorously as any trade tool. The DGT 'Core Skills (Training Methodology)' document lists classroom furniture, an LCD/LED projector or interactive flat panel, a multimedia computer, a UPS, a white board, a projector screen, a digital video camera with tripod, an air conditioner for the computer room, wall charts, transparencies and DVDs, a printer with scanner, a steel cupboard and a first aid box. The syllabus additionally requires the use and maintenance of the OHP, the digital camera, the LCD projector and the smart board, and the preparation of charts, transparencies, slides, posters and mock-ups. The items below record what an instructor trainee must be able to identify, operate, care for and use safely.

TM-001 Black board and white board

Also called: Chalk board; dry-wipe white board 6 x 4 ft

Identify	A writing surface with a frame and a tray for chalk or markers and a duster. The white board has a low-glare coated steel or laminate surface.
Operate	Plan the board layout before the session (title, diagram area, key points), write large and legibly, rule guide lines lightly, and clean at the end. Use the board for the sequence of a demonstration, not for long text.
Safety / precautions	Chalk dust is a respiratory irritant and aggravates asthma — use dust-free chalk and a damp cloth. Marker pens must be the correct low-odour dry-wipe type; permanent markers ruin the surface and their solvent vapour is a hazard in a closed room.
Care / calibration	Clean the white board with the specified cleaner only, replace the duster, keep the tray dry, and resurface a scratched board.
Trades	All CITS trades
PPE	None

TM-002 Flannel board, magnetic board and display boards

Also called: Flannel graph; magnetic board; bulletin board; chart stand

Identify	A fabric-covered board to which cut-outs with a rough backing adhere, a steel board that holds magnetic strips and symbols, and a pin board for notices and charts.
Operate	Prepare the cut-outs in advance, place them progressively as the explanation develops, and build the diagram in front of the class so the sequence becomes visible. This is the cheapest way to demonstrate a process.
Safety / precautions	Pins and staples are a puncture hazard — keep a magnet or a pin box and never leave pins on the floor. Do not mount a display board where it can fall on the class.
Care / calibration	Store the cut-outs flat in labelled envelopes, keep the boards dusted, and check the frame fixings.
Trades	All CITS trades
PPE	None

TM-003 Overhead projector (OHP) and transparencies

Also called: OHP with a fresnel lens and a lamp; acetate transparency sheets; marker pens

TM-003 Overhead projector (OHP) and transparencies

Identify	A box with a lamp, a condenser lens, a glass stage and a focusing head with a mirror, projecting a transparency onto a screen. Still listed in the syllabus even though it is largely superseded by digital projection.
Operate	Place the transparency on the stage the right way up, focus the head, adjust the mirror angle, and reveal information progressively by covering part of the sheet with paper. Switch off when not in use.
Safety / precautions	The lamp and the stage become very hot — never touch them, never place papers on the stage, and allow cooling before moving the projector. Route the mains cable where it cannot be tripped over.
Care / calibration	Clean the fresnel lens and the stage glass with a soft cloth, replace the lamp correctly, and store transparencies flat and interleaved.
Trades	All CITS trades
PPE	None

TM-004 LCD / LED projector and screen

Also called: Multimedia projector 3,000–4,000 lumens; tripod or wall screen; HDMI and VGA cabling

Identify	A projector with an input panel (HDMI, VGA, USB), a lens with a zoom and focus ring, a remote control and ventilation grilles; a screen in a matte white material.
Operate	Place the projector square to the screen at the correct throw distance, connect the source, select the input, focus and correct keystone, and then switch off the room lighting to the level that keeps notes legible. Use the blank or freeze function rather than switching off when you want attention.
Safety / precautions	Never look into the lens while it is on, and never allow the projector to run with the lamp cover removed — the lamp is under pressure and its housing is hot. Do not block the vents; a projector that overheats is a fire risk. Suspend the projector from a rated mount and secure the ceiling fixing.
Care / calibration	Clean the filter and the lens, let the fan run through its cool-down cycle before switching off at the mains, replace lamps or modules per the hour counter, and store the remote without batteries if it will not be used.
Trades	All CITS trades
PPE	None

TM-005 Interactive flat panel / smart board

Also called: IFP with touch overlay; interactive whiteboard with projector; visualiser

Identify	A large touch panel with HDMI and USB inputs, an on-board computer or an OPS module, a stylus and a wireless mirroring option; the visualiser is a document camera that displays an object or a page on the screen.
Operate	Calibrate the touch, connect the source, use the annotation tools over a diagram, save the annotated board as a PDF for the class, and mirror a tablet or phone for a live demonstration. Use the visualiser to show a small tool or a measurement to the whole class.
Safety / precautions	The panel is heavy and must be mounted on a rated wall fixing. Do not use a sharp object on the surface — a scratched panel is an expensive loss. The panel runs on mains voltage; do not open the rear cover.
Care / calibration	Clean with a soft dry or slightly damp cloth and the approved cleaner, keep the vents clear, update the firmware, and shut down properly.
Trades	All CITS trades
PPE	None

TM-006 Multimedia computer, UPS and printer-scanner

Also called: Desktop or laptop with projector output; UPS; multifunction printer

Identify	A computer with HDMI/VGA output, a UPS with a battery and a load display, and a printer with a scanner platen and a document feeder. Listed in the CITS tool list as a multimedia computer with accessories, a UPS and a printer with scanner.
Operate	Prepare the presentation and the digital handouts, connect to the projector, and print the job sheets, operation sheets and assessment sheets. Scan trainee work and record it.
Safety / precautions	Keep liquids away from the equipment, do not open the power supply, and do not defeat the UPS alarm. Toner is an irritant — vacuum with a toner-safe cleaner, and never use a domestic vacuum.
Care / calibration	Back up teaching material, update the software, clean the scanner glass, and replace the UPS battery on schedule.
Trades	All CITS trades
PPE	None

TM-007 Digital video camera with tripod

Also called: Camcorder; DSLR or mirrorless with video; smartphone gimbal; tripod

Identify	A camera with a lens, a recording medium, a microphone input and a tripod bush; a tripod with a pan-and-tilt head and a quick-release plate.
Operate	Frame the demonstration to show the hands and the work, set the focus manually for close work, record in good light, and use the footage for micro-teaching review and for trainees who missed a session.
Safety / precautions	Do not film in a way that distracts an operator of a machine; never place a tripod in a walkway; obtain consent before filming trainees; and do not record in areas where personal information is visible.
Care / calibration	Charge and remove the battery, clean the lens, store the card separately, and keep the camera in a dry box.
Trades	All CITS trades
PPE	None

TM-008 Charts, posters, models, mock-ups and specimens

Also called: Wall chart; flip chart; poster; scaled model; cut-away mock-up; sectioned specimen; component board

Identify	Prepared two- or three-dimensional teaching aids: a printed chart, a flip chart on a stand, a working model or a cut-away of a machine element, a mounted board of components with labels.
Operate	Introduce the aid at the point in the lesson where it resolves a difficulty, point to the specific part, pass the specimen round for handling where it is safe to do so, and then remove the aid so that attention returns to the work.
Safety / precautions	A cut-away or sectioned specimen may have sharp edges — deburr and mount it behind a guard. Models with moving parts must not be powered in front of the class without a guard. Display boards must be fixed so they cannot fall.
Care / calibration	Store flat and dust-free, repair promptly, and update when the curriculum changes.
Trades	All CITS trades
PPE	None

TM-009 Written instructional material templates

Also called: Job sheet; operation sheet; information sheet; assignment sheet; experiment sheet; final job check sheet; lesson plan; demonstration plan

Identify	Standard pro-forms: the job sheet states the job, drawing, tools, material, procedure and safety; the operation sheet details a single operation step by step; the information sheet carries the theory; the check sheet the acceptance criteria.
Operate	Prepare the material from the learning outcome, list the tools and the safety precautions, break the procedure into numbered steps, and include a self-check with the tolerances. Use the same structure across the trade so that trainees learn one format.
Safety / precautions	Every job and operation sheet must carry its own safety precautions and the PPE required — this is the administrative control that makes the practical session safe before it begins.
Care / calibration	Version-control the templates, keep a master set, and review them when the curriculum is revised.
Trades	All CITS trades
PPE	None

TM-010 Trainee progress and assessment records

Also called: Daily diary; progress chart; theory and practical record; assessment sheet; competence log

Identify	Registers and charts that record attendance, sessional marks, job completion, assessment results and remediation.
Operate	Record at the time of the activity, not later; chart the progress so that a trainee can see their own trend; and use the record to plan remediation before the All India Trade Test.
Safety / precautions	The record is also the evidence that safety training and PPE issue happened, which matters in any incident investigation. Keep it accurate and current.
Care / calibration	Store in a locked cupboard, back up any digital record, and retain per the NCVT retention rule.
Trades	All CITS trades
PPE	None

9. Assessing Tool Competence — Instructor Checklists

9.1 Why a checklist

Tool competence has three parts and all three must be assessed: identification and selection, operation to specification, and safe working. A trainee who can produce a job but cannot identify a damaged tool, or who works quickly with a missing guard, is not yet competent. The checklists below are generic; a trade-specific version should add the tolerance and the operations from the curriculum.

9.2 Identification and selection

Ask the trainee to name the tool and its parts, state its specification and range, choose the correct tool for a stated job from a set, explain why the others are unsuitable, identify damage or wear, and check that the calibration status is valid. Typical prompt: ‘select the instrument you would use to measure the depth of this blind hole to 0.02 mm, and show me why the vernier caliper is the right choice here.’

9.3 Operation to specification

Observe the set-up, the technique, the parameter selection, the measurement discipline and the finish. The trainee should state the speed, feed or setting before starting and explain why. Insist on measuring the work with the correct instrument and on recording the result against the tolerance.

9.4 Safe working

Observe PPE, guards, workholding, isolation, housekeeping before and after the job, tool condition checks, and the response to a deliberately introduced fault such as a frayed cable or a missing guard. A single unsafe act should stop the assessment and trigger re-demonstration, not a mark deduction.

9.5 Recording and remediation

Record each element as competent or not yet competent with a short note, give the trainee the record, and schedule re-assessment after targeted practice. Keep the record as evidence for the sessional marks and for the institute's training records.

9.6 Generic tool competence checklist

Phase	What the assessor looks for
1. Preparation	Correct PPE worn; work area clear; tool clean and undamaged; calibration valid; job sheet and drawing available; material identified
2. Identification	Names the tool and its parts; states specification, range and least count; explains the principle; selects the correct tool for the job
3. Setting up	Workheld securely in a vice, clamp or fixture; machine guards in place; speed, feed or pressure selected and justified; tool set to the correct position
4. Operation	Correct technique and sequence; hands clear of the cutting line; workpiece measured at the right stages; coolant or lubricant used correctly
5. Safety behaviour	No guard bypass; no gloves near rotating parts; chuck key removed; isolation used where required; chips cleared with a brush; correct lifting technique
6. Inspection	Correct instrument used; reading recorded against the tolerance; visual check for burrs, cracks and finish
7. Close-down	Machine stopped and isolated; tool cleaned, oiled and returned; swarf and waste disposed of correctly; work area left clean; breakage or defect reported
8. Knowledge	Explains the hazard, the control and what to do in an emergency; identifies the safety signs and the location of first aid and fire equipment

Table 9.1 — Instructor checklist for a practical tool assessment.

10. Trade-to-Tool Group Index

Use this index to move from a trade to the groups that hold its tools. Every trade also uses the universal items in Group 1 (hand tools), the relevant instruments from Group 2, and the safety and first aid content of Chapter 3.

Trade family	Tool groups	Notes
Fitter, Turner, Machinist, Tool & Die Maker, Operator Advanced Machine Tools	H, M, C, T, W	Full bench, metrology, machine tool and joining coverage; CMM and slip gauges at advanced level
Welder, Fabricator, Sheet Metal, Forger, Foundryman	H, M, C, W	Welding, cutting, forming and inspection; add T for fabrication machine tools
Electrician, Wireman, Lift & Escalator	H, M, E, B	Electrical hand tools, instruments and access equipment; add C for drilling and chasing
Electronics Mechanic, Computer Hardware, ICT, COPA, DTP, Multimedia	H, M, X, E	Fine hand tools, ESD control, test instruments and ICT cabling
Mechanic Motor Vehicle, Two Wheeler, Diesel, Auto Electrical, Agricultural Machinery	H, M, C, A, E	Lifting, measuring, diagnostic and special service tools
Refrigeration & AC, Plumber, Pump Operator, Chemical Plant trades	H, M, C, W, R, E	Pressure, vacuum, refrigerant, pipework and fluid instrumentation
Mason, Carpenter, Painter, Draughtsman (Civil), Surveyor, Interior Design	H, M, B, C	Setting out, forming, finishing, access equipment and wood/masonry machines
Dress Making, Fashion Design, Surface Ornamentation, Textile trades	S, H, M	Sewing, cutting, pressing, embroidery, loom and wet processing
Printing, Photography, DTP	S, H, M, X	Press, guillotine, camera and image processing; add B for finishing
Food Production, Bakery, Catering, Food Processing, Dairying	S, M, H	Kitchen, bakery and preservation plant; add R for pumps and valves
Cosmetology, Spa, Health Sanitary, Hospital House Keeping, Old Age Care	S, H, M	Beauty, sterilisation, cleaning, patient handling and infection control
Firemen, Fire Technology, Health Safety & Environment	S, M, B	Fire equipment, first aid, PPE, access equipment and monitoring instruments
CITS — all trades	TM, plus the groups of the parent trade	Training aids and assessment tools are added to the trade's own tool set

Table 10.1 — Trade families mapped to tool groups.

11. Glossary and Bibliography

11.1 Glossary of abbreviations and technical terms

Term	Meaning
AITT	All India Trade Test — the NCVT examination at the end of a CTS course
CITS	Crafts Instructor Training Scheme — the one-year instructor training programme at NSTIs
CTS	Craftsmen Training Scheme — the ITI trade training scheme
DGT	Directorate General of Training, Ministry of Skill Development & Entrepreneurship
NCVT	National Council for Vocational Training
NSTI	National Skill Training Institute (formerly Central Training Institute)
NCIC	National Craft Instructor Certificate — awarded on passing CITS
NTC / NAC	National Trade Certificate / National Apprenticeship Certificate
NSQF	National Skills Qualification Framework — levels 1 to 10
NOS	National Occupational Standard — the unit-level competence statement
QP	Qualification Pack — the NSQF document that groups NOS for a job role
TM	Training Methodology — the common CITS subject on teaching and assessment
POT	Principles of Teaching — the theory component of Training Methodology
VCS / WCS	Vocational Calculation & Science / Workshop Calculation & Science
OJT	On-the-Job Training — 150 hours of industry attachment or a group project in CITS
PPE	Personal Protective Equipment
RCD / ELCB / RCCB	Residual current device / earth leakage circuit breaker / residual current circuit breaker — protective devices that trip on earth leakage
MCB / MCCB	Miniature / moulded case circuit breaker — overcurrent protection
LOTO	Lockout and tagout — the isolation procedure before maintenance
5S	Sort, Set in order, Shine, Standardise, Sustain — the housekeeping method in the syllabus
Least count	The smallest value an instrument can read, for example 0.02 mm on a vernier caliper
Go / No-Go	A limit gauge that must pass on one limit and must not pass on the other
Wringing	The molecular adhesion that holds slip gauges together in a stack
Swarf	Chips and turnings produced by machining
Snag / pinch point	A point where a body part can be caught between two parts, one of them moving
Kickback	The violent ejection of work by a rotating blade or cutter
Auto-darkening	A welding filter that darkens automatically when the arc strikes
Flashback arrester	A device that stops a flame travelling back into a gas hose or cylinder
Micron (vacuum)	One thousandth of a millimetre of mercury — used for evacuation levels
ESD	Electrostatic discharge — the static discharge that destroys electronic components
Tolerance	The permitted variation in a dimension, for example ± 0.02 mm
Fit	The relationship between the sizes of a hole and a shaft, from clearance to interference
MSDS / SDS	Material (or Safety) Data Sheet — the hazard information for a chemical
LEV	Local exhaust ventilation — extraction at the source of fume or dust

Table 11.1 — Abbreviations and terms used in this document.

11.2 Sources

The compendium was compiled from the following sources. Curriculum documents were used for trade structure, tool names and specifications; standards and manufacturer guidance were used for operating practice and safety.

Source	What it was used for
DGT, Competency Based Curriculum — Fitter (CTS, NSQF Level 4), Version 3.0, dgt.gov.in	Trade structure, hours, learning outcomes, tolerances, trainee tool kit and Annexure-I tool list
DGT, Competency Based Curriculum — Fitter (CITS, NSQF Level 5 and 6), dgt.gov.in and bharatskills.gov.in	CITS structure, trade technology outcomes, detailed Fitter CITS tool and equipment list (items 1–146 plus machinery)
DGT, Competency Based Curriculum — Electrician (CTS, NSQF Level 5), cstaricalcutta.gov.in and dgt.gov.in	Electrician trainee tool kit, shop tools, test instruments, starters and machines
DGT, Core Skills (Training Methodology) — CITS, cstaricalcutta.gov.in	Training Methodology outcomes and the official tools and equipment list for TM
DGT, Core Skills (Workshop Calculation & Science, Engineering Drawing, Soft Skills and Training Methodology) — CITS, NSQF Level 6, cstaricalcutta.gov.in	Engineering Technology and TM subject structure and tool lists
DGT / NCVT trade curriculum index, bharatskills.gov.in and dgt.gov.in	The full CTS and CITS trade list from which the trade table in Chapter 2 was compiled
NSTI websites (Trivandrum, Noida, Indore, Panipat, Vadodara, Mumbai, Bengaluru, Ludhiana, Allahabad, Patna, Agartala), *.dgt.gov.in	Current CTS and CITS offerings, NSQF levels, duration, eligibility, pass criteria and attendance rules
DGT, Craftsmen Training Scheme and Crafts Instructor Training Scheme scheme documents	Scheme history, network size, admission rules, certification and the AITT process
IS / BIS standards for hand tools, abrasive wheels, PPE and electrical accessories; IEC 60900 for insulated tools	Safety requirements quoted for insulated tools, guards and PPE
ISO 9001 / ISO 45001 and OSHA guidance on machine guarding, lockout/tagout and hazard communication	Hierarchy of control, guarding principles and safe working rules
Manufacturer operating manuals for machine tools, welding sets, refrigeration equipment and training aids	Parameter ranges, care routines and specific safety instructions

Table 11.2 — Principal sources.

12. Limitations and How to Keep This Document Current

12.1 It is a compilation, not a standard

This document has no statutory force. Where an institute's own safety manual, a manufacturer's instructions, a State Council syllabus or a statutory requirement differs, that document prevails. Parameters quoted here — speeds, pressures, torques, temperature ranges, clearances — are indicative and must be confirmed against the specific machine and material before use in training or in assessment.

12.2 Currency of the curriculum

DGT revises curricula periodically (the Fitter CTS is at version 3.0 and the Electrician at a revised CTS edition, while several CITS documents are at version 2.0 with newer revisions in circulation). New trades such as Drone Technician, Additive Manufacturing, Semiconductor Technician and Artificial Intelligence Programming Assistant have been introduced recently and their tool lists continue to evolve. Before using this compendium for an indent or an audit, check the current curriculum PDF for the specific trade on dgt.gov.in or bharatskills.gov.in.

12.3 Coverage

The register covers the tools that a trainee and an instructor must be able to identify, operate and use safely. Highly specialised equipment that appears in a single advanced trade (for example marine engine test beds, mining machinery or semiconductor fabrication trainers) is represented at class level. Regional and State Council variants may add items.

12.4 Maintenance of this document

Suggested cycle: review annually; update the tool register when a curriculum revision is notified; add a new group when a new trade introduces a new class of tool; and verify the Annexure-I references each year. Record each revision with a date and a short note so that instructors can see what changed.

12.5 Using it in an ITI

Three practical uses: as a teaching aid in the first-week tool identification sessions; as a job-sheet annexure, quoting the tool code so that the trainee knows exactly which tool and which precautions apply; and as a store and audit checklist for Annexure-I compliance, calibration and PPE issue.

Appendix A — Alphabetical Index of Entries

Every catalogued entry, alphabetically by name, with its code and group. Use the code to locate the entry in Part B and to quote it in a job sheet, store indent or inspection checklist.

Entry	Code	Group
Abrasive cut-off saw (chop saw)	C-008	Group C
Additive manufacturing (3D printer) and accessory tools	T-015	Group T
Adhesive, sealant and mechanical fastening tools	W-013	Group W
Air handling, duct and airflow instruments	R-008	Group R
Anti-static wrist strap, mat and ESD station	X-001	Group X
Arbor press and broaching press	T-012	Group T
Arc welding machine (transformer, inverter and rectifier)	W-001	Group W
Auto electrical tools — alternator, starter and charging system	A-012	Group A
Bakery equipment	S-011	Group S
Ball peen hammer	H-009	Group H

Entry	Code	Group
Bandsaw blade, hacksaw blade and jigsaw blade	C-013	Group C
Bar bending, cutting and steel fixing tools	B-011	Group B
Battery, charger and DC supply	E-013	Group E
Beauty and cosmetology tools	S-014	Group S
Belt sander, disc sander and linisher	C-017	Group C
Bench brush, hand brush and swarf rake	H-025	Group H
Bench grinder and pedestal grinder	C-010	Group C
Bench vice and leg vice	H-014	Group H
Black board and white board	TM-001	Group TM
Bore gauge (dial and telescopic)	M-023	Group M
Brake service tools	A-008	Group A
Brazing and silver soldering equipment	W-010	Group W
Brazing and soldering set for refrigeration	R-005	Group R
Brazing and welding inspection tools	W-012	Group W
Brinell / Rockwell hardness tester	M-026	Group M
C-clamp, toolmaker's clamp and hand vice	H-015	Group H
Cable pulling, jointing and termination tools	E-010	Group E
Carpentry hand tools — saw, plane, chisel, mallet, square	B-005	Group B
Centre lathe (all-geared headstock)	T-001	Group T
Centre punch, dot punch and prick punch	H-008	Group H
Charts, posters, models, mock-ups and specimens	TM-008	Group TM
Clamp meter (tong tester)	E-005	Group E
CNC lathe (turning centre)	T-009	Group T
CNC machining centre (vertical and horizontal)	T-010	Group T
Cold chisel (flat, cross-cut, diamond, round-nose)	H-011	Group H
Combination set	H-004	Group H
Commercial cooking equipment	S-010	Group S
Comparators — mechanical, electrical, pneumatic and stand mounted	M-013	Group M
Compression tester and cylinder leakage tester	A-006	Group A
Compressor service tools	R-007	Group R
Computer assembly and service tools	X-006	Group X
Concrete cutting and surface preparation tools	B-010	Group B
Conduit, bending and threading tools	E-009	Group E
Coolant system, chip conveyor and swarf handling	T-014	Group T
Coordinate measuring machine (CMM)	M-024	Group M
Cross peen and sledge hammer	H-010	Group H
Cutting tools — shears, rotary cutter and straight knife	S-002	Group S
Demolition hand tools	B-013	Group B
Desoldering pump, wick and hot air rework station	X-004	Group X
Dial test indicator with magnetic base	M-012	Group M

Entry	Code	Group
Digital camera, camcorder and tripod (training aid)	X-012	Group X
Digital micrometer and digital dial indicator	M-007	Group M
Digital vernier caliper	M-002	Group M
Digital video camera with tripod	TM-007	Group TM
Display, printer and scanner service tools	X-010	Group X
Dividers and wing compass	H-006	Group H
Drill chuck, key and Morse taper sleeve	C-002	Group C
Dumpy level, auto level and levelling staff	M-027	Group M
Earth tester and earth resistance measurement	E-007	Group E
Electric motors and starters (demonstration set)	E-011	Group E
Electrical safety equipment — LOTO, RCD and insulated mat	E-018	Group E
Electrician's knife and cable stripping tool	E-002	Group E
Electrode holder and welding cables	W-002	Group W
Electronic trainer kits and project boards	X-009	Group X
Embroidery and surface ornamentation tools	S-005	Group S
Engine repair hand tools — ring spanner, socket and puller sets	A-004	Group A
Engineer's try square	H-003	Group H
Feeler (thickness) gauge	M-017	Group M
File card and brush	H-024	Group H
Files — flat, hand, half-round, round, square, triangular, knife-edge and needle	H-012	Group H
Files, rasps and rotary burrs (power)	C-014	Group C
Fire fighting equipment (for all trades)	S-018	Group S
First aid kit and emergency equipment	S-019	Group S
Flannel board, magnetic board and display boards	TM-002	Group TM
Food preparation hand tools and machines	S-012	Group S
Food preservation and processing equipment	S-013	Group S
Formwork, shuttering and propping tools	B-012	Group B
Fuel injection and common-rail service tools	A-007	Group A
Gas cylinders and regulators	W-006	Group W
Gas welding and cutting set (oxy-acetylene / LPG)	W-005	Group W
Glass and glazing tools	B-015	Group B
Grinding machines — surface, cylindrical and tool and cutter	T-008	Group T
Hacksaw frame and blades	H-013	Group H
Hand drill and breast drill	H-023	Group H
Hand reamer, tap wrench and die stock	H-021	Group H
Hand snips, notcher and pneumatic shear	C-022	Group C
Handheld thermometer, infrared thermometer and pyrometer	M-033	Group M
Health care and hospital support equipment	S-016	Group S
Hole saw, annular cutter and step drill	C-006	Group C
Housekeeping and cleaning equipment	S-017	Group S

Entry	Code	Group
Hydraulic press and arbor press	C-023	Group C
Industrial sewing machine	S-001	Group S
Inside micrometer and telescopic (T-) gauge	M-006	Group M
Instrumentation for fluid measurement	R-013	Group R
Insulated hand tools with voltage rating	E-017	Group E
Interactive flat panel / smart board	TM-005	Group TM
Iron, steam press and fusing machine	S-004	Group S
Ladder, platform and insulated access equipment	E-020	Group E
Laser and optical tools (printer, DVD, photo equipment)	X-011	Group X
Laser level, line laser and rotary laser	M-029	Group M
Lathe chucks — three-jaw self-centring and four-jaw independent	T-002	Group T
Lathe cutting tools — HSS and carbide inserts	T-003	Group T
LCD / LED projector and screen	TM-004	Group TM
LCR meter, component tester and logic probe	E-015	Group E
Leak detection equipment	R-006	Group R
Limit gauges — plug, snap, ring and caliper	M-011	Group M
Long-nose plier, flush cutter and tweezers	X-003	Group X
Lubrication and fluid service equipment	A-013	Group A
Machine tool alignment tools	T-013	Group T
Machine vice, angle plate and clamping kit	T-011	Group T
Magnetic base drilling machine	C-007	Group C
Manifold gauge set and hoses	R-001	Group R
Masonry drill, hammer and chisel	B-002	Group B
Measuring tape (steel, retractable)	H-002	Group H
Measuring, levelling and setting-out tools for building	B-004	Group B
Measuring, marking and drafting tools for apparel	S-003	Group S
Megger (insulation resistance tester)	E-006	Group E
Metal cutting chisels, trepanning tool and tube cutter	C-021	Group C
Micrometer, bore gauge and Plastigauge for engine measurement	A-005	Group A
MIG / MAG welding set (GMAW)	W-008	Group W
Milling cutters and arbors	T-006	Group T
Mixing tools — trough, mixer and vibrator	B-003	Group B
Multimedia computer, UPS and printer-scanner	TM-006	Group TM
Multimeter (analogue and digital)	E-004	Group E
Multimeter, logic probe and battery tester	X-005	Group X
Neon tester and voltage detector	E-003	Group E
Network tools — crimper, punch-down, tester and tone generator	X-007	Group X
Oil can, oil stone and abrasive stone	H-026	Group H
Optical profile projector and toolmaker's microscope	M-025	Group M
Oscilloscope and signal generator	E-014	Group E

Entry	Code	Group
Outside micrometer	M-005	Group M
Overhead projector (OHP) and transparencies	TM-003	Group TM
Oxy-fuel cutting torch with nozzles and tip cleaner	W-014	Group W
Painting and decorating tools	B-008	Group B
Photography, image capture and processing equipment	S-009	Group S
Pillar (bench) drilling machine	C-004	Group C
Pipe and fitting tools — pipe vice, wrench, cutter and threading	R-009	Group R
Pipe cutter and pipe threading die set	H-032	Group H
Pipe wrench, pipe vice and chain vice	H-031	Group H
Plastering and rendering machine	B-016	Group B
Pliers set — combination, nose, side cutting and crimping	E-001	Group E
Pliers — combination, long nose, side cutting, water pump and circlip	H-019	Group H
Plumb bob, square, bevel and marking gauge (building and wood)	M-030	Group M
Plumbing hand tools — adjustable wrench, basin wrench, plunger, auger and drain snake	R-010	Group R
Portable angle grinder	C-011	Group C
Portable electric drill	C-003	Group C
Portable power woodworking tools — circular saw, jigsaw, router, planer	B-006	Group B
Power hacksaw and band saw	C-009	Group C
Precision screwdriver and bit set	X-002	Group X
Printing press equipment — offset, digital and screen	S-008	Group S
Pumps, valves and fluid machinery (demonstration)	R-012	Group R
Punches — letter and number punch sets	H-029	Group H
Punches, chisels and hammers for masonry and carpentry	H-033	Group H
Radial arm drill and gang drill	C-005	Group C
Radius gauge and fillet gauge	M-018	Group M
Reamer, counterbore and countersink	C-015	Group C
Refrigerant recovery and charging machine	R-003	Group R
Riveting tools — hand, pneumatic and pop riveter	W-011	Group W
Scaffolding, ladder and working platform	B-009	Group B
Scrapers and hand scrapers	H-022	Group H
Screw extractor (Ezy-out) and stud remover	H-030	Group H
Screw pitch gauge	M-009	Group M
Screw thread micrometer and three-wire method	M-008	Group M
Screwdrivers — slotted, Phillips, Pozidriv, Torx and insulated	H-016	Group H
Scriber (straight and bent)	H-005	Group H
Shaping machine and planing machine	T-007	Group T
Sharpening equipment — oilstone, diamond hone, grinder jig	C-024	Group C
Shearing machine, guillotine and nibbler	C-012	Group C
Signal generators, power supply (bench) and frequency counter	X-008	Group X

Entry	Code	Group
Sine bar and sine centre	M-015	Group M
Skin and nail care equipment	S-015	Group S
Slip gauge (gauge block) set	M-016	Group M
Socket set, torque wrench and Allen keys	H-018	Group H
Solar PV tools and test gear	E-019	Group E
Soldering and desoldering station with ESD protection	E-016	Group E
Soldering irons, soldering station and solder	W-009	Group W
Spanners — open end, ring, combination, box and adjustable	H-017	Group H
Spirit level and precision level	M-021	Group M
Spring calipers — outside, inside and hermaphrodite	H-007	Group H
Steel rule (bench rule)	H-001	Group H
Straight edge and feeler for flatness	M-022	Group M
Surface plate and marking-off table	H-028	Group H
Surface roughness comparator plates	M-020	Group M
Suspension and steering tools	A-009	Group A
Tachometer and stroboscope	M-031	Group M
Tailstock, live centre, dead centre and steady rest	T-004	Group T
Tap, die and thread rolling head	C-016	Group C
Textile machinery — loom, knitting and spinning	S-006	Group S
Textile wet processing equipment	S-007	Group S
Theodolite and total station	M-028	Group M
Thread plug gauge and thread ring gauge (Go / No-Go)	M-010	Group M
TIG welding set (GTAW)	W-007	Group W
Tile cutting and fixing tools	B-014	Group B
Tin snips and aviation shears	H-020	Group H
Torque wrench and angle gauge	A-003	Group A
Trainee progress and assessment records	TM-010	Group TM
Trammel and large-diameter measuring tools	M-034	Group M
Transformer and winding equipment	E-012	Group E
Trolley jack and axle stands	A-001	Group A
Trowel, float and hawk	B-001	Group B
Tube working tools — bender, flare, swage and cutter	R-004	Group R
Twist drill (straight and taper shank)	C-001	Group C
Tyre service equipment	A-010	Group A
Ultrasonic and abrasive jet cutting (advanced)	C-025	Group C
V-block, angle plate, parallel block and surface gauge	H-027	Group H
Vacuum pump and micron gauge	R-002	Group R
Vehicle body repair tools	A-014	Group A
Vehicle diagnostic scanner and oscilloscope	A-011	Group A
Vehicle hoist and service pit	A-002	Group A

Entry	Code	Group
Vernier bevel protractor	M-014	Group M
Vernier caliper (analogue)	M-001	Group M
Vernier depth gauge and depth micrometer	M-004	Group M
Vernier height gauge	M-003	Group M
Vertical milling machine and horizontal milling machine	T-005	Group T
Vibration meter, sound level meter and lux meter	M-032	Group M
Water pressure test pump and leak test equipment	R-011	Group R
Welding and fabrication for pipework	R-014	Group R
Welding helmet, hand shield and filter glass	W-003	Group W
Welding table, positioner, clamps and fixtures	W-004	Group W
Wire gauge and sheet metal gauge	M-019	Group M
Wiring accessories — switches, sockets, holders and MCBs	E-008	Group E
Woodworking circular saw and table saw	C-018	Group C
Woodworking machine tools — circular saw bench, planer and spindle moulder	B-007	Group B
Woodworking planer, thicknesser and jointer	C-019	Group C
Woodworking router and spindle moulder	C-020	Group C
Written instructional material templates	TM-009	Group TM

Table A.1 — 226 entries including the Training Methodology aids.

End of document.